

Analysis and Control of Evolution Equations: a friendly introduction

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Abstract In these notes, the reader is introduced to some of the main results of the theory of abstract evolution equations and their controllability properties under the action of an additive control. The material is almost uniformly distributed among three main topics: the generation of strongly continuous semigroups of bounded linear operators, the solvability of the associated inhomogeneous Cauchy problem, and the basic controllability properties (namely, approximate, null, and exact controllability) of abstract linear control systems in continuous time. These topics were covered in several graduate courses I taught at the University of Rome Tor Vergata and formed the basis of my lectures at the CIME School on "Inverse Problems and Controllability for PDEs in the Direction of Nonlocal Operators", held in Montecatini Terme on June 17-21, 2024, under the direction of Giuseppe Floridia, Paola Loreti, and Masahiro Yamamoto. In order to maximize inclusivity, I have assumed a minimum amount of prerequisites: Lebesgue's integral and the basic theory of Hilbert and Banach spaces. Consequently, in most examples dealing with partial differential equations, I only treat the case of one spatial dimension, where Sobolev functions can be quickly defined. Similarly, to avoid developing Bochner's integral, the inhomogeneous Cauchy problem (and the corresponding control systems) is addressed within Hilbert space settings. I wish to express my gratitude to the directors of the CIME School in Montecatini Terme for giving me this opportunity, to the Ph.D. students and researchers who attended my lectures at CIME, and to all the students who participated in my courses on Control, Dynamics, and Optimization at Tor Vergata. Last but not least, I dedicate these lecture notes to Giuseppe Da Prato, who was a mentor to me in these topics and much more.

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Notation

$\mathbb{R}, \mathbb{N}, \mathbb{Z}$	Real, natural, and integer numbers
$\mathbb{R}^+, \mathbb{R}^{*+}$	$[0, \infty)$ and $(0, \infty)$
$\mathbb{N}^*, \mathbb{Z}^*$	$\mathbb{N} \setminus \{0\}, \mathbb{Z} \setminus \{0\}$
$\lfloor \tau \rfloor, \{\tau\}$	Integer and fractional parts of $\tau \in \mathbb{R}$
$\Re \lambda, \Im \lambda$	Real and imaginary parts of $\lambda \in \mathbb{C}$
$\ \cdot\ $	Norm in a Banach space X or modulus of a real/complex number
$\mathcal{L}(X)$	Space of bounded linear operators on a Banach space X
$C_b(X)$	Bounded uniformly continuous functions on a metric space X
$\text{osc } f(\delta)$	Oscillation of f on sets of diameter δ
$C_b(I; X)$	Bounded, uniformly continuous X -valued functions on I
$C_b^1(I; X)$	Continuously differentiable functions $f \in C_b(I; X)$ such that $f' \in C_b(I; X)$
$L^p(I)$	p -integrable functions on I , $1 \leq p < \infty$
$D(A)$	Domain of operator A
$C(a, b; X)$	Continuous X -valued functions on $[a, b]$ (also denoted by $C([a, b]; X)$)
Π_ω	Half-plane $\{\lambda \in \mathbb{C} : \Re \lambda > \omega\}$
e^{tA}	Exponential of bounded operator A generating a semigroup
$R(\lambda, A)$	Resolvent of A : $(\lambda I - A)^{-1}$
$\sigma(A), \rho(A)$	Spectrum and resolvent set of A
$H^k(\Omega)$	Sobolev space of order k on Ω
$H_0^1(\Omega)$	Sobolev space with zero trace on $\partial\Omega$
$\ f\ _\infty, \ f\ _p$	Supremum norm, L^p -norm
C_0 -semigroup	Strongly continuous semigroup on X

1 Semigroups of bounded linear operators

Let $(X, |\cdot|)$ be a (real or complex) Banach space. We denote by $\mathcal{L}(X)$ the Banach space of all bounded linear operators $\Lambda : X \rightarrow X$ with norm

$$\|\Lambda\| = \sup_{|u| \leq 1} |\Lambda u|.$$

We recall that, for any given $A, B \in \mathcal{L}(X)$, the product AB remains in $\mathcal{L}(X)$ and we have that

$$\|AB\| \leq \|A\| \|B\|. \quad (1)$$

So, $\mathcal{L}(X)$ is a Banach algebra.

Proposition 1 *Let $A \in \mathcal{L}(X)$ be such that $\|A\| < 1$. Then $(I - A)^{-1} \in \mathcal{L}(X)$ and*

$$(I - A)^{-1} = \sum_{n=0}^{\infty} A^n. \quad (2)$$

Proof. We observe that the series on the right-hand side of (2) is totally convergent in $\mathcal{L}(X)$. So,

$$\Lambda := \sum_{n=0}^{\infty} A^n \in \mathcal{L}(X).$$

Moreover,

$$(I - A)\Lambda = \sum_{n=0}^{\infty} (I - A)A^n = I = \sum_{n=0}^{\infty} A^n(I - A) = \Lambda(I - A)$$

which completes the proof. $\square$

1.1 Uniformly continuous semigroups

We now introduce some notions that are fundamental for our approach.

Definition 1 A semigroup of bounded linear operators on X is a map

$$S : [0, \infty) \rightarrow \mathcal{L}(X)$$

with the following properties:

- (a) $S(0) = I$,
- (b) $S(t + s) = S(t)S(s)$ for all $t, s \geq 0$.

We will use the equivalent notation $\{S(t)\}_{t \geq 0}$ and the abbreviated form $S(t)$.

Definition 2 The infinitesimal generator of a semigroup of bounded linear operators $S(t)$ is the map $A : D(A) \subset X \rightarrow X$ defined by

$$\begin{cases} D(A) = \{u \in X : \exists \lim_{t \downarrow 0} \frac{S(t)u - u}{t}\} \\ Au = \lim_{t \downarrow 0} \frac{S(t)u - u}{t} \end{cases} \quad \forall u \in D(A) \quad (3)$$

Exercise 1 Let $A : D(A) \subset X \rightarrow X$ be the infinitesimal generator of a semigroup of bounded linear operators $S(t)$. Prove that

- (a) $D(A)$ is a subspace of X ,
- (b) A is a linear operator.

We begin by studying a special class of semigroups.

Definition 3 A semigroup $S(t)$ of bounded linear operators on X is *uniformly continuous* if

$$\lim_{t \downarrow 0} \|S(t) - I\| = 0.$$

Semigroups enjoy an import asymptotic behaviour, as we show below. We will see in the next section that such a property can be extended to more general families of semigroups.

Proposition 2 Let $S(t)$ be a uniformly continuous semigroup of bounded linear operators. Then there exists $M \geq 1$ and $\omega \in \mathbb{R}$ such that

$$\|S(t)\| \leq M e^{\omega t} \quad \forall t \geq 0.$$

Proof. Let $\tau > 0$ be such that $\|S(t) - I\| \leq 1/2$ for all $t \in [0, \tau]$. Then

$$\|S(t)\| \leq \|I\| + \|S(t) - I\| \leq \frac{3}{2} \quad \forall t \in [0, \tau].$$

Since every $t \geq 0$ can be represented as $t = \lceil t/\tau \rceil \tau + \{t/\tau\} \tau$, we have that

$$\|S(t)\| \leq \|S(\tau)\|^{\lceil t/\tau \rceil} \|S(\{t/\tau\} \tau)\| \leq \left(\frac{3}{2}\right)^{\lceil t/\tau \rceil + 1} \leq \left(\frac{3}{2}\right)^{\frac{t}{\tau} + 1} = M e^{\omega t}$$

with $M = 3/2$ and $\omega = \log(3/2)/\tau$. □

By applying the above theorem it is easy to deduce the following result.

Corollary 1 A semigroup $S(t)$ is uniformly continuous if and only if

$$\lim_{s \rightarrow t} \|S(s) - S(t)\| = 0 \quad \forall t \geq 0.$$

We will now discuss an important example of semigroups.

Example 1 Let $A \in \mathcal{L}(X)$. Then

$$e^{tA} := \sum_{n=0}^{\infty} \frac{t^n}{n!} A^n$$

is a uniformly continuous semigroup of bounded linear operators on X . More precisely, the following properties hold.

(a) $\sum_{n=0}^{\infty} \frac{t^n}{n!} A^n$ converges for all $t \geq 0$ and $e^{tA} \in \mathcal{L}(X)$.

Proof. Indeed, the series is totally convergent in $\mathcal{L}(X)$ because

$$\sum_{n=0}^{\infty} \left\| \frac{t^n}{n!} A^n \right\| \leq \sum_{n=0}^{\infty} \frac{t^n}{n!} \|A\|^n < \infty.$$

The conclusion follows. $\square$

(b) $e^{(t+s)A} = e^{tA} e^{sA}$ for all $s, t \geq 0$.

Proof. We have that

$$\begin{aligned} e^{(t+s)A} &= \sum_{n=0}^{\infty} \frac{(t+s)^n}{n!} A^n = \sum_{n=0}^{\infty} \frac{A^n}{n!} \sum_{k=0}^n \binom{n}{k} t^k s^{n-k} \\ &= \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{t^k A^k}{k!} \frac{s^{n-k} A^{n-k}}{(n-k)!} \end{aligned}$$

where the last term coincides with the Cauchy product of the two series giving e^{tA} and e^{sA} . $\square$

(c) $Ae^{tA} = e^{tA}A$ for all $t \geq 0$.

(d) $\|e^{tA} - I\| = \left\| \sum_{n=1}^{\infty} \frac{t^n}{n!} A^n \right\| \leq t\|A\|e^{t\|A\|}$ for all $t \geq 0$.

(e) $\left\| \frac{e^{tA} - I}{t} - A \right\| = \left\| \sum_{n=2}^{\infty} \frac{t^{n-1}}{n!} A^n \right\| \leq t\|A\|^2 e^{t\|A\|}$ for all $t \geq 0$.

Notice that property (e) shows that A is the infinitesimal generator of e^{tA} .

In fact, being the exponential of a bounded operator characterizes all uniformly continuous semigroups, as we show next.

Theorem 1 *For any linear operator $A : D(A) \subset X \rightarrow X$ the following properties are equivalent:*

- (a) A is the infinitesimal generator of a uniformly continuous semigroup,
- (b) $D(A) = X$ and $A \in \mathcal{L}(X)$.

Proof. Example 1 shows that (b) $\Rightarrow$ (a). Let us prove that (a) $\Rightarrow$ (b). Let $\tau > 0$ be fixed such that

$$\left\| I - \frac{1}{\tau} \int_0^{\tau} S(t) dt \right\| < 1.$$

Then the bounded linear operator $\int_0^{\tau} S(t) dt$ is invertible. For all $h > 0$ we have that

$$\begin{aligned} \frac{S(h) - I}{h} \int_0^{\tau} S(t) dt &= \frac{1}{h} \left(\int_0^{\tau} S(t+h) dt - \int_0^{\tau} S(t) dt \right) \\ &= \frac{1}{h} \left(\int_h^{\tau+h} S(t) dt - \int_0^{\tau} S(t) dt \right) = \frac{1}{h} \left(\int_{\tau}^{\tau+h} S(t) dt - \int_0^h S(t) dt \right). \end{aligned}$$

Hence

$$\begin{array}{ccc} \frac{S(h)-I}{h} & = & \frac{1}{h} \left(\int_{\tau}^{\tau+h} S(t)dt - \int_0^h S(t)dt \right) \left(\int_0^{\tau} S(t)dt \right)^{-1} \\ \downarrow & h \downarrow 0 & \downarrow \\ A & = & (S(\tau) - I) \left(\int_0^{\tau} S(t)dt \right)^{-1}. \end{array}$$

This shows that $A \in \mathcal{L}(X)$. $\square$

The above theorem ensures that, if S is a uniformly continuous semigroup with infinitesimal generator $A \in \mathcal{L}(X)$, then for any $u_0 \in X$ the function $u(t) = S(t)u_0$ is of class $C^1([0, \infty[; X)$ and solves the Cauchy problem

$$\begin{cases} u'(t) = Au(t) & t > 0 \\ u(0) = u_0. \end{cases} \quad (4)$$

Moreover, $S(t) = e^{tA}$ as our next result shows.

Proposition 3 *Problem (4) has a unique solution given by $u(t) = e^{tA}u_0$.*

Proof. The fact that $u(t) = e^{tA}u_0$ solves (4) follows from Example 1. We proceed to prove that u is the unique solution of (4). Let $v \in C^1([0, \infty[; X)$ be another solution of the above problem. Fix any $t > 0$ and set $U(s) = e^{(t-s)A}v(s)$ for all $s \in [0, t]$. Then

$$U'(s) = -Ae^{(t-s)A}v(s) + e^{(t-s)A}Av(s) = 0 \quad \forall s \in [0, t].$$

Therefore, U is constant on $[0, t]$ by Corollary 8 of Appendix A. So, $v(t) = U(t) = U(0) = e^{tA}u_0$. $\square$

The following class of problems can be solved by the above existence and uniqueness result.

Example 2 Consider the integral equation

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) = \int_0^1 k(x, y)u(t, y) dy & t > 0 \\ u(0, x) = u_0(x) \end{cases} \quad (5)$$

where $k \in L^2([0, 1] \times [0, 1])$ and $u_0 \in L^2(0, 1)$. Problem (5) can be recast in the abstract form (4) taking $X = L^2(0, 1)$ and

$$Au(x) = \int_0^1 k(x, y)u(y) dy \quad \text{for a.e. } x \in (0, 1).$$

Then Proposition 3 insures that (5) has a unique solution $u \in C^1([0, \infty[; X)$ given by $u(t) = e^{tA}u_0$.

1.2 Strongly continuous semigroups

There are important examples of semigroups that do not fall into the class we examined in the previous section. Our next example is a case in point.

Example 3 (Translations on $\mathbb{R}$) Let $C_b(\mathbb{R})$ be the Banach space of all bounded uniformly continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ with the uniform norm

$$\|f\|_\infty = \sup_{x \in \mathbb{R}} |f(x)|.$$

For any $t \in \mathbb{R}_+$ define

$$(S(t)f)(x) = f(x+t) \quad \forall f \in C_b(\mathbb{R}).$$

The following holds true.

1. $S(t)$ is a semigroup of bounded linear operators on $C_b(\mathbb{R})$.
2. $S(t)$ fails to be uniformly continuous.

Proof. For any $n \in \mathbb{N}$ the function

$$f_n(x) = e^{-nx^2} \quad (x \in \mathbb{R})$$

belongs to $C_b(\mathbb{R})$ and has norm equal to 1. Therefore, for any $t > 0$

$$\|S(t) - I\| \geq \|S(t)f_n - f_n\|_\infty = \sup_{x \in \mathbb{R}} |e^{-n(x+t)^2} - e^{-nx^2}| \geq 1 - e^{-nt^2}.$$

Since this is true for any n , we have that $\|S(t) - I\| \geq 1$. □

3. For all $f \in C_b(\mathbb{R})$ we have that $\|S(t)f - f\|_\infty \rightarrow 0$ as $t \downarrow 0$.

Proof. Indeed,

$$\|S(t)f - f\|_\infty = \sup_{x \in \mathbb{R}} |f(x+t) - f(x)| \leq \text{osc}_f(t) \xrightarrow{t \downarrow 0} 0$$

This completes the proof. □

Motivated by the above example, we now introduce a notion that will be in use throughout these notes.

Definition 4 A semigroup $S(t)$ of bounded linear operators on X is called *strongly continuous* (or of *class C_0* , or even a *C_0 -semigroup*) if

$$\lim_{t \downarrow 0} S(t)u = u \quad \forall u \in X. \quad (6)$$

First, we show that strongly continuous semigroups have a similar asymptotic behaviour as the one observed in Proposition 2.

Theorem 2 *Let $S(t)$ be a C_0 -semigroup of bounded linear operators on X . Then there exist $\omega \geq 0$ and $M \geq 1$ such that*

$$\|S(t)\| \leq M e^{\omega t} \quad \forall t \geq 0. \quad (7)$$

Proof. We first prove the following:

$$\exists \tau > 0 \quad \text{and} \quad M \geq 1 \quad \text{such that} \quad \|S(t)\| \leq M \quad \forall t \in [0, \tau]. \quad (8)$$

We argue by contradiction assuming there exists a sequence $t_n \downarrow 0$ such that $\|S(t_n)\| \geq n$ for all $n \geq 1$. Then, the principle of uniform boundedness implies that, for some $u \in X$,

$$\sup_{n \geq 1} \|S(t_n)u\| = \infty,$$

in contrast with (6), since $S(t_n)u \rightarrow u$ as $n \rightarrow \infty$.

Now, given $t \in \mathbb{R}_+$, let $n \in \mathbb{N}$ and $\delta \in [0, \tau]$ be such that

$$t = n\tau + \delta.$$

Then, in view of (8),

$$\|S(t)\| = \|S(\delta)S(\tau)^n\| \leq M \cdot M^n = M \cdot (M^{1/\tau})^{n\tau} \leq M \cdot (M^{1/\tau})^t$$

which yields (7) with $\omega = \frac{\log M}{\tau}$. $\square$

As in the case of uniformly continuous semigroups, Theorem 2 has the following immediate consequence.

Corollary 2 *Let $S(t)$ be a C_0 -semigroup of bounded linear operators on X . Then for every $u \in X$ the map $t \mapsto S(t)u$ is continuous from $\mathbb{R}_+$ into X .*

We now introduce specific classes of strongly continuous semigroups.

Definition 5 A C_0 -semigroup of bounded linear operators on X is called *uniformly bounded* if $S(t)$ satisfies (7) with $\omega = 0$. If, in addition, $M = 1$, we say that $S(t)$ is a *contraction semigroup*.

Exercise 2 Prove that the translation semigroup of Example 3 satisfies

$$\|S(t)\| = 1 \quad \forall t \geq 0.$$

So, $S(t)$ is a contraction semigroup.

Exercise 3 For any fixed $p \geq 1$, let $X = L^p(\mathbb{R})$ and define, $\forall f \in X$,

$$(S(t)f)(x) = f(x+t) \quad \forall x \in \mathbb{R}, \forall t \geq 0. \quad (9)$$

Prove that S is C_0 -semigroup which fails to be uniformly continuous.

Solution. Suppose S is uniformly continuous and let $\tau > 0$ be such that $\|S(t) - I\| < 1/2$ for all $t \in [0, \tau]$. Then by taking $f_n(x) = n^{1/p} \chi_{[0, 1/n]}(x)$ for $p < \infty$ and $n > 1/\tau$ we have that $\|f_n\| = 1$ and

$$\|S(\tau)f_n - f_n\| = \left(\int_{\mathbb{R}} n |\chi_{[0, 1/n]}(x + \tau) - \chi_{[0, 1/n]}(x)|^p dx \right)^{\frac{1}{p}} = 2^{1/p}. \quad \square$$

The next exercise shows that any uniformly bounded semigroup can be recast into a contraction one by a suitable change of norm.

Exercise 4 Given a uniformly bounded C_0 -semigroup, $\|S(t)\| \leq M$, define

$$|u|_S = \sup_{t \geq 0} |S(t)u|, \quad \forall u \in X. \quad (10)$$

Show that:

- (a) $|\cdot|_S$ is a norm on X ,
- (b) $|u| \leq |u|_S \leq M|u|$ for all $u \in X$, and
- (c) S is a contraction semigroup with respect to $|\cdot|_S$.

1.3 The infinitesimal generator of a C_0 -semigroup

The following results derives some of the main properties of the infinitesimal generator of a C_0 -semigroup of bounded linear operators on X .

Theorem 3 Let $A : D(A) \subset X \rightarrow X$ be the infinitesimal generator of a C_0 -semigroup of bounded linear operators on X , denoted by $S(t)$. Then the following properties hold true.

- (a) For all $t \geq 0$

$$\lim_{h \downarrow 0} \frac{1}{h} \int_t^{t+h} S(s)u \, ds = S(t)u \quad \forall u \in X.$$

- (b) For all $t \geq 0$ and $u \in X$

$$\int_0^t S(s)u \, ds \in D(A) \quad \text{and} \quad A \int_0^t S(s)u \, ds = S(t)u - u.$$

- (c) $D(A)$ is dense in X .
- (d) For all $u \in D(A)$ and $t \geq 0$ we have that $S(t)u \in D(A)$, $t \mapsto S(t)u$ is continuously differentiable, and

$$\frac{d}{dt} S(t)u = AS(t)u = S(t)Au.$$

- (e) For all $u \in D(A)$ and all $0 \leq s \leq t$ we have that

$$S(t)u - S(s)u = \int_s^t S(\tau)Au \, d\tau = \int_s^t AS(\tau)u \, d\tau.$$

Proof. All integrals are to be understood in the Cauchy sense.

- (a) This point is an immediate consequence of the strong continuity of S .
- (b) For any $t \geq h > 0$ we have that

$$\begin{aligned} \frac{S(h) - I}{h} \left(\int_0^t S(s)u \, ds \right) &= \frac{1}{h} \int_0^t (S(h+s) - S(s))u \, ds \\ &= \frac{1}{h} \left(\int_h^{t+h} S(s)u \, ds - \int_0^t S(s)u \, ds \right) \\ &= \frac{1}{h} \left(\int_t^{t+h} S(s)u \, ds - \int_0^h S(s)u \, ds \right). \end{aligned}$$

Therefore, by (a),

$$\lim_{h \downarrow 0} \frac{S(h) - I}{h} \left(\int_0^t S(s)u \, ds \right) = S(t)u - u$$

which proves (b).

- (c) This point follows from (a) and (b).
- (d) For all $u \in D(A)$, $t \geq 0$, and $h > 0$ we have that

$$\frac{S(h) - I}{h} S(t)u = S(t) \frac{S(h) - I}{h} u \rightarrow S(t)Au \quad \text{as } h \downarrow 0.$$

Therefore, $S(t)u \in D(A)$ and $AS(t)u = S(t)Au = \frac{d^+}{dt} S(t)u$. In order to prove the existence of the left derivative, observe that for all $0 < h < t$

$$\frac{S(t-h)u - S(t)u}{-h} = S(t-h) \frac{S(h) - I}{h} u.$$

Moreover, by (7),

$$\begin{aligned} \left| S(t-h) \frac{S(h) - I}{h} u - S(t)Au \right| &\leq |S(t-h)| \cdot \left| \frac{S(h) - I}{h} u - S(h)Au \right| \\ &\leq M e^{\omega t} \left| \frac{S(h) - I}{h} u - S(h)Au \right| \xrightarrow{h \downarrow 0} 0. \end{aligned}$$

So,

$$\frac{S(t-h)u - S(t)u}{-h} \rightarrow S(t)Au = AS(t)u \quad \text{as } h \downarrow 0.$$

This shows that the left and right derivatives coincide.

(e) This point follows from (d).

The proof is complete. $\square$

Exercise 5 Show that the infinitesimal generator of the C_0 -semigroup of left translations on $\mathbb{R}$ we introduced in Example 3 is given by

$$\begin{cases} D(A) = C_b^1(\mathbb{R}) \\ Af = f' \end{cases} \quad \forall f \in D(A).$$

Solution. For any $f \in C_b^1(\mathbb{R})$ we have that

$$\left| \frac{S(t)f - f}{t} - f' \right|_\infty = \sup_{x \in \mathbb{R}} \left| \frac{f(x+t) - f(x)}{t} - f'(x) \right| \leq \text{osc}_{f'}(t) \xrightarrow{t \downarrow 0} 0$$

Therefore, $C_b^1(\mathbb{R}) \subset D(A)$ and $Af = f'$ for all $f \in C_b^1(\mathbb{R})$.

Conversely, let $f \in D(A)$. Then, $Af \in C_b(\mathbb{R})$ and

$$\sup_{x \in \mathbb{R}} \left| \frac{f(x+t) - f(x)}{t} - Af(x) \right| \xrightarrow{t \downarrow 0} 0. \quad (11)$$

So, the right derivative $f'_+(x)$ exists for all $x \in \mathbb{R}$ and equals $Af(x)$. Moreover, for any $t > 0$ we have that

$$\begin{aligned} \sup_{x \in \mathbb{R}} \left| \frac{f(x-t) - f(x)}{-t} - Af(x) \right| &\leq \sup_{x \in \mathbb{R}} \left| \frac{f(x) - f(x-t)}{t} - Af(x-t) \right| \\ &\quad + \sup_{x \in \mathbb{R}} |Af(x-t) - Af(x)| \xrightarrow{t \downarrow 0} 0 \end{aligned}$$

So, the left derivative $f'_-(x)$ also exists for all $x \in \mathbb{R}$ and equals $Af(x)$. Thus, $D(A) \subset C_b^1(\mathbb{R})$ and $Af = f'$. $\square$

Exercise 6 Let $A : D(A) \subset X \rightarrow X$ be the infinitesimal generator of a uniformly bounded semigroup $\|S(t)\| \leq M$. Prove the *Laundau-Kolmogorov inequality*:

$$|Au|^2 \leq 4M^2 |u| |A^2u| \quad \forall u \in D(A^2), \quad (12)$$

where

$$\begin{cases} D(A^2) = \{u \in D(A) : Au \in D(A)\} \\ A^2u = A(Au), \quad \forall u \in D(A^2). \end{cases} \quad (13)$$

Solution. Assume $M = 1$. For any $u \in D(A^2)$ and all $t \geq 0$ we have

$$\begin{aligned} \int_0^t (t-s)S(s)A^2u \, ds &= [(t-s)S(s)Au]_{s=0}^{s=t} + \int_0^t S(s)Au \, ds \\ &= -tAu + [S(s)u]_{s=0}^{s=t} = -tAu + S(t)u - u. \end{aligned}$$

Therefore, for all $t > 0$,

$$\begin{aligned} |Au| &\leq \frac{1}{t} |S(t)u - u| + \frac{1}{t} \int_0^t (t-s) |S(s)A^2u| ds \\ &\leq \frac{2}{t} |u| + \frac{t}{2} |A^2u|. \end{aligned} \quad (14)$$

If $A^2u = 0$, then the above inequality yields $Au = 0$ by letting $t \rightarrow \infty$. So, (12) is true in this case. On the other hand, for $A^2u \neq 0$ the function of t on the right-hand side of (14) attains its minimum at

$$t_0 = \frac{2|u|^{1/2}}{|A^2u|^{1/2}}.$$

By taking $t = t_0$ in (14) we obtain (12) once again. $\square$

? Question

How to treat the case of $M \neq 1$? *Hint*: recall Exercise 4.

Exercise 7 Use the Landau-Kolmogorov inequality to deduce the interpolation inequality

$$|f'|_\infty^2 \leq 4 |f|_\infty |f''|_\infty \quad \forall f \in C_b^2(\mathbb{R}).$$

1.4 Closed operators

We recall that $X \times X$ is a Banach space with norm

$$\|(u, v)\| = |u| + |v| \quad \forall (u, v) \in X \times X.$$

Definition 6 An operator $A : D(A) \subset X \rightarrow X$ is said to be *closed* if its graph

$$\text{graph}(A) := \{(u, v) \in X \times X : u \in D(A), v = Au\}$$

is a closed subset of $X \times X$.

The following characterisation of closed operators is straightforward.

Proposition 4 *The linear operator $A : D(A) \subset X \rightarrow X$ is closed if and only if, for any sequence $\{u_n\} \subset D(A)$, the following holds:*

$$\begin{cases} u_n \rightarrow u \\ Au_n \rightarrow v \end{cases} \implies u \in D(A) \quad \text{and} \quad Au = v. \quad (15)$$

Example 4 In the Banach space $X = C_b(\mathbb{R})$, the linear operator

$$\begin{cases} D(A) = C_b^1(\mathbb{R}) \\ Af = f' \end{cases} \quad \forall f \in D(A).$$

is closed. Indeed, for any sequence $\{f_n\} \subset C_b^1(\mathbb{R})$ such that

$$\begin{cases} f_n \rightarrow f & \text{in } C_b(\mathbb{R}) \\ f'_n \rightarrow g & \text{in } C_b(\mathbb{R}), \end{cases}$$

we have that $f \in C_b^1(\mathbb{R})$ and $f' = g$.

Example 5 In the Banach space $X = C([0, 1])$ with the uniform norm, the linear operator

$$\begin{cases} D(A) = C^1([0, 1]) \\ (Af)(x) = f'(0) \end{cases} \quad \forall x \in [0, 1]$$

fails to be closed. Indeed, for any $n \geq 1$ let

$$f_n(x) = \frac{\sin(nx)}{n} \quad x \in [0, 1].$$

Then

$$\begin{cases} D(A) \ni f_n \rightarrow 0 & \text{in } C_b(\mathbb{R}) \\ Af_n = 1 & \forall n \geq 1, \end{cases}$$

in contrast with (15).

Exercise 8 Prove that if $A : D(A) \subset X \rightarrow X$ is a closed operator and $B \in \mathcal{L}(X)$, then $A + B : D(A) \subset X \rightarrow X$ is also closed. What about BA ?

Exercise 9 Prove that, if $A : D(A) \subset X \rightarrow X$ is a closed operator and $f \in C([a, b]; D(A))$, where $D(A)$ is endowed with the graph norm, then

$$A \int_a^b f(t) dt = \int_a^b Af(t) dt. \quad (16)$$

Solution. Let $\pi_n = \{t_i^n\}_{i=0}^{i_n} \in \Pi(a, b)$ be such that $\text{diam}(\pi_n) \rightarrow 0$ and let $\sigma_n = \{s_i^n\}_{i=1}^{i_n} \in \Sigma(\pi_n)$. Then

$$\begin{cases} S_{\pi_n}^{\sigma_n}(f) = \sum_{i=1}^{i_n} f(s_i^n)(t_i^n - t_{i-1}^n) \longrightarrow \int_a^b f(t) dt \\ AS_{\pi_n}^{\sigma_n}(f) = \sum_{i=1}^{i_n} Af(s_i^n)(t_i^n - t_{i-1}^n) \longrightarrow \int_a^b Af(t) dt \end{cases} \quad (n \rightarrow \infty)$$

in X . Therefore, by Proposition 4,

$$\int_a^b f(t) dt \in D(A)$$

and (16) holds. $\square$

Our next goal is to show that the infinitesimal generator is closed.

Proposition 5 *The infinitesimal generator of a C_0 -semigroup $S(t)$ is a closed operator.*

Proof. Let $A : D(A) \subset X \rightarrow X$ be the infinitesimal generator of $S(t)$ and let $\{u_n\} \subset D(A)$ be as in (15). By Theorem 3-(d) we have that, for all $t \geq 0$,

$$S(t)u_n - u_n = \int_0^t S(s)Au_n ds.$$

Hence, taking the limit as $n \rightarrow \infty$ and dividing by t , we obtain

$$\frac{S(t)u - u}{t} = \frac{1}{t} \int_0^t S(s)v ds.$$

Passing to the limit as $t \downarrow 0$, we conclude that $Au = v$. $\square$

Remark 1 From Proposition 5 it follows that the domain $D(A)$ of the infinitesimal generator of a C_0 -semigroup is a Banach space with the *graph norm*

$$|u|_{D(A)} = |u| + |Au| \quad \forall u \in D(A).$$

Exercise 10 Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with boundary of class C^2 . Define

$$\begin{cases} D(A) = H^2(\Omega) \cap H_0^1(\Omega) \\ Au = \Delta u \end{cases} \quad \forall u \in D(A),$$

where $H^2(\Omega)$ and $H_0^1(\Omega)$ denote the usual Sobolev spaces of functions with square summable weak derivatives of order two and one, respectively, and null trace in the latter case (for more on these spaces see, for instance, [14] and [24]).

Prove that A is a closed operator on the Hilbert space $X = L^2(\Omega)$.

Solution. Let $u_i \in H^2(\Omega) \cap H_0^1(\Omega)$ be such that

$$\begin{cases} u_i \rightarrow u \\ \Delta u_i \rightarrow v \end{cases} \quad \text{in } L^2(\Omega).$$

By elliptic regularity (see, for instance, [24]), we have that

$$\|u_i - u_j\|_{H^2(\Omega)} \leq C \|\Delta u_i - \Delta u_j\|_2$$

for some constant $C > 0$. Hence, $\{u_i\}$ is a Cauchy sequence in the Hilbert space $H^2(\Omega) \cap H_0^1(\Omega)$. So, $u \in H^2(\Omega) \cap H_0^1(\Omega)$ and $\Delta u = v$. $\square$

Given a closed operator $A : D(A) \subset X \rightarrow X$, let us consider the Cauchy problem with initial datum $u_0 \in X$

$$\begin{cases} u'(t) = Au(t) & t > 0 \\ u(0) = u_0. \end{cases} \quad (17)$$

Definition 7 A classical solution of problem (17) is a function

$$u \in C([0, \infty); X) \cap C^1((0, \infty); X) \cap C((0, \infty); D(A))^1$$

such that $u(0) = u_0$ and $u'(t) = Au(t)$ for all $t > 0$.

Our next result ensures the existence and uniqueness of a classical solution to (17) for initial data in $D(A)$, provided A is the infinitesimal generator of a C_0 -semigroup of bounded linear operators on X .

Proposition 6 Let $A : D(A) \subset X \rightarrow X$ be the infinitesimal generator of a C_0 -semigroup of bounded linear operators on X , $S(t)$. Then, for every $u_0 \in D(A)$, problem (17) has a unique classical solution given by $u(t) = S(t)u_0$ for all $t \geq 0$.

Proof. The fact that $u(t) = S(t)u_0$ satisfies (17) is point (d) of Theorem 3. To show that u is the unique solution of the problem let v be any other classical solution of (17), fix $t > 0$, and set

$$U(s) = S(t-s)v(s), \quad \forall s \in [0, t].$$

Then, for all $s \in]0, t[$ and $0 < |h| < \min\{s, t-s\}$ we have that

$$\begin{aligned} & \frac{U(s+h) - U(s)}{h} - S(t-s)v'(s) + AS(t-s)v(s) \\ &= S(t-s-h) \frac{v(s+h) - v(s)}{h} - S(t-s)v'(s) \\ & \quad + \left(\frac{S(t-s-h) - S(t-s)}{h} + AS(t-s) \right) v(s). \end{aligned}$$

Now, point (d) of Theorem 3 immediately yields

$$\lim_{h \rightarrow 0} \frac{S(t-s-h) - S(t-s)}{h} v(s) = -AS(t-s)v(s).$$

Moreover,

$$\begin{aligned} & S(t-s-h) \frac{v(s+h) - v(s)}{h} - S(t-s)v'(s) \\ &= S(t-s-h) \left(\frac{v(s+h) - v(s)}{h} - v'(s) \right) \\ & \quad + (S(t-s-h) - S(t-s))v'(s), \end{aligned}$$

¹ Here $D(A)$ is regarded as a Banach space with the graph norm.

where

$$(S(t-s-h) - S(t-s))v'(s) \xrightarrow{h \rightarrow 0} 0$$

by the strong continuity of $S(t)$, while

$$\begin{aligned} \left| S(t-s-h) \left(\frac{v(s+h) - v(s)}{h} - v'(s) \right) \right| \\ \leq M e^{\omega(t-s-h)} \left| \frac{v(s+h) - v(s)}{h} - v'(s) \right| \xrightarrow{h \rightarrow 0} 0 \end{aligned}$$

in view of (7). Therefore,

$$U'(s) = -AS(t-s)v(s) + S(t-s)Av(s) = 0, \quad \forall s \in]0, T[.$$

So, U is constant and $u(t) = U(t) = U(0) = v(t)$. $\square$

Exercise 11 Let $S(t)$ and $T(t)$ be C_0 -semigroups with infinitesimal generators

$$A : D(A) \subset X \rightarrow X \quad \text{and} \quad B : D(B) \subset X \rightarrow X,$$

respectively. Show that

$$A = B \implies S(t) = T(t) \quad \forall t \geq 0.$$

Example 6 (Transport equation in $C_b(\mathbb{R})$) Returning to the left-translation semigroup on $C_b(\mathbb{R})$ of Example 3, by Proposition 6 and Exercise 5 we conclude that for each $f \in C_b^1(\mathbb{R})$ the unique solution of the problem

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) = \frac{\partial u}{\partial x}(t, x) & (t, x) \in \mathbb{R}_+ \times \mathbb{R} \\ u(0, x) = f(x) & x \in \mathbb{R} \end{cases}$$

is given by $u(t, x) = f(x + t)$.

1.5 Resolvent and spectrum

Let $A : D(A) \subset X \rightarrow X$ be a closed operator on a complex Banach space X .

Definition 8 The *resolvent set* of A , $\rho(A)$, is the set of all $\lambda \in \mathbb{C}$ such that $\lambda I - A : D(A) \rightarrow X$ is bijective. The set $\sigma(A) = \mathbb{C} \setminus \rho(A)$ is called the *spectrum* of A . For any $\lambda \in \rho(A)$ the linear operator

$$R(\lambda, A) := (\lambda I - A)^{-1} : X \rightarrow X$$

is called the *resolvent* of A .

Example 7 On $X = C([0, 1])$ with the uniform norm consider the linear operator $A : D(A) \subset X \rightarrow X$ defined by

$$\begin{cases} D(A) = C^1([0, 1]) \\ Af = f', \quad \forall f \in D(A), \end{cases}$$

which is closed (compare to Example 4). Then $\sigma(A) = \mathbb{C}$ because for any $\lambda \in \mathbb{C}$ the function $f_\lambda(x) = e^{\lambda x}$ satisfies

$$\lambda f_\lambda(x) - f'_\lambda(x) = 0 \quad \forall x \in [0, 1].$$

On the other hand, for the closed operator A_0 defined by

$$\begin{cases} D(A_0) = \{f \in C^1([0, 1]) : f(0) = 0\} \\ A_0 f = f', \quad \forall f \in D(A_0), \end{cases}$$

we have that $\sigma(A_0) = \emptyset$. Indeed, for any $g \in X$ the problem

$$\begin{cases} \lambda f(x) - f'(x) = g(x) & x \in [0, 1] \\ f(0) = 0 \end{cases}$$

admits the unique solution

$$f(x) = - \int_0^x e^{\lambda(x-s)} g(s) dx \quad (x \in [0, 1])$$

which belongs to $D(A_0)$.

We shall now derive basic properties of the resolvent operator.

Proposition 7 (Properties of $R(\lambda, A)$) *Let $A : D(A) \subset X \rightarrow X$ be a closed operator on a complex Banach space X . Then the following holds true.*

(a) $R(\lambda, A) \in \mathcal{L}(X)$ for any $\lambda \in \rho(A)$.

(b) For any $\lambda \in \rho(A)$

$$AR(\lambda, A) = \lambda R(\lambda, A) - I. \quad (18)$$

(c) The resolvent identity holds:

$$R(\lambda, A) - R(\mu, A) = (\mu - \lambda)R(\lambda, A)R(\mu, A) \quad \forall \lambda, \mu \in \rho(A). \quad (19)$$

(d) For any $\lambda, \mu \in \rho(A)$

$$R(\lambda, A)R(\mu, A) = R(\mu, A)R(\lambda, A). \quad (20)$$

Proof. Let $\lambda, \mu \in \rho(A)$.

(a) Since A is closed, so is $\lambda I - A$ and also $R(\lambda, A) = (\lambda I - A)^{-1}$. So, $R(\lambda, A) \in \mathcal{L}(X)$ by the closed graph theorem.

(b) This point follows from the definition of $R(\lambda, A)$.

(c) By (18) we have that

$$[\lambda R(\lambda, A) - AR(\lambda, A)]R(\mu, A) = R(\mu, A)$$

and

$$R(\lambda, A)[\mu R(\mu, A) - AR(\mu, A)] = R(\lambda, A).$$

Since $AR(\lambda, A) = R(\lambda, A)A$ on $D(A)$, (19) follows.

(d) Apply (19) to compute

$$\begin{aligned} R(\lambda, A) - R(\mu, A) &= (\mu - \lambda)R(\lambda, A)R(\mu, A) \\ R(\mu, A) - R(\lambda, A) &= (\lambda - \mu)R(\mu, A)R(\lambda, A). \end{aligned}$$

Adding the above identities side by side yields the conclusion.

The proof is complete. $\square$

The resolvent operator is smooth on the resolvent set, as we show next.

Theorem 4 (Analyticity of $R(\lambda, A)$) *Let $A : D(A) \subset X \rightarrow X$ be a closed operator on a complex Banach space X . Then the resolvent set $\rho(A)$ is open in $\mathbb{C}$ and for any $\lambda_0 \in \rho(A)$ we have that*

$$|\lambda - \lambda_0| < \frac{1}{\|R(\lambda_0, A)\|} \implies \lambda \in \rho(A). \quad (21)$$

Moreover, the resolvent $R(\lambda, A)$ is given by the (Neumann) series

$$R(\lambda, A) = \sum_{n=0}^{\infty} (\lambda_0 - \lambda)^n R(\lambda_0, A)^{n+1}. \quad (22)$$

Consequently, $\lambda \mapsto R(\lambda, A)$ is analytic on $\rho(A)$ and

$$\frac{d^n}{d\lambda^n} R(\lambda, A) = (-1)^n n! R(\lambda, A)^{n+1} \quad \forall n \in \mathbb{N}. \quad (23)$$

Proof. For all $\lambda \in \mathbb{C}$ and $\lambda_0 \in \rho(A)$ we have that

$$\lambda I - A = \lambda_0 I - A + (\lambda - \lambda_0)I = [I - (\lambda_0 - \lambda)R(\lambda_0, A)](\lambda_0 I - A).$$

This operator is bijective if and only if $[I - (\lambda_0 - \lambda)R(\lambda_0, A)]$ is invertible, which is the case if λ satisfies (21). Then

$$R(\lambda, A) = R(\lambda_0, A)[I - (\lambda_0 - \lambda)R(\lambda_0, A)]^{-1} = \sum_{n=0}^{\infty} (\lambda_0 - \lambda)^n R(\lambda_0, A)^{n+1}.$$

The analyticity of $R(\lambda, A)$ and (23) follows from (22). $\square$

Finally, we give a criterion that will be useful to identify the infinitesimal generator of a given C_0 -semigroup of bounded linear operators on X .

Proposition 8 *Let $A : D(A) \subset X \rightarrow X$ and $B : D(B) \subset X \rightarrow X$ be closed linear operators in X and suppose that A extends B (or, A is an extension of B), that is,*

$$D(B) \subset D(A) \quad \text{and} \quad Au = Bu \quad \forall x \in D(B).$$

If $\rho(A) \cap \rho(B) \neq \emptyset$, then $A = B$.

Proof. It suffices to show that $D(A) \subset D(B)$. Fix any $\lambda \in \rho(A) \cap \rho(B)$ and let $u \in D(A)$. Define

$$v = \lambda u - Au \quad \text{and} \quad w = R(\lambda, B)v.$$

Then $w \in D(B)$ and

$$\lambda w - Bw = v = \lambda u - Au.$$

Since $B \subset A$, we have that

$$\lambda w - Bw = \lambda w - Aw.$$

Thus, $(\lambda I - A)(u - w) = 0$ and, being $\lambda I - A$ invertible, we conclude that $u = w$. So, $u \in D(B)$. $\square$

1.6 The Hille-Yosida generation theorem

So far, all properties we have deduced for the infinitesimal generator of a C_0 -semigroup of bounded linear operators on X required the knowledge of the semigroup itself. Now, we would like to address the following:

? Question

How to decide whether a given operator $A : D(A) \subset X \rightarrow X$ is the infinitesimal generator of some C_0 -semigroup of bounded linear operators on X ?

A full answer to such a question is provided by a celebrated result by Hille and Yosida. As a preliminary, we need to show that the Laplace transform of a semigroup gives the resolvent of its infinitesimal generator.

Theorem 5 (Integral representation of $R(\lambda, A)$) *Let $A : D(A) \subset X \rightarrow X$ be the infinitesimal generator of a C_0 -semigroup of bounded linear operators on X , $S(t)$, and let $M \geq 1$ and $\omega \in \mathbb{R}$ be such that*

$$\|S(t)\| \leq M e^{\omega t} \quad \forall t \geq 0. \quad (24)$$

Then $\rho(A)$ contains the half-plane

$$\Pi_\omega = \{\lambda \in \mathbb{C} : \Re \lambda > \omega\} \quad (25)$$

and

$$R(\lambda, A)u = \int_0^\infty e^{-\lambda t} S(t)u \, dt \quad \forall u \in X, \forall \lambda \in \Pi_\omega. \quad (26)$$

Proof. We have to prove that, given any $\lambda \in \Pi_\omega$ and $u \in X$, the equation

$$\lambda v - Av = u \quad (27)$$

has a unique solution $v \in D(A)$ given by the right-hand side of (26).

Existence: observe that $v := \int_0^\infty e^{-\lambda t} S(t)u \, dt \in X$ because $\Re \lambda > \omega$. Moreover, for all $h > 0$,

$$\begin{aligned} \frac{S(h)v - v}{h} &= \frac{1}{h} \left\{ \int_0^\infty e^{-\lambda t} S(t+h)u \, dt - \int_0^\infty e^{-\lambda t} S(t)u \, dt \right\} \\ &= \frac{1}{h} \left\{ e^{\lambda h} \int_h^\infty e^{-\lambda t} S(t)u \, dt - \int_0^\infty e^{-\lambda t} S(t)u \, dt \right\} \\ &= \frac{e^{\lambda h} - 1}{h} v - \frac{e^{\lambda h}}{h} \int_0^h e^{-\lambda t} S(t)u \, dt. \end{aligned}$$

So

$$\lim_{h \downarrow 0} \frac{S(h)v - v}{h} = \lambda v - u$$

which in turn yields that $v \in D(A)$ and (27) holds true.

Uniqueness: let $v \in D(A)$ be a solution of (27). Then

$$\begin{aligned} \int_0^\infty e^{-\lambda t} S(t)u \, dt &= \int_0^\infty e^{-\lambda t} S(t)(\lambda v - Av) \, dt \\ &= \lambda \int_0^\infty e^{-\lambda t} S(t)v \, dt - \int_0^\infty e^{-\lambda t} \frac{d}{dt} S(t)v \, dt = v \end{aligned}$$

which implies that v is given by (26). $\square$

Remark 2 An easy consequence of Theorem 5 is that $S(t)R(\lambda, A) = R(\lambda, A)S(t)$ for any $t \geq 0$ and $\lambda \in \Pi_\omega$. Indeed, by (26) we have that

$$\begin{aligned} S(t)R(\lambda, A)u &= \int_0^\infty e^{-\lambda s} S(t)S(s)u \, ds \\ &= \int_0^\infty e^{-\lambda s} S(s)S(t)u \, ds = R(\lambda, A)S(t)u \quad \forall u \in X \end{aligned}$$

Example 8 (Right-translation semigroup on $[0, \infty)$) On the real Banach space

$$X = \{f \in C_b([0, \infty)) : f(0) = 0\}$$

with the uniform norm, consider the right-translation semigroup

$$(S(t)f)(x) = \begin{cases} f(x-t) & x > t \\ 0 & x \in [0, t] \end{cases} \quad \forall x, t \geq 0.$$

It is easy to check that S is a C_0 -semigroup on X with $\|S(t)\| = 1$ for all $t \geq 0$. In order to characterize its infinitesimal generator A , let us consider the operator $B : D(B) \subset X \rightarrow X$ defined by

$$\begin{cases} D(B) = \{f \in X : f' \in X\} \\ Bf = -f', \quad \forall f \in D(B). \end{cases}$$

We claim that:

(i) $B \subset A$

Proof. Let $f \in D(B)$. Then, for all $x, t \geq 0$ we have

$$\frac{(S(t)f)(x) - f(x)}{t} = \begin{cases} -\frac{f(x)}{t} = -f'(x_t), & 0 \leq x \leq t \\ \frac{f(x-t) - f(x)}{t} = -f'(x_t) & x \geq t \end{cases}$$

with $0 \leq x - x_t \leq t$. Therefore

$$\sup_{x \geq 0} \left| \frac{(S(t)f)(x) - f(x)}{t} + f'(x) \right| \leq \sup_{|x-y| \leq t} |f'(x) - f'(y)| \rightarrow 0 \quad \text{as } t \downarrow 0$$

because f' is uniformly continuous. $\square$

(ii) $1 \in \rho(B)$

Proof. For any $g \in X$ the unique solution f of the problem

$$\begin{cases} f \in D(B) \\ f(x) + f'(x) = g(x) \quad \forall x \geq 0 \end{cases}$$

is given by

$$f(x) = \int_0^x e^{s-x} g(s) ds \quad (x \geq 0). \quad \square$$

Since $1 \in \rho(A)$ by Theorem 5, Proposition 8 ensures that $A = B$.

We are now ready to derive the aforementioned characterization of the infinitesimal generator of a C_0 -semigroup of bounded linear operators on X .

Theorem 6 (Hille-Yosida generation theorem) *Let $M \geq 1$ and $\omega \in \mathbb{R}$. For a linear operator $A : D(A) \subset X \rightarrow X$ the following properties are equivalent:*

(a) *A is the infinitesimal generator of a C_0 -semigroup, $S(t)$, such that*

$$\|S(t)\| \leq M e^{\omega t} \quad \forall t \geq 0. \quad (28)$$

(b) *A is closed, $D(A)$ is dense in X , and*

$$\rho(A) \supseteq \Pi_\omega = \{\lambda \in \mathbb{C} : \Re \lambda > \omega\} \quad (29)$$

$$\|R(\lambda, A)^k\| \leq \frac{M}{(\Re \lambda - \omega)^k} \quad \forall k \geq 1, \forall \lambda \in \Pi_\omega \quad (30)$$

Proof of (a) $\Rightarrow$ (b) The fact that A is closed, $D(A)$ is dense in X , and (29) holds true has already been proved, see Theorem 3-(c), Proposition 5, and Theorem 5. In order to prove (30) observe that, by using (26) to compute the k -th derivative of the resolvent of A , we obtain

$$\frac{d^k}{d\lambda^k} R(\lambda, A)u = (-1)^k \int_0^\infty t^k e^{-\lambda t} S(t)u \, dt \quad \forall u \in X, \forall \lambda \in \Pi_\omega.$$

Therefore,

$$\left\| \frac{d^k}{d\lambda^k} R(\lambda, A) \right\| \leq M \int_0^\infty t^k e^{-(\Re \lambda - \omega)t} \, dt = \frac{M k!}{(\Re \lambda - \omega)^{k+1}}$$

where the integral is easily computed by induction. The conclusion follows recalling (23). $\square$

The proof that (b) $\Rightarrow$ (a) will require a longer reasoning. We begin with some preliminary results that are interesting in their own right.

Lemma 1 *Let $A : D(A) \subset X \rightarrow X$ be as in point (b) of Theorem 6. Then:*

(i) *For all $u \in X$*

$$\lim_{n \rightarrow \infty} nR(n, A)u = u. \quad (31)$$

(ii) *The Yosida Approximation A_n of A , defined as*

$$A_n = nAR(n, A) \quad (n > \omega) \quad (32)$$

is a sequence of bounded operators on X which satisfies

$$A_n A_m = A_m A_n \quad \forall n, m > \omega \quad (33)$$

and

$$\lim_{n \rightarrow \infty} A_n u = Au \quad \forall u \in D(A). \quad (34)$$

(iii) *Set $\omega_+ = \max\{\omega, 0\}$. For all $m, n > 2\omega_+$, $u \in D(A)$, and $t \geq 0$ we have that*

$$\|e^{tA_n}\| \leq M e^{\frac{n\omega t}{n-\omega}} \leq M e^{2\omega_+ t} \quad (35)$$

$$|e^{tA_n}u - e^{tA_m}u| \leq M^2 t e^{2\omega_+ t} |A_n u - A_m u|. \quad (36)$$

Consequently, for all $u \in D(A)$ the sequence $u_n(t) := e^{tA_n}u$ is Cauchy in $C([0, T]; X)$ for any $T > 0$.

Proof. Set $\omega_+ = \max\{\omega, 0\}$. Throughout the proof, the indices m, n are positive integers for which the resolvents under consideration are defined.

Proof of (i). By (18), for every $u \in D(A)$ and $n > \omega$,

$$\begin{aligned} |nR(n, A)u - u| &= |AR(n, A)u| \\ &= |R(n, A)Au| \leq \frac{M|Au|}{n - \omega} \longrightarrow 0 \quad \text{as } n \rightarrow \infty. \end{aligned}$$

Here we used (30) with $k = 1$. Moreover,

$$\|nR(n, A)\| \leq \frac{Mn}{n - \omega} \leq 2M \quad \forall n > 2\omega_+.$$

Since $D(A)$ is dense in X , these estimates imply (31) for every $u \in X$. Indeed, given $\varepsilon > 0$, choose $u_\varepsilon \in D(A)$ such that $|u_\varepsilon - u| < \varepsilon$. For $n > 2\omega_+$,

$$\begin{aligned} |nR(n, A)u - u| &\leq |nR(n, A)(u - u_\varepsilon)| + |nR(n, A)u_\varepsilon - u_\varepsilon| + |u_\varepsilon - u| \\ &\leq (2M + 1)\varepsilon + \frac{M|Au_\varepsilon|}{n - \omega}. \end{aligned}$$

Taking the upper limit as $n \rightarrow \infty$ and then letting $\varepsilon \downarrow 0$ proves (31).

Proof of (ii). The resolvent identity (18) gives

$$A_n = nAR(n, A) = n^2R(n, A) - nI \quad (n > \omega). \quad (37)$$

Thus $A_n \in \mathcal{L}(X)$. By (20), the resolvents $R(n, A)$ and $R(m, A)$ commute. Therefore,

$$\begin{aligned} A_n A_m &= [n^2R(n, A) - nI][m^2R(m, A) - mI] \\ &= [m^2R(m, A) - mI][n^2R(n, A) - nI] = A_m A_n. \end{aligned}$$

Finally, for every $u \in D(A)$, (31) yields

$$A_n u = nR(n, A)Au \longrightarrow Au \quad \text{as } n \rightarrow \infty.$$

This proves (34).

Proof of (iii). By (37),

$$e^{tA_n} = e^{-nt} \sum_{k=0}^{\infty} \frac{n^{2k} t^k R(n, A)^k}{k!} \quad (t \geq 0, n > \omega).$$

Using (30) and $M \geq 1$, we obtain

$$\begin{aligned} \|e^{tA_n}\| &\leq M e^{-nt} \sum_{k=0}^{\infty} \frac{n^{2k} t^k}{k!(n - \omega)^k} \\ &= M \exp\left(\frac{n\omega}{n - \omega} t\right). \end{aligned}$$

For $n > 2\omega_+$,

$$\frac{n\omega}{n-\omega} \leq 2\omega_+.$$

Indeed, if $\omega \leq 0$, the left-hand side is nonpositive; if $\omega > 0$, then $n/(n-\omega) \leq 2$. Consequently,

$$\|e^{tA_n}\| \leq M \exp\left(\frac{n\omega}{n-\omega}t\right) \leq Me^{2\omega_+t},$$

which proves (35).

Next, fix $u \in D(A)$ and let $u_n(t) = e^{tA_n}u$. Then

$$\begin{cases} (u_n - u_m)'(t) = A_n(u_n - u_m)(t) + (A_n - A_m)u_m(t), \\ (u_n - u_m)(0) = 0. \end{cases}$$

Variation of constants for the bounded operator A_n gives

$$\begin{aligned} e^{tA_n}u - e^{tA_m}u &= \int_0^t e^{(t-s)A_n}(A_n - A_m)e^{sA_m}u \, ds \\ &= \int_0^t e^{(t-s)A_n}e^{sA_m}(A_n - A_m)u \, ds, \end{aligned} \tag{38}$$

where the second equality follows from (33). Hence, for $m, n > 2\omega_+$,

$$\begin{aligned} |e^{tA_n}u - e^{tA_m}u| &\leq M^2 \int_0^t e^{2\omega_+(t-s)}e^{2\omega_+s}|A_nu - A_mu| \, ds \\ &= M^2te^{2\omega_+t}|A_nu - A_mu|. \end{aligned}$$

This proves (36). In particular, for every $T > 0$,

$$\sup_{0 \leq t \leq T} |e^{tA_n}u - e^{tA_m}u| \leq M^2Te^{2\omega_+T}|A_nu - A_mu|.$$

By (34), the right-hand side tends to zero as $m, n \rightarrow \infty$. Thus $\{e^{tA_n}u\}_n$ is Cauchy in $C([0, T]; X)$. $\square$

Exercise 12 Use a density argument to prove that, for every $u \in X$ and every $T > 0$, the sequence $\{e^{tA_n}u\}_n$ is Cauchy in $C([0, T]; X)$.

Solution. Set $\omega_+ = \max\{\omega, 0\}$. Fix $u \in X$, $T > 0$, and $\varepsilon > 0$. Since $D(A)$ is dense in X , choose $u_\varepsilon \in D(A)$ such that $|u_\varepsilon - u| < \varepsilon$. For all positive integers $m, n > 2\omega_+$ and $t \in [0, T]$, estimate (35) gives

$$\begin{aligned} |e^{tA_n}u - e^{tA_m}u| &\leq |e^{tA_n}(u - u_\varepsilon)| + |(e^{tA_n} - e^{tA_m})u_\varepsilon| + |e^{tA_m}(u_\varepsilon - u)| \\ &\leq |(e^{tA_n} - e^{tA_m})u_\varepsilon| + 2Me^{2\omega_+T}\varepsilon. \end{aligned}$$

By Lemma 1 (iii), the first term tends to zero uniformly on $[0, T]$ as $m, n \rightarrow \infty$. Hence

$$\limsup_{m, n \rightarrow \infty} \sup_{0 \leq t \leq T} |e^{tA_n}u - e^{tA_m}u| \leq 2Me^{2\omega_+T}\varepsilon.$$

Since $\varepsilon > 0$ is arbitrary, the conclusion follows. $\square$

Proof of (b) $\Rightarrow$ (a) Set $\omega_+ = \max\{\omega, 0\}$. By Lemma 1 and Exercise 12, for every $u \in X$ the limit

$$S(t)u = \lim_{n \rightarrow \infty} e^{tA_n}u \quad (t \geq 0) \quad (39)$$

exists uniformly on every bounded interval $[0, T]$. Consequently, $t \mapsto S(t)u$ is continuous and $S(0)u = u$.

For every $t \geq 0$, the map $S(t)$ is linear. Moreover, the sharper estimate in (35) gives

$$|e^{tA_n}u| \leq M \exp\left(\frac{n\omega}{n-\omega}t\right) |u|.$$

Passing to the limit as $n \rightarrow \infty$, we obtain

$$|S(t)u| \leq M e^{\omega t} |u| \quad \forall u \in X, \quad t \geq 0.$$

Thus $S(t) \in \mathcal{L}(X)$ and $\|S(t)\| \leq M e^{\omega t}$.

To verify the semigroup property, fix $s, t \geq 0$ and $u \in X$. For all sufficiently large n ,

$$\begin{aligned} |e^{tA_n} e^{sA_n} u - S(t)S(s)u| &\leq \|e^{tA_n}\| |e^{sA_n} u - S(s)u| \\ &\quad + |(e^{tA_n} - S(t))S(s)u| \\ &\leq M e^{2\omega_+ t} |e^{sA_n} u - S(s)u| \\ &\quad + |(e^{tA_n} - S(t))S(s)u|. \end{aligned}$$

Both terms tend to zero as $n \rightarrow \infty$. Since $e^{tA_n} e^{sA_n} = e^{(t+s)A_n}$, we conclude that

$$S(t+s)u = S(t)S(s)u.$$

Therefore $S(t)$ is a C_0 -semigroup on X with the required growth bound.

We now identify its infinitesimal generator. Fix $u \in D(A)$ and $T > 0$. For $t \in [0, T]$,

$$\begin{aligned} \left| \frac{d}{dt} e^{tA_n} u - S(t)Au \right| &\leq |e^{tA_n}(A_n u - Au)| + |(e^{tA_n} - S(t))Au| \\ &\leq M e^{2\omega_+ T} |A_n u - Au| + |(e^{tA_n} - S(t))Au|. \end{aligned}$$

By (34) and (39), the right-hand side tends to zero uniformly on $[0, T]$. Passing to the limit in

$$e^{tA_n} u - u = \int_0^t e^{sA_n} A_n u \, ds,$$

we obtain

$$S(t)u - u = \int_0^t S(s)Au \, ds.$$

Since $s \mapsto S(s)Au$ is continuous,

$$\frac{d}{dt}S(t)u = S(t)Au \quad \forall u \in D(A), \quad t \geq 0, \quad (40)$$

where the derivative at $t = 0$ is understood from the right.

Let $B : D(B) \subset X \rightarrow X$ be the infinitesimal generator of $S(t)$. Taking $t = 0$ in (40) shows that $A \subset B$. Moreover, $\Pi_\omega \subset \rho(A)$ by assumption (b), while $\Pi_\omega \subset \rho(B)$ by Theorem 5 and the growth bound just established. Thus Proposition 8 yields $A = B$. $\square$

Remark 3 The above proof shows that condition (b) in Theorem 6 can be relaxed as follows:

(b') A is closed, $D(A)$ is dense in X , and

$$\rho(A) \supseteq]\omega, \infty[\quad (41)$$

$$\|R(n, A)^k\| \leq \frac{M}{(n - \omega)^k} \quad \forall k \geq 1, \forall n > \omega. \quad (42)$$

Remark 4 When $M = 1$, the countably many bounds in condition (b) follow from (30) for $k = 1$, that is,

$$\|R(\lambda, A)\| \leq \frac{1}{\Re \lambda - \omega} \quad \forall k \geq 1, \forall \lambda \in \Pi_\omega.$$

Example 9 (Parabolic equations in $L^2(\Omega)$) Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with boundary of class C^2 . Define

$$\begin{cases} D(A) = H^2(\Omega) \cap H_0^1(\Omega) \\ Au = \sum_{i,j=1}^n D_i(a_{ij}D_ju) + \sum_{i=1}^n b_i D_iu + cu \quad \forall u \in D(A). \end{cases}$$

where

(H1) $a_{ij} \in C^1(\overline{\Omega})$ and $a_{ij} = a_{ji}$ for all $i, j = 1, \dots, n$. Moreover, there exists $\theta > 0$ such that

$$\sum_{i,j=1}^n a_{ij}(x) \xi_i \xi_j \geq \theta |\xi|^2 \quad \forall \xi \in \mathbb{R}^n, \quad \forall x \in \Omega.$$

(H2) $b_i \in L^\infty(\Omega)$ for all $i = 1, \dots, n$ and $c \in L^\infty(\Omega)$.

In order to apply the Hille-Yosida theorem to show that A is the infinitesimal generator of a C_0 -semigroup $S(t)$ on $L^2(\Omega)$, one can check that the following assumptions are satisfied.

1. $D(A)$ is dense in $L^2(\Omega)$.
[This is a known property of Sobolev spaces (see, for instance, [24].)]
2. A is a closed operator.
Proof. Let $u_k \in D(A)$ be such that

$$u_k \xrightarrow{k \rightarrow \infty} u \quad \text{and} \quad Au_k \xrightarrow{k \rightarrow \infty} f.$$

Then, for all $h, k \geq 1$ we have that $v_{hk} := u_h - u_k$ satisfies

$$\begin{cases} \sum_{i,j=1}^n D_i(a_{ij} D_j v_{hk}) + \sum_{i=1}^n b_i D_i v_{hk} + c v_{hk} =: f_{hk} & \text{in } \Omega \\ v_{hk} = 0 & \text{on } \partial\Omega. \end{cases}$$

So, elliptic regularity insures that

$$\|v_{hk}\|_{H^2(\Omega)} \leq C(\|f_{hk}\|_2 + \|v_{hk}\|_2)$$

for some constant $C > 0$. The above inequality implies that $\{u_k\}$ is a Cauchy sequence in $D(A)$ and this yields $f = Au$. $\square$

3. $\exists \omega \in \mathbb{R}$ such that $\rho(A) \supset]\omega, \infty[$.

[This follows from elliptic theory (see, for instance, [24]).]

4. $\|R(\lambda, A)\| \leq \frac{1}{\lambda - \omega}$ for all $k \geq 1$ and $\lambda > \omega$.

[This follows from elliptic theory (see, for instance, [24]).]

Then, for any $u_0 \in H^2(\Omega) \cap H_0^1(\Omega)$, the function $u(t, x) = (S(t)u_0)(x)$ is the unique solution of the initial-boundary value problem

$$\begin{cases} \frac{\partial u}{\partial t} = \sum_{i,j=1}^n D_i(a_{ij} D_j u) + \sum_{i=1}^n b_i D_i u + cu & \text{in }]0, \infty[\times \Omega \\ u = 0 & \text{on }]0, \infty[\times \partial\Omega \\ u(0, x) = u_0(x) & x \in \Omega. \end{cases}$$

in the class

$$C^1([0, \infty); L^2(\Omega)) \cap C([0, \infty); H^2(\Omega) \cap H_0^1(\Omega)).$$

Example 10 On

$$X = \{u \in C([0, \pi]) : u(0) = 0 = u(\pi)\}$$

with the uniform norm, consider the linear operator $A : D(A) \subset X \rightarrow X$ defined by

$$\begin{cases} D(A) = \{u \in C^2([0, \pi]) : u(0) = u(\pi) = 0 = u''(0) = u''(\pi)\} \\ Au = u'', \quad \forall u \in D(A). \end{cases}$$

Let us show that A generates a C_0 -semigroup of contractions on X and derive the initial-boundary value problem which is solved by such a semigroup.

We will check that all the assumptions of the Hille Yosida theorem are satisfied. Let $I = [0, \pi]$.

1. A is a closed operator: first of all, we claim that

$$\|u'\|_\infty \leq \pi \|u''\|_\infty \quad \forall u \in D(A). \quad (43)$$

Indeed, for any $x, y \in [0, \pi]$ we have that

$$u'(y) = u'(x) + \int_x^y u''(s) ds.$$

Therefore, by integrating with respect to x over $[0, \pi]$ we obtain

$$\pi u'(y) = \int_0^\pi dx \int_x^y u''(s) ds \quad \forall y \in [0, \pi].$$

So,

$$\begin{aligned} \pi |u'(y)| &= \left| \int_0^\pi dx \int_x^y u''(s) ds \right| \\ &\leq \int_0^\pi \left| \int_x^y u''(s) ds \right| dx \leq \pi^2 |u''|_\infty \quad \forall y \in [0, \pi], \end{aligned}$$

which in turn yields (43). Now, take any sequence $\{u_k\} \subset D(A)$ such that

$$u_k \xrightarrow{k \rightarrow \infty} u \quad \text{and} \quad Au_k \xrightarrow{k \rightarrow \infty} v.$$

Then, in view of (43), $\{u_k\}$ converges in $C^2([0, \pi])$. Therefore, $u \in D(A)$ and $Au = v$.

2. $D(A)$ is dense in X : this can be proved by a well-known regularising technique based on convolutions. Let $\rho : \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$\rho(x) = \begin{cases} C e^{\frac{1}{x^2-1}} & |x| < 1 \\ 0 & |x| \geq 1 \end{cases}$$

where

$$\frac{1}{C} = \int_{-1}^1 e^{\frac{1}{x^2-1}} dx.$$

For any $\varepsilon > 0$ define

$$\rho_\varepsilon(x) = \frac{1}{\varepsilon} \rho\left(\frac{x}{\varepsilon}\right) \quad (x \in \mathbb{R})$$

Then

$$\{\rho_\varepsilon\}_{\varepsilon>0} \subset C_c^\infty(\mathbb{R}), \quad \rho_\varepsilon \geq 0, \quad \text{spt}(\rho_\varepsilon) \subset [-\varepsilon, \varepsilon], \quad \int_{\mathbb{R}} \rho_\varepsilon(x) dx = 1.$$

Lemma 2 For any $f \in C_c((0, \pi))$

$$\rho_\varepsilon * f(x) := \int_{\mathbb{R}} \rho_\varepsilon(y) f(x-y) dy \xrightarrow{\varepsilon \downarrow 0} f(x) \quad \text{uniformly on } [0, \pi]$$

Proof. For $\varepsilon > 0$ sufficiently small we have that $\text{spt}(\rho_\varepsilon * f) \subset (0, \pi)$ and

$$\sup_{x \in [0, \pi]} |\rho_\varepsilon * f(x) - f(x)| \leq \int_{-\varepsilon}^{\varepsilon} \rho_\varepsilon(y) |f(x-y) - f(x)| dy \xrightarrow{\varepsilon \downarrow 0} 0$$

because f is uniformly continuous on $[0, \pi]$. $\square$

We can now complete the argument to show that $D(A)$ is dense in X . Fix any $f \in X$. For any $\sigma > 0$ there exists $\delta > 0$ such that

$$x \in [0, 2\delta] \cup [\pi - 2\delta, \pi] \implies |f(x)| < \sigma.$$

Let us set

$$f^\sigma(x) = \begin{cases} 0 & 0 \leq x \leq \delta \\ \frac{x-\delta}{\delta} f(2\delta) & \delta \leq x \leq 2\delta \\ f(x) & 2\delta \leq x \leq \pi - 2\delta \\ \frac{\pi-\delta-x}{\delta} f(\pi - 2\delta), & \pi - 2\delta < x \leq \pi - \delta. \\ 0 & \pi - \delta \leq x \leq \pi. \end{cases}$$

Then $|f^\sigma - f|_\infty < 2\sigma$. So, by taking σ and then ε small enough, we can make $|f^\sigma_\varepsilon - f|_\infty$ as small as we wish. Since $f^\sigma_\varepsilon \in C_c^\infty((0, \pi)) \subset D(A)$, the latter set is dense in X .

3. $\rho(A) \supset]0, \infty[$: fix any $\lambda > 0$; we have to show that for any $f \in X$ the problem

$$\begin{cases} u \in D(A) \\ \lambda u(x) - u''(x) = f(x), \quad x \in (0, \pi) \end{cases} \quad (44)$$

has a unique solution. Once a solution $\bar{u}$ of the linear differential equation above is known, all solutions have the form

$$u(x) = c_1 e^{x\sqrt{\lambda}} + c_2 e^{-x\sqrt{\lambda}} + \bar{u}(x) \quad (x \in (0, \pi))$$

for suitable constants $c_1, c_2 \in \mathbb{R}$. We claim that there is a unique pair of constants c_1, c_2 such that the corresponding solution satisfies the boundary conditions

$$u(0) = 0 = u(\pi) \quad (45)$$

(this is enough to guarantee that $u \in D(A)$). Indeed, the linear system

$$\begin{cases} c_1 + c_2 + \bar{u}(0) = 0 \\ c_1 e^{\pi\sqrt{\lambda}} + c_2 e^{-\pi\sqrt{\lambda}} + \bar{u}(\pi) = 0 \end{cases}$$

has a unique solution because

$$\begin{vmatrix} 1 & 1 \\ e^{\pi\sqrt{\lambda}} & e^{-\pi\sqrt{\lambda}} \end{vmatrix} \neq 0.$$

4. $\|R(\lambda, A)\| \leq \frac{1}{\lambda}$ for all $\lambda > 0$: fix any $f \in X$ and let $u = R(\lambda, A)f$. Let $x_0 \in [0, \pi]$ be such that $|u(x_0)| = |u|_\infty$. If $u(x_0) > 0$, then $x_0 \in (0, \pi)$ is a maximum point of u . So, $u''(x_0) \leq 0$ and we have that

$$\lambda|u|_\infty = \lambda u(x_0) \leq \lambda u(x_0) - u''(x_0) = f(x_0) \leq |f|_\infty.$$

On the other hand, if $u(x_0) < 0$, then $x_0 \in (0, \pi)$ once again and x_0 is a minimum point of u . Thus, $u''(x_0) \geq 0$ and

$$\lambda|u|_\infty = -\lambda u(x_0) \leq -\lambda u(x_0) + u''(x_0) = -f(x_0) \leq |f|_\infty.$$

In any case, we have that $\lambda|u|_\infty \leq |f|_\infty$.

In conclusion, the Hille-Yosida theorem ensures that A is the infinitesimal generator of a strongly continuous semigroup of contractions on X , $S(t)$. Therefore, for any $u_0 \in D(A)$, the initial-boundary value problem

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) = \frac{\partial^2 u}{\partial x^2}(t, x) & t \geq 0, x \in (0, \pi) \\ u(t, 0) = 0 = u(t, \pi) & \frac{\partial^2 u}{\partial x^2}(t, 0) = 0 = \frac{\partial^2 u}{\partial x^2}(t, \pi) \quad (t > 0) \\ u(0, x) = u_0(x) & x \in [0, \pi] \end{cases} \quad (46)$$

has a unique classical solution given by $u(t, x) = (S(t)u_0)(x)$. Finally, we observe that the boundary condition

$$\frac{\partial^2 u}{\partial x^2}(t, 0) = 0 = \frac{\partial^2 u}{\partial x^2}(t, \pi) \quad (t > 0)$$

is redundant as it follows from $u(t, 0) = 0 = u(t, \pi)$ and the equation in (46). $\square$

Remark 5 Observe that the linear operator $A : D(A) \subset X \rightarrow X$ defined by

$$\begin{cases} D(A) = \{ u \in C^2([0, \pi]) : u(0) = u(\pi) = 0 \} \\ Au = u'', \quad \forall u \in D(A). \end{cases}$$

is not densely defined on $C([0, \pi])$.

Example 11 On $X = L^2(0, \pi)$ consider the operator defined by

$$\begin{cases} D(A) = H^2(0, \pi) \cap H_0^1(0, \pi) \\ Au(x) = \frac{d^2 u}{dx^2}(x) \quad x \in [0, \pi] \text{ a.e.} \end{cases} \quad (47)$$

We are going to apply Theorem 6 to show that A generates a C_0 -semigroup of contractions on X and derive the initial-boundary value problem which is solved by such a semigroup.

1. A is a closed operator: as we did for Example 10, we claim that

$$\|u'\|_2 \leq \pi \|u''\|_2 \quad \forall u \in D(A). \quad (48)$$

Indeed, taking a representative of u of class $C^1([0, \pi])$ we have that

$$u'(y) = u'(x) + \int_x^y u''(s) ds \quad \forall x, y \in [0, \pi].$$

Therefore, by integrating with respect to x over $[0, \pi]$ as above, we obtain

$$\pi u'(y) = \int_0^\pi dx \int_x^y u''(s) ds \quad \forall y \in [0, \pi].$$

So, by Hölder's inequality

$$\begin{aligned} \pi |u'(y)| &= \left| \int_0^\pi dx \int_x^y u''(s) ds \right| \\ &\leq \int_0^\pi \left| \int_x^y u''(s) ds \right| dx \leq \pi^{3/2} \|u''\|_2 \quad \forall y \in [0, \pi], \end{aligned}$$

which in turn yields (48). Now, take any sequence $\{u_k\} \subset D(A)$ such that

$$u_k \xrightarrow{k \rightarrow \infty} u \quad \text{and} \quad Au_k \xrightarrow{k \rightarrow \infty} v \quad \text{in} \quad L^2(0, \pi).$$

In view of (48), $\{u_k\}$ converges in $H^2(0, \pi)$. Consequently, $u_k \xrightarrow{k \rightarrow \infty} u$ uniformly on $[0, \pi]$ and, since $u_k(0) = 0 = u_k(\pi)$, we have that $u \in H_0^1(0, \pi)$. Therefore, $u \in D(A)$ and $Au = v$.

2. Since $C_c^\infty((0, \pi)) \subset D(A)$, we have that $D(A)$ is dense in X .
3. $(0, \infty) \subset \rho(A)$: fix $\lambda > 0$ and $f \in L^2(0, \pi)$. We will show that

$$\lambda u(x) - u''(x) = f(x) \quad (0 < x < \pi \text{ a.e.}) \quad (49)$$

has a unique solution $u^f \in D(A)$.

Let

$$\phi_n(x) = \sqrt{\frac{2}{\pi}} \sin(nx), \quad n \geq 1.$$

These functions form a complete orthonormal system in $L^2(0, \pi)$. Write

$$f = \sum_{n=1}^{\infty} f_n \phi_n \quad \text{in } L^2(0, \pi), \quad f_n = \sqrt{\frac{2}{\pi}} \int_0^\pi f(x) \sin(nx) dx.$$

By Parseval's identity,

$$\|f\|_2^2 = \sum_{n=1}^{\infty} |f_n|^2.$$

If $u \in D(A)$ solves (49), integration by parts against ϕ_n gives

$$(\lambda + n^2)u_n = f_n, \quad u_n = \langle u, \phi_n \rangle, \quad n \geq 1.$$

Hence there is at most one solution, necessarily given by

$$u^f(x) = \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} \frac{f_n}{\lambda + n^2} \sin(nx). \quad (50)$$

To verify existence, observe that

$$\sum_{n=1}^{\infty} n^4 |u_n|^2 = \sum_{n=1}^{\infty} \frac{n^4}{(\lambda + n^2)^2} |f_n|^2 \leq \sum_{n=1}^{\infty} |f_n|^2 < \infty.$$

By orthogonality of the sine and cosine functions, the partial sums in (50) are therefore Cauchy in $H^2(0, \pi)$. Their limit belongs to $H^2(0, \pi) \cap H_0^1(0, \pi)$, since each partial sum vanishes at both endpoints. Passing to the limit in the equations satisfied by the partial sums proves (49). Thus $\lambda I - A$ is bijective for every $\lambda > 0$, which proves $(0, \infty) \subset \rho(A)$.

4. $\|R(\lambda, A)\| \leq \frac{1}{\lambda}$ for all $\lambda > 0$: multiplying both members of (49) by u and integrating over $[0, \pi]$ one obtains, for all $\lambda > 0$,

$$\lambda \int_0^\pi |u(x)|^2 dx + \int_0^\pi |u'(x)|^2 dx = \int_0^\pi f(x)u(x) dx$$

which yields

$$\|u^f\|_2 \leq \frac{1}{\lambda} \|f\|_2 \quad \forall \lambda > 0.$$

By applying the Hille-Yosida theorem we conclude that A is the infinitesimal generator of a strongly continuous semigroup of contractions on X , $S(t)$. Therefore, for any $u_0 \in H^2(0, \pi) \cap H_0^1(0, \pi)$, the initial-boundary value problem

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) = \frac{\partial^2 u}{\partial x^2}(t, x) & (t, x) \in \mathbb{R}_+ \times]0, \pi[\\ u(t, 0) = 0 = u(t, \pi) & t \geq 0 \\ u(0, x) = u_0(x) & x \in [0, \pi] \end{cases}$$

has a unique classical solution given by $u(t, x) = (S(t)u_0)(x)$. $\square$

Exercise 13 Let $S(t)$ be the C_0 -semigroup generated by operator A in (47). Prove that, for any $f \in L^2(0, \pi)$, $S(t)f$ is given by

$$(S(t)f)(x) = \int_0^\pi K(t, x, y) f(y) dy, \quad \forall t > 0, \quad x \in]0, \pi[\text{ a.e.,}$$

where

$$K(t, x, y) = \frac{2}{\pi} \sum_{k=1}^{\infty} e^{-k^2 t} \sin(kx) \sin(ky), \quad t > 0. \quad (51)$$

Proof. Let us consider the Fourier series of f :

$$f(x) = \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} f_n \sin(nx) = \frac{2}{\pi} \sum_{n=1}^{\infty} \left(\int_0^{\pi} \sin(ny) f(y) dy \right) \sin(nx) \quad (x \in [0, \pi])$$

Let us set

$$\lambda_n = -n^2 \quad \phi_n(x) = \sqrt{\frac{2}{\pi}} \sin(nx) \quad (n \geq 1, x \in [0, \pi])$$

Then

$$\phi_n \in D(A) \quad \text{and} \quad A\phi_n = \lambda_n \phi_n \quad (n \geq 1)$$

Since $f = \sum_{n=1}^{\infty} f_n \phi_n$ we have that $S(t)f = \sum_{n=1}^{\infty} f_n S(t)\phi_n$. Now, from the proof of the Hille-Yosida theorem it follows that

$$S(t)\phi_n = \lim_{k \rightarrow \infty} e^{tA_k} \phi_n = \lim_{k \rightarrow \infty} \sum_{j=0}^{\infty} \frac{t^j}{j!} A_k^j \phi_n$$

where $A_k = kAR(k, A)$ is the Yosida approximation of A . Since $(kI - A) \frac{\phi_n}{k - \lambda_n} = \phi_n$, we have that

$$R(k, A)\phi_n = \frac{\phi_n}{k - \lambda_n}$$

Therefore,

$$A_k \phi_n = kAR(k, A)\phi_n = \frac{k\lambda_n}{k - \lambda_n} \phi_n \quad \text{and} \quad A_k^j \phi_n = \left(\frac{k\lambda_n}{k - \lambda_n} \right)^j \phi_n$$

Finally,

$$\begin{aligned} S(t)\phi_n &= \lim_{k \rightarrow \infty} \sum_{j=0}^{\infty} \frac{t^j}{j!} \left(\frac{k\lambda_n}{k - \lambda_n} \right)^j \phi_n \\ &= \lim_{k \rightarrow \infty} e^{\frac{k\lambda_n}{k - \lambda_n} t} \phi_n = e^{\lambda_n t} \phi_n = e^{-n^2 t} \phi_n. \end{aligned}$$

In conclusion, $S(t)f = \sum_{n=1}^{\infty} e^{-n^2 t} f_n \phi_n$, which yields in turn

$$\begin{aligned} (S(t)f)(x) &= \frac{2}{\pi} \sum_{n=1}^{\infty} e^{-n^2 t} \left(\int_0^{\pi} \sin(ny) f(y) dy \right) \sin(nx) \\ &= \frac{2}{\pi} \int_0^{\pi} \sum_{n=1}^{\infty} e^{-n^2 t} \sin(nx) \sin(ny) f(y) dy = \int_0^{\pi} K(t, x, y) f(y) dy \end{aligned}$$

where K , given by (51), is called the *heat kernel* on $[0, \pi]$. Finally, strong continuity allows to recover the value at $t = 0$:

$$S(t)f \longrightarrow f \quad \text{in } L^2(0, \pi) \quad \text{as } t \downarrow 0,$$

thus completing the proof. $\square$

1.7 Asymptotic behaviour of C_0 -semigroups

Let $S(t)$ be a C_0 -semigroup of bounded linear operators on X . Then Theorem 2 ensures the existence of constants $\omega \geq 0$ and $M \geq 1$ such that $S(t)$ satisfies an exponential growth condition of the form $\|S(t)\| \leq Me^{\omega t}$. In this section, we are interested in establishing the “smallest” constant ω such that such an asymptotic behaviour holds true. This is the reason that motivates the introduction of the growth bound of a C_0 -semigroup of bounded linear operators on X and our next proposition.

Definition 9 The extended real number

$$\omega_0(S) = \inf_{t>0} \frac{\log \|S(t)\|}{t}, \quad (52)$$

with the convention $\log 0 = -\infty$, is called the *type* or *growth bound* of $S(t)$.

Proposition 9 The growth bound of S satisfies

$$\omega_0(S) = \lim_{t \rightarrow \infty} \frac{\log \|S(t)\|}{t} \in [-\infty, \infty). \quad (53)$$

Moreover, for every real $\eta > \omega_0(S)$, there exists $M_\eta \geq 1$ such that

$$\|S(t)\| \leq M_\eta e^{\eta t} \quad \forall t \geq 0. \quad (54)$$

In particular, if $\omega_0(S) = -\infty$, then (54) holds for every $\eta \in \mathbb{R}$.

Proof. The exponential bound (7) implies $\omega_0(S) < +\infty$. Fix any real $\eta > \omega_0(S)$. By (52), there exists $\tau > 0$ such that

$$\|S(\tau)\| \leq e^{\eta \tau}.$$

For every $t \geq 0$, write

$$t = n\tau + \delta, \quad n \in \{0, 1, 2, \dots\}, \quad 0 \leq \delta < \tau.$$

Using the semigroup property and (7), we obtain

$$\begin{aligned} \|S(t)\| &\leq \|S(\delta)\| \|S(\tau)^n\| \\ &\leq Me^{\omega \delta} e^{\eta n \tau} \\ &= Me^{(\omega - \eta)\delta} e^{\eta t} \\ &\leq M_\eta e^{\eta t}, \end{aligned}$$

where

$$M_\eta = M \exp(\max\{\omega - \eta, 0\}\tau).$$

This proves (54).

Consequently,

$$\limsup_{t \rightarrow \infty} \frac{\log \|S(t)\|}{t} \leq \eta \quad \text{for every real } \eta > \omega_0(S).$$

If $\omega_0(S)$ is finite, letting $\eta \downarrow \omega_0(S)$ and recalling that

$$\frac{\log \|S(t)\|}{t} \geq \omega_0(S) \quad (t > 0)$$

proves (53). If $\omega_0(S) = -\infty$, the upper bound holds for every real η , so the limit is $-\infty$. $\square$

A related notion, referred to the infinitesimal generator rather than the semigroup, is that of the spectral bound.

Definition 10 For any operator $A : D(A) \subset X \rightarrow X$ we define the *spectral bound* of A as

$$s(A) = \sup\{ \Re \lambda : \lambda \in \sigma(A) \}.$$

Here we use the convention $\sup \emptyset = -\infty$. An immediate consequence of the integral representation of the resolvent is the following.

Corollary 3 Let $S(t)$ be a C_0 -semigroup on X with infinitesimal generator A . Then

$$-\infty \leq s(A) \leq \omega_0(S) < +\infty.$$

Proof. Combining Theorem 5 with (54), we obtain

$$\Pi_\eta \subset \rho(A) \quad \text{for every real } \eta > \omega_0(S).$$

If $\omega_0(S)$ is finite, this implies $s(A) \leq \eta$ for every $\eta > \omega_0(S)$, and hence $s(A) \leq \omega_0(S)$. If $\omega_0(S) = -\infty$, the above inclusion holds for every real η . Therefore $\rho(A) = \mathbb{C}$, so $\sigma(A) = \emptyset$ and $s(A) = -\infty$. $\square$

The above two bounds may coincide or be different, they may be finite or infinite. We now give examples where all these possibilities occur, beginning with the semigroup of the right translations on a bounded interval.

Example 12 For fixed $T > 0$ and $p \geq 1$ let $X = L^p(0, T)$ and

$$(S(t)f)(x) = \begin{cases} f(x-t) & x \in [t, T] \\ 0 & x \in [0, t) \end{cases} \quad \forall x \in [0, T], \forall t \geq 0.$$

Then S is a C_0 -semigroup of bounded linear operators on X which satisfies $\|S(t)\| \leq 1$ for all $t \geq 0$. Moreover, observe that S is *nilpotent*, that is, we have $S(t) \equiv 0$, $\forall t \geq T$.

Deduce that $\omega_0(S) = -\infty$. So, the spectral bound of the infinitesimal generator of $S(t)$ also equals $-\infty$. $\square$

Our next example falls into the case of $-\infty < s(A) = \omega_0(S)$.

Example 13 In the Banach space

$$X = C_b(\mathbb{R}_+; \mathbb{C}),$$

with the uniform norm, the left-translation semigroup

$$(S(t)f)(x) = f(x+t) \quad \forall x, t \geq 0$$

is a C_0 -semigroup of contractions on X which satisfies $\|S(t)\| = 1$ (*Exercise*). Therefore

$$\omega_0(S) = 0.$$

The infinitesimal generator of $S(t)$ is given by

$$\begin{cases} D(A) = C_b^1(\mathbb{R}_+; \mathbb{C}) \\ Af = f' \end{cases} \quad \forall f \in D(A).$$

By Theorem 5 we have that

$$\rho(A) \supset \{\lambda \in \mathbb{C} : \Re \lambda > 0\}.$$

We claim that

$$\sigma(A) \supset \{\lambda \in \mathbb{C} : \Re \lambda \leq 0\}.$$

Indeed, for any $\lambda \in \mathbb{C}$ the function $f_\lambda(x) := e^{\lambda x}$ satisfies $\lambda f - f' = 0$. Moreover, $f_\lambda \in D(A)$ for $\Re \lambda \leq 0$. Therefore

$$s(A) = 0. \quad \square$$

Finally, we give an example where $s(A) < \omega_0(S)$.

Example 14 Let us denote by $C_0(\mathbb{R}_+; \mathbb{C})$ the Banach space of all continuous functions $f : \mathbb{R}_+ \rightarrow \mathbb{C}$ such that

$$\lim_{x \rightarrow \infty} f(x) = 0$$

with the uniform norm. Let X be the space of all functions $f \in C_0(\mathbb{R}_+; \mathbb{C})$ such that

$$\|f\| := \sup_{x \in \mathbb{R}_+} |f(x)| + \int_0^\infty |f(x)| e^x dx < \infty.$$

Exercise 14 Prove that $(X, \|\cdot\|)$ is a Banach space.

Once again, the left-translation semigroup

$$(S(t)f)(x) = f(x+t) \quad \forall x, t \geq 0$$

is a C_0 -semigroup of contractions on X . Indeed, for all $t \geq 0$

$$\begin{aligned} \|S(t)f\| &= \sup_{x \in \mathbb{R}_+} |f(x+t)| + \int_0^\infty |f(x+t)|e^x dx \\ &\leq \sup_{x \in \mathbb{R}_+} |f(x)| + e^{-t} \int_0^\infty |f(x)|e^x dx. \end{aligned}$$

Exercise 15 Prove that $\|S(t)\| = 1$ for all $t \geq 0$

Therefore

$$\omega_0(S) = 0.$$

The infinitesimal generator of $S(t)$ is given by

$$\begin{cases} D(A) = \{f \in X : f' \in X\} \\ Af = f' \end{cases} \quad \forall f \in D(A).$$

For any $\lambda \in \mathbb{C}$ the function $f_\lambda(x) := e^{\lambda x}$ satisfies $\lambda f - f' = 0$ and $f_\lambda \in D(A)$ for $\Re \lambda < -1$. So,

$$s(A) \geq -1. \quad (55)$$

We claim that

$$\rho(A) \supset \{\lambda \in \mathbb{C} : \Re \lambda > -1\}. \quad (56)$$

Indeed, a change of variables shows that, for any $g \in X$, the function

$$f(x) = \int_0^\infty e^{-\lambda t} (S(t)g)(x) dt = \int_0^\infty e^{-\lambda t} g(x+t) dt \quad (x \geq 0)$$

satisfies $\lambda f - f' = g$. Consequently, if we show that $f \in X$, then $f \in D(A)$ follows and so $\lambda \in \rho(A)$. To check that $f \in X$ observe that, for all $x \geq 0$,

$$\begin{aligned} |f(x)| &\leq \int_0^\infty |e^{-\lambda t} g(x+t)| dt \\ &= \int_0^\infty e^{-t \Re \lambda} |g(x+t)| e^{x+t} e^{-x-t} dt \\ &= e^{-x} \int_0^\infty e^{-t(1+\Re \lambda)} |g(x+t)| dt \\ &\leq e^{-x} \int_x^\infty e^s |g(s)| ds \end{aligned} \quad (57)$$

which insures that $f \in C_0(\mathbb{R}_+; \mathbb{C})$. Furthermore, by (57) we compute

$$\begin{aligned}
\int_0^\infty |f(x)|e^x dx &\leq \int_0^\infty dx \int_0^\infty e^{-t(1+\Re \lambda)} e^{x+t} |g(x+t)| dt \\
&= \int_0^\infty e^{-t(1+\Re \lambda)} dt \int_0^\infty e^{x+t} |g(x+t)| dx \\
&\leq \int_0^\infty e^{-t(1+\Re \lambda)} dt \int_0^\infty e^\tau |g(\tau)| d\tau < \infty.
\end{aligned}$$

From (55) and (56) it follows that $s(A) = -1 < 0 = \omega_0(S)$. $\square$

Exercise 16 Let $S(t)$ be the C_0 -semigroup on $L^2(\Omega)$ associated with the initial-boundary value problem

$$\begin{cases} \frac{\partial u}{\partial t} = \Delta u & \text{in }]0, \infty[\times \Omega \\ u = 0 & \text{on }]0, \infty[\times \partial\Omega \\ u(0, x) = u_0(x) & x \in \Omega \end{cases} \quad (58)$$

Show that $\omega_0(S) < 0$.

Solution. We know from Example 9 that the infinitesimal generator of $S(t)$ is the operator A defined by

$$\begin{cases} D(A) = H^2(\Omega) \cap H_0^1(\Omega) \\ Au = \Delta u \end{cases} \quad \forall u \in D(A).$$

For $u_0 \in D(A)$, let $u(t, x) = (S(t)u_0)(x)$. Then u satisfies (58). So,

$$\frac{d}{dt} \left(\frac{1}{2} \int_\Omega |u(t, x)|^2 dx \right) = - \int_\Omega |Du(t, x)|^2 dx \quad \forall t > 0.$$

Moreover, by Poincaré's inequality we have that

$$\int_\Omega |u(t, x)|^2 dx \leq c(\Omega) \int_\Omega |Du(t, x)|^2 dx.$$

Therefore,

$$\frac{d}{dt} |u(t)|^2 \leq - \frac{2}{c(\Omega)} |u(t)|^2$$

which ensures, by Gronwall's lemma, that

$$|u(t)| \leq e^{-t/c(\Omega)} |u_0| \quad \forall t > 0.$$

By a density argument, one concludes that the above inequality holds true for any $u_0 \in L^2(\Omega)$, so that $\omega_0(S) \leq -1/c(\Omega)$. $\square$

1.8 Strongly continuous groups

The notion of C_0 -semigroup of bounded linear operators on X is directly connected with solutions of forward-in-time evolution equations. If the equation is well-posed both forward and backward, then the natural notion of solution is that of a group.

Definition 11 A strongly continuous group, or a C_0 -group, of bounded linear operators on X is a map $G : \mathbb{R} \rightarrow \mathcal{L}(X)$ with the following properties:

- (a) $G(0) = I$ and $G(t + s) = G(t)G(s)$ for all $t, s \in \mathbb{R}$,
- (b) for all $u \in X$

$$\lim_{t \rightarrow 0} G(t)u = u. \quad (59)$$

The definition of infinitesimal generator naturally extends to groups.

Definition 12 The infinitesimal generator of a C_0 -group of bounded linear operators on X , $G(t)$, is the map $A : D(A) \subset X \rightarrow X$ defined by

$$\begin{cases} D(A) = \{u \in X : \exists \lim_{t \rightarrow 0} \frac{S(t)u - u}{t}\} \\ Au = \lim_{t \rightarrow 0} \frac{S(t)u - u}{t} \end{cases} \quad \forall u \in D(A)$$

The Hille-Yosida generation theorem takes the following form for groups.

Theorem 7 Let $M \geq 1$ and $\omega \geq 0$. For a linear operator $A : D(A) \subset X \rightarrow X$ the following properties are equivalent:

- (a) A is the infinitesimal generator of a C_0 -group, $G(t)$, such that

$$\|G(t)\| \leq Me^{\omega|t|} \quad \forall t \in \mathbb{R}. \quad (60)$$

- (b) A and $-A$ are the infinitesimal generators of C_0 -semigroups, $S_+(t)$ and $S_-(t)$ respectively, satisfying

$$\|S_{\pm}(t)\| \leq Me^{\omega t} \quad \forall t \geq 0. \quad (61)$$

- (c) A is closed, $D(A)$ is dense in X , and

$$\rho(A) \supseteq \{\lambda \in \mathbb{C} : |\Re \lambda| > \omega\} \quad (62)$$

$$\|R(\lambda, A)^k\| \leq \frac{M}{(|\Re \lambda| - \omega)^k} \quad \forall k \geq 1, \forall |\Re \lambda| > \omega \quad (63)$$

Proof. (a) $\Rightarrow$ (b) For $t \geq 0$, define

$$S_+(t) = G(t), \quad S_-(t) = G(-t).$$

These are strongly continuous semigroups satisfying

$$\|S_{\pm}(t)\| \leq Me^{\omega t} \quad \forall t \geq 0.$$

Denote their infinitesimal generators by A_+ and A_- , respectively. We prove that $A_+ = A$ and $A_- = -A$.

By the definition of the group generator, for every $u \in D(A)$,

$$\lim_{h \downarrow 0} \frac{S_+(h)u - u}{h} = Au.$$

Thus $D(A) \subseteq D(A_+)$ and $A_+u = Au$ on $D(A)$. Conversely, if $u \in D(A_+)$, then

$$\frac{G(-h)u - u}{-h} = G(-h) \frac{G(h)u - u}{h} \rightarrow A_+u \quad (h \downarrow 0),$$

where we used the strong continuity and local uniform boundedness of $G(-h)$. Hence the two-sided limit defining Au exists and equals A_+u . Therefore, $D(A_+) = D(A)$ and $A_+ = A$.

Similarly, for every $u \in D(A)$,

$$\frac{S_-(h)u - u}{h} = \frac{G(-h)u - u}{h} \rightarrow -Au \quad (h \downarrow 0).$$

Thus $D(A) \subseteq D(A_-)$ and $A_-u = -Au$ on $D(A)$. Conversely, if $u \in D(A_-)$, then

$$\frac{G(h)u - u}{h} = -G(h) \frac{G(-h)u - u}{h} \rightarrow -A_-u \quad (h \downarrow 0).$$

Consequently, $u \in D(A_+) = D(A)$ and $Au = -A_-u$. This proves that $D(A_-) = D(A)$ and $A_- = -A$.

(b) $\Rightarrow$ (c) By the Hille-Yosida theorem we conclude that A is closed, $D(A)$ is dense in X , and

$$\begin{aligned} \rho(A) &\supseteq \Pi_\omega = \{\lambda \in \mathbb{C} : \Re \lambda > \omega\} \\ \|R(\lambda, A)^k\| &\leq \frac{M}{(\Re \lambda - \omega)^k} \quad \forall k \geq 1, \forall \lambda \in \Pi_\omega. \end{aligned}$$

Since

$$\lambda I + A = -(-\lambda I - A), \quad (64)$$

we have that $\lambda I + A$ if and only if so is $-\lambda I - A$. So, $-\rho(A) = \rho(-A) \supseteq \Pi_\omega$, or

$$\rho(A) \supseteq -\Pi_\omega = \{\lambda \in \mathbb{C} : \Re \lambda < -\omega\}.$$

Moreover,

$$\|R(-\lambda, A)^k\| = \|R(\lambda, -A)^k\| \leq \frac{M}{(\Re \lambda - \omega)^k} \quad \forall k \geq 1, \forall \lambda \in \Pi_\omega.$$

Lemma 3 *Let A and $S_\pm(t)$ be as in point (b) above. Then*

- (i) $S_+(t)S_-(s) = S_-(s)S_+(t)$ for all $s, t \geq 0$,
- (ii) $S_+(t)^{-1} = S_-(t)$ for all $t \geq 0$.

Proof. Recall that

$$S_+(t) = \lim_{n \rightarrow \infty} e^{tA_n}, \quad S_-(t) = \lim_{n \rightarrow \infty} e^{tB_n}$$

where

$$A_n = nAR(n, A), \quad B_n = -nAR(n, -A) = nAR(-n, A)$$

are the Yosida approximations of A and $-A$, respectively. Since A_n and B_m commute in view of (20), so do e^{tA_n} and e^{tB_m} and (i) holds true.

Consequently,

$$S(t) := S_+(t)S_-(t) \quad (t \geq 0)$$

is also a C_0 -semigroup. Indeed, for all $t, s \geq 0$ we have that

$$\begin{aligned} S(t+s) &= S_+(t+s)S_-(t+s) \\ &= S_+(t)S_+(s)S_-(t)S_-(s) = S_+(t)S_-(t)S_+(s)S_-(s) = S(t)S(s). \end{aligned}$$

Moreover, for all $u \in D(A) = D(-A)$, we have that

$$\frac{S(t)u - u}{t} = S_+(t) \frac{S_-(t)u - u}{t} + \frac{S_+(t)u - u}{t} \xrightarrow{t \downarrow 0} -Au + Au = 0.$$

So, $\frac{d}{dt} S(t)u = 0$ for all $t \geq 0$. Hence, $S(t)u = u$ for all $t \geq 0$ and $u \in D(A)$. By density, $S(t)u = u$ for all $x \in X$, which yields $S_+(t)^{-1} = S_-(t)$. $\square$

(c) $\Rightarrow$ (a) Recalling (64), by the Hille-Yosida theorem it follows that $\pm A$ is the infinitesimal generator of a C_0 -semigroup, $S_{\pm}(t)$, satisfying (61). For all $u \in X$ define

$$G(t)u = \begin{cases} S_+(t)u & (t \geq 0) \\ S_-(-t)u & (t < 0). \end{cases}$$

We deduce that (59) and (60) hold true, and A is the infinitesimal generator of $G(t)$. Let us check the group property $G(t+s) = G(t)G(s)$. Such a property immediately follows from the semigroup property of $S_{\pm}$ if t and s have the same sign. Let now $t \geq 0$ and all $s \leq 0$ be such that $t+s \geq 0$. Recalling point (ii) of Lemma 3, we have

$$G(t)G(s) = S_+(t)S_-(-s) = S_+(t+s)S_+(-s)S_+(-s)^{-1} = G(t+s).$$

The case of $t+s \leq 0$ is similar. $\square$

1.9 Additional problems

Exercise 17 Let S be C_0 -semigroup of bounded linear operators on X and let $K \subset X$ be compact. Prove that for every $t_0 \geq 0$

$$\lim_{t \rightarrow t_0} \sup_{u \in K} |S(t)u - S(t_0)u| = 0. \quad (65)$$

Solution. We may assume $S \in \mathcal{G}(M, 0)$ for some $M > 0$ without loss of generality. Let $t_0 > 0$ and fix any $\varepsilon > 0$. Since K is totally bounded, there exist $u_1, \dots, u_{N_\varepsilon} \in X$ such that

$$K \subset \bigcup_{n=1}^{N_\varepsilon} B\left(u_n, \frac{\varepsilon}{M}\right).$$

Moreover, there exists $\tau > 0$ such that

$$|t - t_0| < \tau \implies |S(t)u_n - S(t_0)u_n| < \varepsilon \quad \forall n = 1, \dots, N_\varepsilon.$$

Thus, for all $|t - t_0| < \tau$ we have that, if $u \in K$ is such that $u \in B(u_n, \frac{\varepsilon}{M})$, then

$$\begin{aligned} |S(t)u - S(t_0)u| &\leq |S(t)u - S(t)u_n| + |S(t)u_n - S(t_0)u_n| + |S(t_0)u_n - S(t_0)u| \\ &\leq 2M|u - u_n| + \varepsilon < 3\varepsilon. \end{aligned}$$

So, the limit of $|S(t)u - S(t_0)u|$ as $t \rightarrow t_0$ is uniform on K . $\square$

Exercise 18 Let $A : D(A) \subset X \rightarrow X$ be a closed operator satisfying (30) but suppose $D(A)$ fails to be dense in X . In the Banach space $Y := \overline{D(A)}$, define the operator B , called the *part of A in Y* , by

$$\begin{cases} D(B) = \{u \in D(A) : Au \in Y\} \\ Bu = Au \quad \forall u \in D(B). \end{cases}$$

Prove that B is the infinitesimal generator of a C_0 -semigroup on Y .

Solution. $R(\lambda, A)(Y) \subset D(B)$ for all $\lambda \in \mathbb{C}$ such that $\Re \lambda > \omega$. Indeed, owing to (18) for all $u \in D(A)$ we have that

$$\lim_{n \rightarrow \infty} nR(n, A)u = \lim_{n \rightarrow \infty} \{R(n, A)Au + u\} = u. \quad (66)$$

Since $\|nR(n, A)\|$ is bounded, (66) holds true for all $u \in Y$. Hence, $D(B)$ is dense in Y . So, B satisfies in Y all the assumptions of Theorem 6. $\square$

Exercise 19 Let X be a Banach space and let $A : D(A) \subset X \rightarrow X$ be the infinitesimal generator of a uniformly bounded semigroup. Define, for $n \geq 1$,

$$D(A^n) := \{u \in D(A^{n-1}) : A^{n-1}u \in D(A)\}.$$

(i) Prove the following extension of the Landau-Kolmogorov inequality (12):

$$|A^k u| \leq (2M)^{k(n-k)} |A^n u|^{\frac{k}{n}} |u|^{\frac{n-k}{n}} \quad \forall u \in D(A^n), \quad \forall 0 \leq k \leq n \quad (67)$$

Solution: proceed by induction. The conclusion is trivial for $n = 1$. Assume (67) holds true for n and let $u \in D(A^{n+1})$. Then, in view of (12), we have that

$$\begin{aligned} |A^n u| &\leq 2M |A^{n+1} u|^{\frac{1}{2}} |A^{n-1} u|^{\frac{1}{2}} \\ &\leq 2M |A^{n+1} u|^{\frac{1}{2}} \left((2M)^{n-1} |A^n u|^{\frac{n-1}{n}} |u|^{\frac{1}{n}} \right)^{\frac{1}{2}} \\ &= (2M)^{\frac{n+1}{2}} |A^{n+1} u|^{\frac{1}{2}} |A^n u|^{\frac{n-1}{2n}} |u|^{\frac{1}{2n}}. \end{aligned}$$

Therefore,

$$|A^n u| \leq (2M)^n |A^{n+1} u|^{\frac{n}{n+1}} |u|^{\frac{1}{n+1}}, \quad (68)$$

which is (67) for $n + 1$ with $k = n$. Now, suppose $0 \leq k < n$. Then, by our inductive assumption and (68),

$$\begin{aligned} |A^k u| &\leq (2M)^{k(n-k)} |A^n u|^{\frac{k}{n}} |u|^{\frac{n-k}{n}} \\ &\leq (2M)^{k(n-k)} \left((2M)^n |A^{n+1} u|^{\frac{n}{n+1}} |u|^{\frac{1}{n+1}} \right)^{\frac{k}{n}} |u|^{\frac{n-k}{n}} \\ &= (2M)^{k(n+1-k)} |A^{n+1} u|^{\frac{k}{n+1}} |u|^{\frac{n+1-k}{n+1}}. \end{aligned}$$

The proof is complete. $\square$

(ii) Using (67), prove that for every $n \geq 1$:

(a) A^n is a closed operator.

Solution: proceed by induction. The conclusion is trivial for $n = 1$. Assume it holds true for n and let $\{u_k\} \subset D(A^{n+1})$ be such that

$$u_k \rightarrow u \quad \& \quad A^{n+1} u_k \rightarrow v \quad (k \rightarrow \infty).$$

Applying (68) to $w_k := A^n u_k \in D(A)$ we obtain

$$|w_k - w_h| \leq (2M)^n |A^{n+1}(u_k - u_h)|^{\frac{n}{n+1}} |u_k - u_h|^{\frac{1}{n+1}} \rightarrow 0 \quad (h, k \rightarrow \infty)$$

Therefore, for some $w \in X$,

$$w_k \rightarrow w \quad \& \quad A w_k \rightarrow v \quad (k \rightarrow \infty).$$

Since A is closed, we conclude that

$$w \in D(A) \quad \& \quad A w = v \quad (k \rightarrow \infty).$$

Then, by our inductive assumption, $u \in D(A^n)$ and $A^n u = w$, which implies in turn

$$u \in D(A^{n+1}) \quad \& \quad A^{n+1} u = A w = v \quad (k \rightarrow \infty). \quad \square$$

(b) $D(A^n)$ is dense in X for every $n \geq 1$.

Solution of (ii)(b): for $n = 1$ the conclusion follows from Theorem 3. Let the conclusion be true for some $n \geq 1$ and fix any $v \in X$. Then, for any $\varepsilon > 0$ there exists $u_\varepsilon \in D(A^n)$ such that $|u_\varepsilon - v| < \varepsilon$. Moreover, recalling point (a),

$$A^n \left(\frac{1}{t} \int_0^t S(s) u_\varepsilon ds \right) = \frac{1}{t} \int_0^t S(s) A^n u_\varepsilon ds$$

Since

$$\frac{1}{t} \int_0^t S(s) A^n u_\varepsilon ds \in D(A) \quad \forall t > 0$$

we conclude that

$$\frac{1}{t} \int_0^t S(s) u_\varepsilon ds \in D(A^{n+1}) \quad \forall t > 0.$$

Moreover, there exists $t_\varepsilon > 0$ such that

$$\left| \frac{1}{t_\varepsilon} \int_0^{t_\varepsilon} S(s) u_\varepsilon ds - v \right| \leq \left| \frac{1}{t_\varepsilon} \int_0^{t_\varepsilon} S(s) u_\varepsilon ds - u_\varepsilon \right| + |u_\varepsilon - v| < 2\varepsilon. \quad \square$$

Exercise 20 Let $(X, |\cdot|)$ be a separable Banach space and let $A : D(A) \subset X \rightarrow X$ be a closed operator with $\rho(A) \neq \emptyset$. Prove that $(D(A), |\cdot|_{D(A)})$ is also separable.

Solution. Let $\{u_n\}_{n \in \mathbb{N}}$ be dense in X and let $\lambda_0 \in \rho(A)$. Fix any $v \in D(A)$ and set $w = \lambda_0 v - Av$. For arbitrary $\varepsilon > 0$ let $u_\varepsilon = u_{n_\varepsilon}$ be such that $|w - u_\varepsilon| < \varepsilon$. Then

$$|v - R(\lambda_0, A)u_\varepsilon| = |R(\lambda_0, A)(w - u_\varepsilon)| \leq \|R(\lambda_0, A)\|\varepsilon.$$

Moreover,

$$\begin{aligned} |Av - AR(\lambda_0, A)u_\varepsilon| &= |AR(\lambda_0, A)(w - u_\varepsilon)| \\ &\leq |\lambda_0 R(\lambda_0, A)(w - u_\varepsilon)| + |w - u_\varepsilon| \leq (|\lambda_0| \|R(\lambda_0, A)\| + 1)\varepsilon. \end{aligned}$$

This exercise that $\{R(\lambda_0, A)u_n\}_{n \in \mathbb{N}}$ is dense in $D(A)$. $\square$

2 Dissipative operators

In the previous chapter we have characterised those operators $A : D(A) \subset X \rightarrow X$ that are the infinitesimal generator of a C_0 -semigroup of bounded linear operators on X . However, applying the Hille-Yosida generation theorem requires checking several conditions. In order to facilitate applicability, it is useful to look at special classes of operators for which such conditions become simpler to verify. This is the purpose of the present chapter.

We begin by considering a Hilbert space H with scalar product $\langle \cdot, \cdot \rangle$.

Definition 13 We say that an operator $A : D(A) \subset H \rightarrow H$ is *dissipative* if

$$\Re \langle Au, u \rangle \leq 0 \quad \forall u \in D(A). \quad (69)$$

Example 15 In $H = L^2(\mathbb{R}_+; \mathbb{C})$ consider the operator

$$\begin{cases} D(A) = H^1(\mathbb{R}_+; \mathbb{C}) \\ Af(x) = f'(x) \quad x \in \mathbb{R}_+ \text{ a.e.} \end{cases}$$

Then

$$2\Re \langle Af, f \rangle = 2\Re \left(\int_0^\infty f'(x) \overline{f(x)} dx \right) = \int_0^\infty \frac{d}{dx} |f(x)|^2 dx = -|f(0)|^2 \leq 0.$$

So, A is dissipative.

The following result provides a characterization of dissipative operators that will allow us to extend this notion to Banach spaces.

Proposition 10 *An operator $A : D(A) \subset H \rightarrow H$ is dissipative if and only if*

$$|(\lambda I - A)u| \geq \lambda |u| \quad \forall u \in D(A) \quad \text{and} \quad \forall \lambda > 0. \quad (70)$$

Proof. Let A be dissipative. Then, for every $u \in D(A)$ we have that

$$|(\lambda I - A)u|^2 = \lambda^2 |u|^2 - 2\lambda \Re \langle Au, u \rangle + |Au|^2 \geq \lambda^2 |u|^2 \quad \forall \lambda > 0.$$

Conversely, suppose A satisfies (70). Then for every $\lambda > 0$ and $u \in D(A)$

$$\lambda^2 |u|^2 - 2\lambda \Re \langle Au, u \rangle + |Au|^2 = |(\lambda I - A)u|^2 \geq \lambda^2 |u|^2$$

So, $2\lambda \Re \langle Au, u \rangle \leq |Au|^2$ which in turn yields (69) as $\lambda \rightarrow \infty$. $\square$

We now define the announced extension.

Definition 14 We say that an operator $A : D(A) \subset X \rightarrow X$ is *dissipative* if

$$|(\lambda I - A)u| \geq \lambda |u| \quad \forall u \in D(A) \quad \text{and} \quad \forall \lambda > 0. \quad (71)$$

Remark 6 It follows from (71) that, if A is dissipative, then

$$\lambda I - A : D(A) \rightarrow X$$

is injective for all $\lambda > 0$.

Dissipative operators are particularly important in the theory of C_0 -semigroup of bounded linear operators on X because they automatically satisfy some of the assumptions of the Hille-Yosida theorem .

Proposition 11 *Let $A : D(A) \subset X \rightarrow X$ be dissipative. If*

$$\exists \lambda_0 > 0 \quad \text{such that} \quad (\lambda_0 I - A)D(A) = X, \quad (72)$$

then the following properties hold:

- (a) $\lambda_0 \in \rho(A)$ and $\|R(\lambda_0, A)\| \leq 1/\lambda_0$,
- (b) A is closed,
- (c) $(\lambda I - A)D(A) = X$ and $\|R(\lambda, A)\| \leq 1/\lambda$ for all $\lambda > 0$.

Proof. We observe that point (a) follows from Remark 6 and inequality (71). As for point (b), we note that, since $R(\lambda_0, A)$ is closed, $\lambda_0 I - A$ is also closed, and therefore A is closed.

Let us prove (c). By point (a), the set

$$\Lambda = \{\lambda \in]0, \infty[: (\lambda I - A)D(A) = X\}$$

is contained in $\rho(A)$ which is open in $\mathbb{C}$. This implies that Λ is also open. Let us show that Λ is closed: let $\lambda_n \rightarrow \lambda > 0$ and fix any $v \in X$. There exists $u_n \in D(A)$ such that

$$\lambda_n u_n - A u_n = v. \quad (73)$$

From (70) it follows that $|u_n| \leq |v|/\lambda_n \leq C$ for some $C > 0$. Again by (70),

$$\begin{aligned} \lambda_m |u_n - u_m| &\leq |\lambda_m(u_n - u_m) - A(u_n - u_m)| \\ &\leq |\lambda_m - \lambda_n| |u_n| + |\lambda_n u_n - A u_n - (\lambda_m u_m - A u_m)| \\ &\leq C |\lambda_m - \lambda_n|. \end{aligned}$$

Therefore $\{u_n\}$ is a Cauchy sequence. Let $x \in X$ be such that $u_n \rightarrow u$. Then $A u_n \rightarrow \lambda u - v$ by (73). Since A is closed, $u \in D(A)$ and $\lambda u - A u = v$. This shows that $\lambda I - A$ is surjective and implies that $\lambda \in \Lambda$. Thus, Λ is both open and closed in $]0, \infty[$. Moreover, $\Lambda \neq \emptyset$ because $\lambda_0 \in \Lambda$. So, $\Lambda =]0, \infty[$. The inequality $\|R(\lambda, A)\| \leq 1/\lambda$ is a consequence of dissipativity. $\square$

2.1 Maximal dissipative operators

Motivated by Proposition 11, we now introduce a class of dissipative operators which is most relevant to the generation of a semigroup.

Definition 15 A dissipative operator $A : D(A) \subset X \rightarrow X$ is called *maximal dissipative* if $\lambda_0 I - A$ is surjective for some $\lambda_0 > 0$ (hence, for all $\lambda > 0$).

Remark 7 Let $A : D(A) \subset X \rightarrow X$ be a maximal dissipative operator and let $\bar{A} \supset A$ be a dissipative extension of A . Then:

- (i) $\bar{A}$ is maximal dissipative ($\lambda I - \bar{A}$ is surjective since so is $\lambda I - A$);

(ii) $\bar{A} = A$ (since both $\rho(A)$ and $\rho(\bar{A})$ contain $]0, \infty[$).

Theorem 8 *Let X be a reflexive Banach space. If $A : D(A) \subset X \rightarrow X$ is a maximal dissipative operator, then $D(A)$ is dense in X .*

Proof. We give the proof for a Hilbert space. The case of a reflexive Banach space is treated in exercises 21 to 24.

Let $v \in X$ be such that $\langle v, u \rangle = 0$ for all $u \in D(A)$. We will show that $v = 0$, or

$$\langle v, w \rangle = 0 \quad \forall w \in X.$$

Since $(I - A)$ is surjective, the above is equivalent to

$$0 = \langle v, u - Au \rangle \quad \forall u \in D(A).$$

So, we need to prove that

$$\langle v, u \rangle = 0 \quad \forall u \in D(A) \implies \langle v, Au \rangle = 0 \quad \forall u \in D(A). \quad (74)$$

Let $u \in D(A)$. Since $nI - A$ is onto, there exists a sequence $\{u_n\} \subset D(A)$ such that

$$nu = nu_n - Au_n \quad \forall n \geq 1. \quad (75)$$

Since $Au_n = n(u_n - u) \in D(A)$, we have that $u_n \in D(A^2)$ and

$$Au = Au_n - \frac{1}{n} A^2 u_n \quad \text{or} \quad Au_n = \left(I - \frac{1}{n} A\right)^{-1} Au.$$

Since $\|(I - \frac{1}{n} A)^{-1}\| \leq 1$ by (70), the above identity yields $|Au_n| \leq |Au|$. So, by (75) we obtain

$$|u_n - u| \leq \frac{1}{n} |Au|.$$

Therefore, $u_n \rightarrow u$. Moreover, since $\{Au_n\}$ is bounded, there is a subsequence Au_{n_k} such that $Au_{n_k} \rightharpoonup w$. Since A is closed, $\text{graph}(A)$ is a closed subspace of $X \times X$. Then, $\text{graph}(A)$ is also weakly closed and we have that $w = Au$. Therefore,

$$\langle v, Au \rangle = \lim_{k \rightarrow \infty} \langle v, Au_{n_k} \rangle = \lim_{k \rightarrow \infty} n_k \langle v, u_{n_k} - u \rangle$$

and (74) follows from the vanishing of the rightmost term above. $\square$

Example 16 We now show that the above density may fail in a general Banach space. On $X = C([0, 1])$ with the uniform norm $\|\cdot\|_\infty = \|\cdot\|_{\infty, [0, 1]}$, consider the linear operator $A : D(A) \subset X \rightarrow X$ defined by

$$\begin{cases} D(A) = \{u \in C^1([0, 1]) : u(0) = 0\} \\ Au(x) = -u'(x) \end{cases} \quad \forall x \in [0, 1].$$

Then, for all $\lambda > 0$ and $f \in X$ we have that the equation $\lambda u - Au = f$ has the unique solution $u \in D(A)$ given by

$$u(x) = \int_0^x e^{\lambda(y-x)} f(y) dy \quad (x \in [0, 1])$$

Therefore, $\lambda I - A$ is onto. Moreover,

$$\lambda |u(x)| \leq \int_0^x \lambda e^{\lambda(y-x)} \|f\|_\infty dy = (1 - e^{-\lambda x}) \|f\|_\infty \leq \|\lambda u - Au\|_\infty.$$

So, A is maximal dissipative. On the other hand, $D(A)$ is not dense in X because all functions in $D(A)$ vanish at $x = 0$.

Exercise 21 We recall that the duality set of a point $x \in X$ is defined as

$$\Phi(x) = \{\phi \in X^* : \langle x, \phi \rangle = |x|^2 = \|\phi\|^2\}. \quad (76)$$

Observe that the Hahn-Banach theorem ensures $\Phi(x) \neq \emptyset$.

We also recall that, for all $x \in X$,

$$\partial|x| = \{\phi \in X^* : |x+h| - |x| \geq \langle h, \phi \rangle, \forall x, h \in X\}. \quad (77)$$

Prove that

$$\Phi(x) = x\partial|x| = \{\psi \in X^* : \psi = |x|\phi, \phi \in \partial|x|\}.$$

Exercise 22 Prove that, for any operator $A : D(A) \subset X \rightarrow X$ the following properties are equivalent:

- (a) A is dissipative,
- (b) for all $x \in D(A)$ there exists $\phi \in \Phi(x)$ such that $\Re \langle Ax, \phi \rangle \leq 0$.

Exercise 23 Let $A : D(A) \subset X \rightarrow X$ be the infinitesimal generator of a C_0 -semigroup of contractions. Prove that, for all $x \in D(A)$,

$$\Re \langle Ax, \phi \rangle \leq 0 \quad \forall \phi \in \Phi(x).$$

Exercise 24 Mimic the proof of Theorem 8 to treat the general case of a reflexive Banach space.

Theorem 9 (Lumer-Phillips 1) Let $A : D(A) \subset X \rightarrow X$ be a densely defined linear operator. Then the following properties are equivalent:

- (a) A is the infinitesimal generator of a C_0 -semigroup of contractions,
- (b) A is maximal dissipative.

Proof. (a) $\Rightarrow$ (b) In view of Theorem 5, we have that $]0, \infty[\subset \rho(A)$. So,

$$(\lambda I - A)D(A) = X \quad \forall \lambda > 0$$

Moreover, by the Hille-Yosida theorem for all $\lambda > 0$ and $v \in X$ we have that $\lambda |R(\lambda, A)v| \leq |v|$ or, setting $u = R(\lambda, A)v$,

$$\lambda|u| \leq |(\lambda I - A)u| \quad \forall u \in D(A).$$

So, A is maximal dissipative.

$(b) \Rightarrow (a)$ We have that:

- (i) $D(A)$ is dense by hypothesis,
- (ii) A is closed by Proposition 11-(b),
- (iii) $]0, \infty[\subset \rho(A)$ and $\|R(\lambda, A)\| \leq 1/\lambda$ for all $\lambda > 0$ by Proposition 11-(c).

The conclusion follows by the Hille-Yosida theorem. $\square$

Example 17 (Wave equation in $L^2(\Omega)$) Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with boundary of class C^2 . For any given $u_0 \in H^2(\Omega) \cap H_0^1(\Omega)$ and $u_1 \in H_0^1(\Omega)$, consider the problem

$$\begin{cases} \frac{\partial^2 u}{\partial t^2}(t, x) = \Delta u & \text{in }]0, \infty[\times \Omega \\ u = 0 & \text{on }]0, \infty[\times \partial\Omega \\ u(0, x) = u_0(x), \quad \frac{\partial u}{\partial t}(0, x) = u_1(x) & x \in \Omega \end{cases} \quad (78)$$

Let H be the Hilbert space $H_0^1(\Omega) \times L^2(\Omega)$ with the scalar product

$$\left\langle \begin{pmatrix} u \\ v \end{pmatrix}, \begin{pmatrix} \bar{u} \\ \bar{v} \end{pmatrix} \right\rangle = \int_{\Omega} (Du(x) \cdot D\bar{u}(x) + v(x)\bar{v}(x)) dx.$$

Define $\mathcal{A} : D(\mathcal{A}) \subset H \rightarrow H$ by

$$\begin{cases} D(\mathcal{A}) = (H^2(\Omega) \cap H_0^1(\Omega)) \times H_0^1(\Omega) \\ \mathcal{A} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ \Delta & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} v \\ \Delta u \end{pmatrix} \end{cases} \quad (79)$$

We will show that $\mathcal{A}$ is the infinitesimal generator of a C_0 -semigroup of contractions on H by checking that $\mathcal{A}$ is maximal dissipative.

Let $\begin{pmatrix} u \\ v \end{pmatrix} \in D(\mathcal{A})$. Then, integrating by parts we obtain

$$\left\langle \mathcal{A} \begin{pmatrix} u \\ v \end{pmatrix}, \begin{pmatrix} u \\ v \end{pmatrix} \right\rangle = \int_{\Omega} (Du(x) \cdot Dv(x) + v(x)\Delta u(x)) dx = 0. \quad (80)$$

So, $\mathcal{A}$ is dissipative.

Now, consider the resolvent equation

$$\begin{cases} \begin{pmatrix} u \\ v \end{pmatrix} \in D(\mathcal{A}) \\ (I - \mathcal{A}) \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} f \\ g \end{pmatrix} \in H \end{cases} \quad (81)$$

which is equivalent to the system

$$\begin{cases} u \in H^2(\Omega) \cap H_0^1(\Omega), & v \in H_0^1(\Omega) \\ u - v = f \in H_0^1(\Omega) \\ v - \Delta u = g \in L^2(\Omega). \end{cases} \quad (82)$$

Using elliptic theory (see, for instance, [24]) one can show that the boundary value problem

$$\begin{cases} u \in H^2(\Omega) \cap H_0^1(\Omega), \\ u - \Delta u = f + g \in L^2(\Omega) \end{cases}$$

has a unique solution. Then, taking $v = u - f \in H_0^1(\Omega)$ we obtain the unique solution of problem (82). So, $\mathcal{A}$ is maximal dissipative and therefore $\mathcal{A}$ is the infinitesimal generator of a C_0 -semigroup of contractions, $S(t)$.

For any $u_0 \in H^2(\Omega) \cap H_0^1(\Omega)$, $u_1 \in H_0^1(\Omega)$, let $u(t)$ ($t \in \mathbb{R}_+$) be the first component of

$$S(t) \begin{pmatrix} u_0 \\ u_1 \end{pmatrix}$$

Then u is the unique solution of problem (78) in the space

$$C^2(\mathbb{R}_+; L^2(\Omega)) \cap C^1(\mathbb{R}_+; H_0^1(\Omega)) \cap C(\mathbb{R}_+; H^2(\Omega) \cap H_0^1(\Omega)).$$

Example 18 Consider the age-structured population model

$$\begin{cases} \frac{\partial u}{\partial t}(t, a) + \frac{\partial u}{\partial a}(t, a) + \mu(a)u(t, a) = 0, & a \in [0, a_1], t \geq 0 \\ u(t, 0) = \int_0^{a_1} \beta(a)u(t, a) da, & t \geq 0 \\ u(0, a) = u_0(a). & a \in [0, a_1]. \end{cases} \quad (83)$$

which was proposed in [29]. Here, $u(t, a)$ is the population density of age a at time t , μ is the mortality rate, β the birth rate, and $a_1 > 0$ is the maximal age. We assume that $\mu, \beta \in C([0, a_1])$, $\mu, \beta \geq 0$, and

$$\int_0^{a_1} \beta(a) e^{-\int_0^a \mu(\rho) d\rho} da < 1. \quad (84)$$

In order to recast problem (83) as an evolution equation in $H = L^2(0, a_1)$, we define the linear operator

$$\begin{cases} D(A) = \left\{ u \in H^1(0, a_1) : u(0) = \int_0^{a_1} \beta(a)u(a) da \right\} \\ Au(a) = -u'(a) - \mu(a)u(a) \quad (a \in [0, a_1] \text{ a.e.}) \end{cases} \quad (85)$$

We now proceed to show the following:

1. A is the infinitesimal generator of a C_0 -semigroup on H .

2. $\rho(A) \supset [0, +\infty)$ and, for any $\lambda > 0$,

$$R(\lambda, A)u(a) = \frac{U(a, 0)}{1 - \int_0^{a_1} \beta(a)U(a, 0)da} \int_0^{a_1} \beta(a) da \int_0^a U(a, s)u(s) ds + \int_0^a U(a, s)u(s) ds, \quad a \in [0, a_1], \quad u \in H, \quad (86)$$

where

$$U(a, s) = e^{-\lambda(a-s) - \int_s^a \mu(\rho) d\rho}, \quad a, s \in [0, a_1]. \quad (87)$$

Proof. Given $\lambda \geq 0$ and $v \in H$ we consider the equation

$$\lambda u - Au = v, \quad (88)$$

which is equivalent to

$$\begin{cases} (\lambda + \mu)u + u' = v, \\ u(0) = \int_0^{a_1} \beta(a)u(a) da. \end{cases} \quad (89)$$

If u is a solution of Eq. (88), then

$$u(a) = U(a, 0)u(0) + \int_0^a U(a, s)v(s) ds, \quad (90)$$

where U is given by Eq. (87). Multiplying Eq. (90) by β and integrating with respect to a over $[0, a_1]$ yields

$$\begin{aligned} u(0) &= \int_0^{a_1} \beta(a)u(a) da \\ &= \left(\int_0^{a_1} \beta(a)U(a, 0) da \right) u(0) + \int_0^{a_1} \beta(a) da \int_0^a U(a, s)v(s) ds. \end{aligned} \quad (91)$$

From Eq. (84), we have

$$\int_0^{a_1} \beta(a)U(a, 0) da < 1, \quad \forall a \in [0, a_1], \quad (92)$$

then, also from Eq. (90),

$$u(0) = \frac{1}{1 - \int_0^{a_1} \beta(a)U(a, 0) da} \int_0^{a_1} \beta(a) da \int_0^a U(a, s)v(s) ds. \quad (93)$$

Consequently, $u(a) = R(\lambda, A)v(a)$ is given by Eq. (86).

Conversely, given $v \in H$, the function

$$u(a) = \frac{U(a, 0)}{1 - \int_0^{a_1} \beta(a) U(a, 0) da} \int_0^{a_1} \beta(a) da \int_0^a U(a, s) v(s) ds + \int_0^a U(a, s) v(s) ds, \quad a \in [0, a_1],$$

fulfills Eq. (88). $\square$

3. For all $u \in D(A)$

$$\langle Au, u \rangle \leq -\frac{1}{2} \int_0^{a_1} u^2(a) \left(2\mu(a) - \int_0^{a_1} \beta^2(s) ds \right) da - \frac{1}{2} u(a_1)^2. \quad (94)$$

Consequently, if

$$2\mu(a) \geq \int_0^{a_1} \beta^2(s) ds, \quad \forall a \in [0, a_1], \quad (95)$$

then A is the infinitesimal generator of a C_0 -semigroup of contractions on H .

Proof. To show Eq. (94), we observe that, for all $u \in D(A)$,

$$\begin{aligned} \langle Au, u \rangle &= - \int_0^{a_1} u'(a) u(a) da - \int_0^{a_1} \mu(a) u^2(a) da \\ &= \frac{1}{2} u(0)^2 - \frac{1}{2} u(a_1)^2 - \int_0^{a_1} \mu(a) u^2(a) da \\ &= \frac{1}{2} \left(\int_0^{a_1} \beta(a) u(a) da \right)^2 - \frac{1}{2} u(a_1)^2 - \int_0^{a_1} \mu(a) u^2(a) da. \end{aligned}$$

So, by Hölder's inequality,

$$\begin{aligned} \langle Au, u \rangle &\leq \frac{1}{2} \int_0^{a_1} \beta^2(a) da \int_0^{a_1} u^2(s) ds - \frac{1}{2} u(a_1)^2 - \int_0^{a_1} \mu(a) u^2(a) da \\ &= -\frac{1}{2} \int_0^{a_1} u^2(a) \left(2\mu(a) - \int_0^{a_1} \beta^2(s) ds \right) da - \frac{1}{2} u(a_1)^2. \end{aligned}$$

This shows that A is maximal dissipative if (95) is satisfied. In this case, the Lumer-Phillips theorem insures that A is the infinitesimal generator of a C_0 -semigroup of contractions.

4. To prove item 1 without the additional assumption (95), set

$$\omega = \frac{1}{2} \int_0^{a_1} \beta(s)^2 ds.$$

Since $\mu \geq 0$, estimate (94) gives

$$\langle (A - \omega I)u, u \rangle \leq 0 \quad \forall u \in D(A).$$

Moreover, by item 2, for every $\lambda > 0$ the operator

$$\lambda I - (A - \omega I) = (\lambda + \omega)I - A$$

maps $D(A)$ onto H . Hence $A - \omega I$ is maximal dissipative. Its domain is therefore dense in H , and the Lumer–Phillips theorem shows that it generates a strongly continuous semigroup of contractions, say $T(t)$. Consequently, A generates the strongly continuous semigroup

$$S(t) = e^{\omega t} T(t), \quad \|S(t)\| \leq e^{\omega t} \quad \forall t \geq 0.$$

This proves item 1. $\square$

When A and $-A$ are maximal dissipative a stronger conclusion holds true.

Corollary 4 *Let $A : D(A) \subset X \rightarrow X$ be a densely defined linear operator. If both A and $-A$ are maximal dissipative, then A is the infinitesimal generator of a C_0 -group, $G(t)$, which satisfies $\|G(t)\| = 1$ for all $t \in \mathbb{R}$.*

Proof. By the Lumer–Phillips theorem, A and $-A$ are infinitesimal generators of C_0 -semigroups of contractions, say $S_+(t)$ and $S_-(t)$, respectively. Therefore, Theorem 7 ensures that A is the infinitesimal generator of a C_0 -group, $G(t)$. Moreover,

$$1 = \|G(t)G(-t)\| \leq \|S_+(t)\| \|S_-(t)\| \leq 1.$$

Hence, $\|G(t)\| = 1$. $\square$

Example 19 (Wave equation (continued)) We return to the wave equation that was studied in Example 17. We proved that operator A , defined in (79), is maximal dissipative. We claim that $-A$ is maximal dissipative as well. Indeed, equation (80) implies that $-A$ is dissipative. Moreover, the resolvent equation for $-A$ takes the form

$$\begin{cases} u \in H^2(\Omega) \cap H_0^1(\Omega), & v \in H_0^1(\Omega) \\ u + v = f \in H_0^1(\Omega) \\ v + \Delta u = g \in L^2(\Omega). \end{cases}$$

So, $v = f - u$ and this yields

$$\begin{cases} u \in H^2(\Omega) \cap H_0^1(\Omega), \\ u - \Delta u = f - g \in L^2(\Omega). \end{cases}$$

The above problem can be uniquely solved arguing exactly as we did for system (82).

Then, by Corollary 4, A is the infinitesimal generator of a C_0 -group, $G(t)$, which satisfies $\|G(t)\| = 1$ for all $t \in \mathbb{R}$. So, for any $u_0 \in H^2(\Omega) \cap H_0^1(\Omega)$, $u_1 \in H_0^1(\Omega)$, the first component $u(t)$ ($t \in \mathbb{R}_+$) of

$$G(t) \begin{pmatrix} u_0 \\ u_1 \end{pmatrix}$$

is the unique solution of the problem

$$\begin{cases} \frac{\partial^2 u}{\partial t^2}(t, x) = \Delta u & \text{in } \mathbb{R} \times \Omega \\ u = 0 & \text{on } \mathbb{R} \times \partial\Omega \\ u(0, x) = u_0(x), \quad \frac{\partial u}{\partial t}(0, x) = u_1(x) & x \in \Omega \end{cases} \quad (96)$$

in the space

$$C^2(\mathbb{R}; L^2(\Omega)) \cap C^1(\mathbb{R}; H_0^1(\Omega)) \cap C(\mathbb{R}; H^2(\Omega) \cap H_0^1(\Omega)).$$

2.2 Adjoint operators and a second Lumer-Phillips theorem

Let $(X, \langle \cdot, \cdot \rangle)$ be a Hilbert space and denote by X^* the topological dual of X . We denote by $j_X : X^* \rightarrow X$ the Riesz isomorphism, which associates with any $\phi \in X^*$ the unique element $j_X(\phi) \in X$ such that

$$\phi(u) = \langle u, j_X(\phi) \rangle \quad \forall u \in X. \quad (97)$$

Given another Hilbert space $(Y, (\cdot, \cdot))$ and a bounded linear operator $\Lambda : X \rightarrow Y$, for any $v \in Y$, define the linear map $\phi_v : X \rightarrow \mathbb{C}$ by

$$\phi_v(u) = (\Lambda u, v) \quad \forall u \in X.$$

Exercise 25 Prove that $\phi_v \in X^*$ for any $v \in Y$.

Definition 16 The *adjoint* of Λ is the bounded linear operator $\Lambda^* : Y \rightarrow X$ defined by

$$\Lambda^* v = j_X \circ \phi_v \quad \forall v \in Y.$$

Observe that Λ^* is characterized by the *adjoint identity*

$$(\Lambda u, v) = \langle u, \Lambda^* v \rangle \quad \forall u \in X, v \in Y.$$

We now proceed to define the adjoint of an unbounded operator. We refer the reader to [27] for the treatment of a general Banach space.

Exercise 26 Let $A : D(A) \subset X \rightarrow X$ be a densely defined linear operator.

1. Prove that the set

$$D(A^*) = \left\{ v \in X \mid \exists C_v \geq 0 : |\langle Au, v \rangle| \leq C_v |u|, \forall u \in D(A) \right\} \quad (98)$$

is a subspace of X .

2. For any $v \in D(A^*)$, define the linear map $\varphi_v : D(A) \rightarrow \mathbb{C}$ by

$$\varphi_v(u) = \langle Au, v \rangle \quad \forall u \in D(A).$$

Prove that φ_v can be uniquely extended to a functional $\phi_v \in X^*$.

Proof. The fact that $D(A^*)$ is a subspace of X is easy to show and will be left to the reader. In order to prove point 2, let $v \in D(A^*)$, fix any $u \in X$, and let $u_n \in D(A)$ be such that $u_n \xrightarrow{n \rightarrow \infty} u$. Then $|\langle A(u_n - u_m), v \rangle| \leq C_v |u_n - u_m|$ which implies that $\langle Au_n, v \rangle = \varphi_v(u_n)$ is a Cauchy sequence in $\mathbb{C}$ and therefore converges as $n \rightarrow \infty$. The map

$$\phi_v(u) := \lim_{n \rightarrow \infty} \varphi_v(u_n) \quad (u \in X), \quad (99)$$

is easily checked to be linear. Moreover,

$$|\phi_v(u)| = \lim_{n \rightarrow \infty} |\langle Au_n, v \rangle| \leq C_v |u| \quad (u \in X)$$

So, $\phi_v \in X^*$. Finally, if $\phi \in X^*$ is any other extension of φ_v , then

$$\phi(u) = \lim_{n \rightarrow \infty} \varphi_v(u_n) = \phi_v(u) \quad (u \in X)$$

This shows that the extension is unique. $\square$

Definition 17 The adjoint of A is the map $A^* : D(A^*) \subset X \rightarrow X$ defined by

$$A^*v = j_X(\phi_v) \quad \forall v \in D(A^*)$$

where $D(A^*)$ is given by (98), j_X by (97), and $\phi_v \in X^*$ is given by (99).

Exercise 27 Prove that, if $A \in \mathcal{L}(X)$, then $A^* \in \mathcal{L}(X)$ and

$$\|A\| = \|A^*\|. \quad (100)$$

Proof. Since $A \in \mathcal{L}(X)$ we have that $D(A^*) = X$ and

$$\langle Au, v \rangle = \langle u, A^*v \rangle \quad \forall u, v \in X.$$

So, by the definition of A^* we have that $\|A^*\| \leq \|A\|$. Moreover, taking $v = Au$ in the above identity, we obtain $|Au|^2 \leq |u| \|A^*\| |Au|$. So, $\|A\| \leq \|A^*\|$. $\square$

Example 20 Let us show that the adjoint of the wave operator A studied in Example 17 and Example 19, is given by $-A$ (in this case, A is said to be *skew-adjoint*). Indeed, for all $(u^0, u^1), (v^0, v^1) \in (H^2(\Omega) \cap H_0^1(\Omega)) \times H_0^1(\Omega)$ we have that

$$\begin{aligned} \left\langle A \begin{pmatrix} u^0 \\ u^1 \end{pmatrix}, \begin{pmatrix} v^0 \\ v^1 \end{pmatrix} \right\rangle &= \int_{\Omega} (Du^1(x) \cdot Dv^0(x) + v^1(x) \Delta u^0(x)) dx \\ &= - \int_{\Omega} (u^1(x) \Delta v^0(x) + Du^0(x) \cdot Dv^1(x)) dx = - \left\langle \begin{pmatrix} u^0 \\ u^1 \end{pmatrix}, A \begin{pmatrix} v^0 \\ v^1 \end{pmatrix} \right\rangle \end{aligned}$$

Therefore, A^* is an extension of $-A$ and then $A^* = -A$ because the resolvent of both operators contain all real numbers $\lambda \neq 0$.

Proposition 12 (properties of A^*) Let $A : D(A) \subset X \rightarrow X$ be a densely defined linear operator. Then the following properties hold.

(i) A satisfies the adjoint identity

$$\langle Au, v \rangle = \langle u, A^*v \rangle \quad \forall u \in D(A), \forall v \in D(A^*). \quad (101)$$

(ii) $A^* : D(A^*) \subset X \rightarrow X$ is a closed linear operator.

(iii) If $\lambda \in \rho(A)$, then $\bar{\lambda} \in \rho(A^*)$ and $R(\bar{\lambda}, A^*) = R(\lambda, A)^*$.

(iv) If A is closed, then $D(A^*)$ is dense in X . If, in addition, $\rho(A) \neq \emptyset$, then $A^{**} = A$.

Proof. (i) Let $v \in D(A^*)$. Then

$$\langle Au, v \rangle = \phi_v(u) = \langle u, j_X(\phi_v) \rangle = \langle u, A^*v \rangle \quad \forall u \in D(A) \quad \square$$

(ii) Let $\{v_n\} \subset D(A^*)$ and $v, w \in X$ be such that

$$\begin{cases} v_n \rightarrow v \\ A^*v_n \rightarrow w \end{cases} \quad (n \rightarrow \infty)$$

Then $\{A^*v_n\}$ is bounded, say $|A^*v_n| \leq C$. So, recalling (101), we have that

$$|\langle Au, v_n \rangle| = |\langle u, A^*v_n \rangle| \leq C|u| \quad \forall u \in D(A)$$

This yields

$$|\langle Au, v \rangle| \leq C|u| \quad \forall u \in D(A)$$

which in turn implies that $v \in D(A^*)$. Moreover

$$\langle Au, v \rangle = \lim_{n \rightarrow \infty} \langle Au, v_n \rangle = \langle u, w \rangle \quad \forall u \in D(A).$$

Thus, $\langle u, A^*v - w \rangle = 0$ for all $u \in D(A)$. Since $D(A)$ is dense, $A^*v = w$.

(iii) First of all, we claim that for any $\lambda \in \mathbb{C}$

$$D((\lambda I - A)^*) = D(A^*) \quad \text{and} \quad (\lambda I - A)^* = \bar{\lambda}I - A^*.$$

Indeed, for any $v \in D(A^*)$ we have that

$$|\langle (\lambda I - A)u, v \rangle| \leq (|\lambda| |v| + C_v)|u| \quad \forall u \in D(A)$$

Hence, $v \in D((\lambda I - A)^*)$ and this shows that $D(A^*) \subset D((\lambda I - A)^*)$. The reverse inclusion can be proved by a similar argument. Moreover, for all $u \in D(A)$, $v \in D(A^*)$ we have that

$$\langle u, (\lambda I - A)^*v \rangle = \langle (\lambda I - A)u, v \rangle = \langle u, (\bar{\lambda}I - A^*)v \rangle.$$

Since $D(A)$ is dense, the last identity implies that $(\lambda I - A)^* = \bar{\lambda}I - A^*$.

Now, let $\lambda \in \rho(A)$. Aiming to prove that $\bar{\lambda} \in \rho(A^*)$, first we show that $\bar{\lambda}I - A^*$ is injective. If $(\bar{\lambda}I - A^*)v = 0$ for some $v \in D(A^*)$, then

$$0 = \langle u, (\bar{\lambda}I - A^*)v \rangle = \langle (\lambda I - A)u, v \rangle \quad \forall u \in D(A).$$

Since $\lambda I - A$ is surjective, the above identity implies that $v = 0$. So, $\bar{\lambda}I - A^*$ is injective. Next, observe that, for all $v \in X$ and $u \in D(A)$,

$$\langle u, v \rangle = \langle R(\lambda, A)(\lambda I - A)u, v \rangle = \langle (\lambda I - A)u, R(\lambda, A)^*v \rangle,$$

yielding $R(\lambda, A)^*v \in D((\lambda I - A)^*) = D(\bar{\lambda}I - A^*) = D(A^*)$ and

$$(\bar{\lambda}I - A^*)R(\lambda, A)^*v = v \quad \forall v \in X. \quad (102)$$

On the other hand, if $u \in X$ and $v \in D(A^*)$, then

$$\langle u, v \rangle = \langle (\lambda I - A)R(\lambda, A)u, v \rangle = \langle R(\lambda, A)u, (\bar{\lambda}I - A^*)v \rangle.$$

Therefore,

$$R(\lambda, A)^*(\bar{\lambda}I - A^*)v = v \quad \forall v \in D(A^*). \quad (103)$$

(102) and (103) imply that $\bar{\lambda} \in \rho(A^*)$ and $R(\bar{\lambda}, A^*) = R(\lambda, A)^*$.

(iv) We argue by contradiction assuming the existence of $u_0 \neq 0$ such that $\langle u_0, v \rangle = 0$ for every $v \in D(A^*)$. Since $(0, u_0) \notin \text{graph}(A)$, which is a closed subspace of $X \times X$, from the Hahn-Banach theorem it follows that there exist $v_1, v_2 \in X$ such that the associated hyperplane in $X \times X$ separates $\text{graph}(A)$ and the point $(0, u_0)$, that is,

$$\langle u, v_1 \rangle - \langle Au, v_2 \rangle = 0 \quad \forall u \in D(A) \quad \text{and} \quad \langle 0, v_1 \rangle - \langle u_0, v_2 \rangle \neq 0$$

But the first identity implies that $v_2 \in D(A^*)$, which in turn yields $\langle u_0, v_2 \rangle = 0$, in contrast with the second equation above. So, $\overline{D(A^*)} = X$.

Finally, for all $u \in D(A)$ we have that

$$|\langle A^*v, u \rangle| = |\langle v, Au \rangle| \leq |Au| |v| \quad \forall v \in D(A^*).$$

Therefore, $D(A) \subset D(A^{**})$ and $A^{**}u = Au$ for all $u \in D(A)$. Moreover, by (iii) we conclude that $\rho(A) \subset \rho(A^{**})$. So, $A = A^{**}$ by Proposition 8. $\square$

In general, $D(A^*)$ may not be dense as our next example shows.

Example 21 Let $X = L^2(\mathbb{R})$ and fix any $f \in L^\infty(\mathbb{R}) \setminus X$ (for instance, $f \equiv 1$). Define

$$D(A) = \left\{ \psi \in X \mid \int_{-\infty}^{+\infty} |f\psi| dx < \infty \right\}$$

Then $D(A)$ is dense in X because it contains all continuous functions with compact support. Fix now $\psi_0 \in X \setminus \{0\}$ and define

$$A\psi = \left(\int_{-\infty}^{+\infty} f\psi dx \right) \psi_0 \quad \forall \psi \in D(A)$$

Then, for all $\phi \in D(A^*)$ and all $\psi \in D(A)$ we have that

$$\int_{-\infty}^{+\infty} \psi A^* \phi \, dx = \langle \psi, A^* \phi \rangle = \langle A\psi, \phi \rangle = \langle \psi_0, \phi \rangle \int_{-\infty}^{+\infty} \psi f \, dx$$

Therefore, $A^* \phi = \langle \psi_0, \phi \rangle f \in X$ if and only if $\langle \psi_0, \phi \rangle = 0$. It follows that $D(A^*) \perp \psi_0$. So, $D(A^*)$ cannot be dense.

We conclude this section with a lemma that will be used in the sequel.

Lemma 4 *Let $A : D(A) \subset X \rightarrow X$ be the infinitesimal generator of a C_0 -semigroup of bounded linear operators on X , $S(t)$. If, $u, w \in X$ satisfy*

$$\langle u, A^* v \rangle = \langle w, v \rangle \quad \forall v \in D(A^*), \quad (104)$$

then $u \in D(A)$ and $Au = w$.

Proof. Fix any $v \in D(A^*)$ and set $v_n = nR(n, A)^* v$ for $n > \omega$, with ω as in (28). Recalling that $R(n, A)^* = R(n, A^*)$ by Proposition 12, we have that

$$\langle u, A^* v_n \rangle = \langle u, nA^* R(n, A^*) v \rangle = \langle u, nR(n, A^*) A^* v \rangle = \langle nAR(n, A)u, v \rangle$$

Moreover, owing to (104),

$$\langle u, A^* v_n \rangle = \langle w, v_n \rangle = \langle nR(n, A)w, v \rangle \quad \forall v \in D(A^*).$$

Consequently, since $D(A^*)$ is dense in X ,

$$\langle nAR(n, A)u, v \rangle = \langle nR(n, A)w, v \rangle \quad \forall v \in X.$$

Therefore, $nR(n, A)u \rightarrow u$ and $nAR(n, A)u = nR(n, A)w \rightarrow w$ as $n \rightarrow \infty$ in view of Lemma 1. Since (A) is closed, the conclusion follows. $\square$

By introducing a dissipativity assumption for the adjoint of A we can replace the maximality requirement in the Lumer-Phillips theorem.

Theorem 10 (Lumer-Phillips 2) *Let $A : D(A) \subset X \rightarrow X$ be a densely defined closed linear operator. If A and A^* are dissipative, then A is the infinitesimal generator of a contraction semigroup on X .*

Proof. In view of Theorem 9, it suffices to show that $]0, \infty[\subset \rho(A)$. Since $\lambda I - A$ is one-to-one for any $\lambda > 0$, one just has to check that

$$(\lambda I - A)D(A) = X \quad \forall \lambda > 0.$$

STEP 1: $(\lambda I - A)D(A)$ is dense in X for every $\lambda > 0$.

Let $v \in X$ be such that

$$\langle \lambda u - Au, v \rangle = 0 \quad \forall u \in D(A).$$

The identity $\langle Au, v \rangle = \lambda \langle u, v \rangle$ yields $v \in D(A^*)$ and the fact that

$$\langle u, \lambda v - A^* v \rangle = 0,$$

first for all $u \in D(A)$ and then, by density, for all $u \in X$. So, $\lambda v - A^* v = 0$. Since, being dissipative, $\lambda I - A^*$ is also one-to-one, we conclude that $v = 0$.

STEP 2: $\lambda I - A$ is surjective for every $\lambda > 0$.

Fix any $v \in X$. By Step 1, there exists $\{u_n\} \subset D(A)$ such that

$$\lambda u_n - Au_n =: v_n \rightarrow v \quad \text{as } n \rightarrow \infty.$$

By (70) we deduce that, for all $n, m \geq 1$,

$$|u_n - u_m| \leq \frac{1}{\lambda} |v_n - v_m|$$

which insures that $\{u_n\}$ is a Cauchy sequence in X . Therefore, there exists $u \in X$ such that

$$\begin{cases} u_n \rightarrow u \\ Au_n = \lambda u_n - v_n \rightarrow \lambda u - v \end{cases} \quad (n \rightarrow \infty)$$

Since A is closed, $u \in D(A)$ and $\lambda u - Au = v$. □

2.3 Self-adjoint operators and Stone's theorem

In order to make further progress we have to better understand the relationship between the adjoint, $S(t)^*$, of a C_0 -semigroup of bounded linear operators on X and the adjoint, A^* , of its infinitesimal generator.

Theorem 11 *Let $S(t)$ be a C_0 -semigroup of bounded linear operators on X with infinitesimal generator $A : D(A) \subset X \rightarrow X$. Then $S(t)^*$ is a C_0 -semigroup of bounded linear operators on X , called the adjoint semigroup, whose infinitesimal generator is A^* , the adjoint of A .*

Proof. The semigroup identities for $S(t)^*$ follow directly from those for $S(t)$. Moreover, by (7) and Exercise 27,

$$\|S(t)^*\| = \|S(t)\| \leq M e^{\omega t} \quad \forall t \geq 0. \quad (105)$$

Increasing ω if necessary, we may assume that $\omega \geq 0$.

We first prove strong continuity. For every $u \in X$ and $v \in D(A^*)$, we have

$$\begin{aligned} \langle u, S(t)^* v - v \rangle &= \langle S(t)u - u, v \rangle \\ &= \left\langle A \int_0^t S(s)u \, ds, v \right\rangle \\ &= \int_0^t \langle S(s)u, A^* v \rangle \, ds. \end{aligned} \quad (106)$$

Consequently,

$$|\langle u, S(t)^*v - v \rangle| \leq Mte^{\omega t} |u| |A^*v|.$$

Taking the supremum over $|u| \leq 1$ yields

$$|S(t)^*v - v| \leq Mte^{\omega t} |A^*v| \quad \forall v \in D(A^*).$$

Thus $S(t)^*v \rightarrow v$ as $t \downarrow 0$ for every $v \in D(A^*)$. Since $D(A^*)$ is dense in X by Proposition 12, the local uniform bound (105) extends this convergence to every $v \in X$. Therefore, $S(t)^*$ is a strongly continuous semigroup.

Let B denote its infinitesimal generator. Now that strong continuity has been established, the vector-valued integral

$$\int_0^t S(s)^* A^* v \, ds$$

is well defined for every $v \in D(A^*)$. Identity (106), valid for all $u \in X$, gives

$$S(t)^*v - v = \int_0^t S(s)^* A^* v \, ds.$$

Hence

$$\frac{S(t)^*v - v}{t} = \frac{1}{t} \int_0^t S(s)^* A^* v \, ds \longrightarrow A^*v \quad (t \downarrow 0).$$

It follows that $A^* \subset B$.

Finally, Theorem 5 and (105) imply $\Pi_\omega \subset \rho(B)$. The same theorem applied to $S(t)$, together with Proposition 12, gives $\Pi_\omega \subset \rho(A^*)$. Thus $\rho(A^*) \cap \rho(B) \neq \emptyset$, and Proposition 8 yields $A^* = B$. $\square$

Of special interest are those operators A which coincide with their adjoint. We next introduce such a class.

Definition 18 A densely defined linear operator $A : D(A) \subset X \rightarrow X$ is called:

(a) *symmetric* if $A \subset A^*$, that is,

$$D(A) \subset D(A^*) \quad \text{and} \quad Au = A^*u \quad \forall u \in D(A).$$

(b) *self-adjoint* if $A = A^*$.

Remark 8 1. Observe that a symmetric operator A is self-adjoint if and only if $D(A^*) \subset D(A)$.

2. In view of Proposition 12, any self-adjoint operator is closed.

3. If $A \in \mathcal{L}(X)$, then A is self-adjoint if and only if A is symmetric. However, if A is unbounded, then symmetry does not yield self-adjointness as our next example shows.

Example 22 In $X = L^2(0, 1; \mathbb{C})$, consider the linear operator

$$\begin{cases} D(A) = H_0^1(0, 1; \mathbb{C}) \\ Au(x) = i u'(x) \quad x \in [0, 1] \text{ a.e.} \end{cases}$$

Then, A is densely defined and symmetric. Indeed, for all $u, v \in D(A)$,

$$\begin{aligned} \langle Au, v \rangle &= i \int_0^1 u'(x) \overline{v(x)} dx \\ &= [iu(x) \overline{v(x)}]_{x=0}^{x=1} - i \int_0^1 u(x) \overline{v'(x)} dx = \langle u, Av \rangle. \end{aligned} \quad (107)$$

On the other hand, A fails to be self-adjoint because, as we show next,

$$D(A^*) \supsetneq H^1(0, 1; \mathbb{C}),$$

so that $D(A) \subsetneq D(A^*)$. Indeed, integrating by parts as in (107), for all $v \in H^1(0, 1; \mathbb{C})$ and $u \in H_0^1(0, 1; \mathbb{C})$ we have that

$$|\langle Au, v \rangle| = \left| -i \int_0^1 u(x) \overline{v'(x)} dx \right| \leq \|u\|_2 \|v'\|_2. \quad \square$$

The next proposition gives a useful sufficient condition for a given operator to be self-adjoint.

Proposition 13 *Let $A : D(A) \subset X \rightarrow X$ be a densely defined closed linear operator such that $\rho(A) \cap \mathbb{R} \neq \emptyset$. If A is symmetric, then A is self-adjoint.*

Proof. We prove that $D(A^*) \subset D(A)$ in two steps. Fix any $\lambda \in \rho(A) \cap \mathbb{R}$.

STEP 1: $\boxed{R(\lambda, A) = R(\lambda, A)^*}$

Since $R(\lambda, A) \in \mathcal{L}(X)$, in view of Exercise 27 it suffices to show that

$$\langle R(\lambda, A)u, v \rangle = \langle u, R(\lambda, A)v \rangle \quad \forall u, v \in X.$$

Fix any $u, v \in X$ and set

$$x = R(\lambda, A)u \quad \text{and} \quad y = R(\lambda, A)v$$

so that $x, y \in D(A)$ and

$$\lambda x - Ax = u \quad \text{and} \quad \lambda y - Ay = v.$$

Since A is symmetric, we have that

$$\langle R(\lambda, A)u, v \rangle = \langle x, v \rangle = \langle x, \lambda y - Ay \rangle = \langle \lambda x - Ax, y \rangle = \langle u, R(\lambda, A)v \rangle.$$

STEP 2: $\boxed{D(A^*) \subset D(A)}$

Let $u \in D(A^*)$ and set $x = \lambda u - A^*u$. Observe that, for all $v \in D(A)$,

$$\langle x, v \rangle = \langle \lambda u - A^*u, v \rangle = \langle u, \lambda v - Av \rangle.$$

Now, take any $w \in X$ and let $v = R(\lambda, A)w$. Then the above identity yields

$$\langle x, R(\lambda, A)w \rangle = \langle u, w \rangle \quad \forall w \in X.$$

So, by Step 1 we conclude that $u = R(\lambda, A)^*x = R(\lambda, A)x \in D(A)$. $\square$

The following is an interesting spectral property of self-adjoint operators.

Proposition 14 *If $A : D(A) \subset X \rightarrow X$ is self-adjoint then*

$$\rho(A) \supset \{\lambda \in \mathbb{C} : \Im \lambda \neq 0\}$$

Consequently, $\sigma(A)$ is real.

Proof. For every $u \in D(A)$ we have that

$$\langle Au, u \rangle = \overline{\langle Au, u \rangle} \in \mathbb{R}. \quad (108)$$

Therefore, $|\langle \lambda u - Au, u \rangle| \geq |\Im \lambda| |u|^2$ which in turn yields

$$|\lambda u - Au| \geq |\Im \lambda| |u| \quad \forall u \in D(A). \quad (109)$$

The last inequality ensures that $\lambda I - A$ is an injective operator with closed range for all $\lambda \in \mathbb{C}$ with $\Im \lambda \neq 0$. Let us show that $(\lambda I - A)D(A)$ is dense in X for any such λ . Suppose there exists $v \neq 0$ such that

$$\langle \lambda u - Au, v \rangle = 0 \quad \forall u \in D(A).$$

Then $v \in D(A^*) = D(A)$ and we have that

$$\langle u, \overline{\lambda}v - Av \rangle = 0 \quad \forall u \in D(A).$$

Since $D(A)$ is dense in X , this implies that $\overline{\lambda}v - Av = 0$. Then, by (108), $\overline{\lambda} = \lambda \in \mathbb{R}$ contradicting $\Im \lambda \neq 0$. $\square$

The following is an immediate consequence of Theorem 10.

Corollary 5 (Lumer-Phillips 3) *Let $A : D(A) \subset X \rightarrow X$ be a densely defined linear operator. If A is self-adjoint and dissipative, then A is the infinitesimal generator of a contraction semigroup on X .*

Example 23 Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with boundary of class C^2 . Define

$$\begin{cases} D(A) = H^2 \cap H_0^1(\Omega; \mathbb{C}) \\ Au(x) = \Delta u(x) - V(x)u(x) \quad x \in \Omega \text{ a.e.} \end{cases} \quad (110)$$

where we assume $V \in L^\infty(\Omega, \mathbb{R})$. Let us check that A is self-adjoint in $L^2(\Omega; \mathbb{C})$. Indeed, integration by parts insures that A is symmetric. So, by Proposition 13, it

suffices to check that $\rho(A) \cap \mathbb{R} \neq \emptyset$. We claim that, for $\lambda \in \mathbb{R}$ large enough, for any $h \in L^2(\Omega; \mathbb{C})$ the problem

$$\begin{cases} w \in H^2 \cap H_0^1(\Omega; \mathbb{C}) \\ (\lambda + V(x))w(x) - \Delta w(x) = h(x) \quad (x \in \Omega \text{ a.e.}) \end{cases} \quad (111)$$

has a unique solution. Equivalently, by setting $f = \Re h$, $g = \Im h \in L^2(\Omega)$ and $u = \Re w$, $v = \Im w$, we have to prove solvability for the boundary value problems

$$\begin{cases} u \in H^2 \cap H_0^1(\Omega) \\ (\lambda + V)u - \Delta u = f \end{cases} \quad \text{and} \quad \begin{cases} v \in H^2 \cap H_0^1(\Omega) \\ (\lambda + V)v - \Delta v = g \end{cases}$$

The latter is a well-established fact in elliptic theory (see, e.g. [24]). On the other hand, operator A fails to be dissipative, in general.

Exercise 28 Prove that operator A in Example 23 is dissipative if

$$\|V\|_\infty \leq 1/C_\Omega,$$

where C_Ω is the Poincaré constant of Ω .

The following property of self-adjoint operators is very useful. We recall that an operator $U \in \mathcal{L}(X)$ is *unitary* if $UU^* = U^*U = I$.

Theorem 12 (Stone) *Let X be a complex Hilbert space. For any densely defined linear operator $A : D(A) \subset X \rightarrow X$ the following properties are equivalent:*

- (a) *A is self-adjoint,*
- (b) *iA is the infinitesimal generator of a C_0 -group of unitary operators.*

Proof. (a) $\Rightarrow$ (b) Since A is self-adjoint, A is closed and we have that

$$\langle Au, u \rangle = \langle u, A^*u \rangle = \langle u, Au \rangle = \overline{\langle Au, u \rangle} \quad \forall u \in D(A).$$

Thus, $\langle Au, u \rangle$ is real, so that

$$\Re \langle iAu, u \rangle = 0 \quad \forall u \in D(A).$$

The above identity implies that both iA and $-iA$ are dissipative operators. Since

$$\langle iAu, v \rangle = i \langle u, Av \rangle = \langle u, -iAv \rangle \quad \forall u, v \in D(A),$$

we have that $(iA)^* = -iA$. So, by Theorem 10 we deduce that $\pm iA$ is the infinitesimal generator of a C_0 -semigroup of contractions that we denote by $e^{\pm iAt}$. Then, by Theorem 7, iA generates a C_0 group, $G(t)$. Such a group is unitary because for any $t \geq 0$ we have that

$$G(t)^{-1} = G(-t) = e^{-iAt} = e^{(iA)^*t} = (e^{iAt})^* = G(t)^*,$$

while, for any $t < 0$,

$$G(t)^{-1} = e^{iA|t|} = e^{iA^*|t|} = e^{(-iA)^*|t|} = (e^{-iA|t|})^* = G(-|t|)^* = G(t)^*. \quad \square$$

(b) $\Rightarrow$ (a) Let iA be the infinitesimal generator of a C_0 -group of unitary operators on X , say $G(t)$. Then, for all $u \in D(A)$, we have that

$$\begin{aligned} iAu &= \lim_{t \rightarrow 0} \frac{G(t)u - u}{t} = -\lim_{t \rightarrow 0} \frac{G(-t)u - u}{t} = -\lim_{t \rightarrow 0} \frac{G(t)^*u - u}{t} = \\ &= -(iA)^*u = iA^*u. \end{aligned}$$

Thus, $u \in D(A^*)$ and $Au = A^*u$. By running the above computation backwards, we conclude that $D(A^*) \subseteq D(A)$. Therefore, A is self-adjoint. $\square$

Example 24 (Schrödinger equation in a bounded domain) Let us consider the initial-boundary value problem

$$\begin{cases} \frac{1}{i} \frac{\partial u}{\partial t}(t, x) = \Delta u(t, x) - V(x)u(t, x) & (t, x) \in \mathbb{R} \times \Omega \\ u(t, x) = 0 & t \in \mathbb{R}, x \in \partial\Omega \\ u(0, x) = u_0(x) & x \in \Omega \end{cases} \quad (112)$$

where $\Omega \subset \mathbb{R}^n$ is a bounded domain with boundary of class C^2 and $V \in L^\infty(\Omega)$. In Example 23, we have already checked that the operator A , defined in (110), is self-adjoint on $L^2(\Omega; \mathbb{C})$. Therefore, by Theorem 12 we conclude that, for any $u_0 \in H^2 \cap H_0^1(\Omega; \mathbb{C})$, problem (112) has a unique solution

$$u \in C^1(\mathbb{R}; L^2(\Omega; \mathbb{C})) \cap C(\mathbb{R}; H^2 \cap H_0^1(\Omega; \mathbb{C})). \quad \square$$

Exercise 29 Prove that the wave group (see examples 17, 19, and 20) is unitary.

Proof. Let us denote by $e^{t\mathcal{A}}$ ($t \in \mathbb{R}$) the strongly continuous group generated by $\mathcal{A}$. Since $\mathcal{A}^* = -\mathcal{A}$ (see Example 20), we have that

$$e^{-t\mathcal{A}} = e^{t\mathcal{A}^*} = (e^{t\mathcal{A}})^* \quad (t \in \mathbb{R})$$

So, $e^{t\mathcal{A}}$ is unitary. $\square$

2.4 The Cauchy problem for a self-adjoint generator

In this section, we will see that the homogeneous Cauchy problem

$$\begin{cases} u'(t) = Au(t) & t > 0 \\ u(0) = u^0 \in X \end{cases} \quad (113)$$

has a classical solution $u \in C([0, T]; X) \cap C^1([0, T]; X) \cap C([0, T]; D(A))$ when A is a self-adjoint and dissipative. Moreover, we will identify an intermediate space for the initial datum to ensure that $u \in H^1(0, T; X) \cap L^2(0, T; D(A))$. We can now prove that (113) has a (unique) classical solution for $u^0 \in X$.

Theorem 13 *Let $A : D(A) \subset X \rightarrow X$ be a self-adjoint dissipative operator and denote by $S(t)$ the C_0 -semigroup of bounded linear operators on X generated by A . Given any $u^0 \in X$, let $u(t) = S(t)u^0$. Then, for any $T > 0$ we have that*

$$u \in C([0, T]; X) \cap C^1([0, T]; X) \cap C([0, T]; D(A)). \quad (114)$$

Moreover, the following inequality holds:

$$4 \int_0^T t |Au(t)|^2 dt - 2T \langle Au(T), u(T) \rangle + |u(T)|^2 \leq |u^0|^2. \quad (115)$$

Observe that all the terms on the left side of (115) are nonnegative, so that each one of them is bounded by $|u^0|^2$.

Proof. For any $n \geq 1$ define

$$u_n^0 = nR(n, A)u^0 \quad \text{and} \quad u_n(t) = S(t)u_n^0.$$

Since $u_n^0 \in D(A)$, we have that $u_n'(t) = Au_n(t)$ for all $t \geq 0$. So ²,

$$\frac{1}{2} \frac{d}{dt} |u_n(t)|^2 = \langle u_n'(t), u_n(t) \rangle = \langle Au_n(t), u_n(t) \rangle$$

Therefore,

$$|u_n(t)|^2 - 2 \int_0^t \langle Au_n(s), u_n(s) \rangle ds = |u_n^0|^2 \quad \forall t \geq 0. \quad (116)$$

Similarly, for all $t \geq 0$ we have that

$$t |u_n'(t)|^2 = t \langle Au_n(t), u_n'(t) \rangle = \frac{1}{2} \frac{d}{dt} (t \langle Au_n(t), u_n(t) \rangle) - \frac{1}{2} \langle Au_n(t), u_n(t) \rangle.$$

Integrating the above identity over $[0, T]$ yields, by (116),

$$\begin{aligned} 2 \int_0^T t |u_n'(t)|^2 dt - T \langle Au_n(T), u_n(T) \rangle &= - \int_0^T \langle Au_n(t), u_n(t) \rangle dt \\ &= \frac{1}{2} (|u_n^0|^2 - |u_n(T)|^2) \end{aligned}$$

or

$$4 \int_0^T t |Au_n(t)|^2 dt - 2T \langle Au_n(T), u_n(T) \rangle + |u_n(T)|^2 \leq |u^0|^2 \quad (117)$$

² Observe that $\langle u_n'(t), u_n(t) \rangle$ is real because A is self-adjoint.

since $\|nR(n, A)\| \leq 1$. The last inequality implies that, for any $\varepsilon \in]0, T[$, $\{u_n\}$ is bounded in $L^2(\varepsilon, T; D(A))$. Therefore, there exists a weakly convergent subsequence $\{u_{n_k}\}$ in $L^2(\varepsilon, T; D(A))$. On the other hand, $u_{n_k} \rightarrow u$ uniformly on $[0, T]$. So, $u \in L^2(\varepsilon, T; D(A))$ for any $\varepsilon \in]0, T[$, which in turn yields $u(t) \in D(A)$ for a.e. $t > 0$. However, as we know, once $u(t)$ enters $D(A)$ it stays there. Thus, $u(t) \in D(A)$ for all $t > 0$ and (114) follows. Moreover, recalling Remark 2 we conclude that

$$Au_n(t) = nAS(t)R(n, A)u^0 = nAR(n, A)u(t) \xrightarrow{n \rightarrow \infty} Au(t) \quad \forall t > 0.$$

Taking the limit as $n \rightarrow \infty$ in (117), by Fatou's lemma we get (115). $\square$

The above result can be used to introduce an intermediate space between X and $D(A)$, namely the interpolation space $[X, D(A)]_{1/2}$, such that $t \mapsto S(t)u_0$ belongs to $H^1(0, T; X) \cap L^2(0, T; D(A))$ whenever $u_0 \in [D(A), X]_{1/2}$. We give a brief account of such a construction, referring the reader to [13] for more on interpolation spaces.

Proposition 15 *Let $A : D(A) \subset X \rightarrow X$ be a self-adjoint dissipative operator and denote by $S(t)$ the C_0 -semigroup of bounded linear operators on X generated by A . Then, for any $u^0 \in X$, the functions*

$$t \mapsto -\langle AS(t)u^0, S(t)u^0 \rangle$$

and

$$t \mapsto -\frac{1}{t} \int_0^t \langle AS(s)u^0, S(s)u^0 \rangle ds$$

are both nonincreasing on $]0, \infty[$.

Proof. Since $S(t)u^0 \in D(A)$ for every $t > 0$ by Theorem 13, we have that

$$0 \leq 2|AS(t)u^0|^2 = \frac{d}{dt} \langle AS(t)u^0, S(t)u^0 \rangle \quad \forall t > 0.$$

This shows that $t \mapsto -\langle AS(t)u^0, S(t)u^0 \rangle$ is nonincreasing on $]0, \infty[$. The other conclusion is a consequence of the general fact which is proven below. $\square$

Lemma 5 *Let $f \geq 0$ be a nonincreasing function on $]0, \infty[$. Then $t \mapsto \frac{1}{t} \int_0^t f(s) ds$ is also nonincreasing on $]0, \infty[$.*

Proof. Observe that for any $0 < t < t'$ we have that

$$f(t') \leq \frac{1}{(t' - t)} \int_t^{t'} f(s) ds \leq f(t) \leq \frac{1}{t} \int_0^t f(s) ds. \quad (118)$$

This yields

$$\begin{aligned}
\frac{1}{t'} \int_0^{t'} f(s) ds &= \frac{1}{t} \int_0^t f(s) ds + \left(\frac{1}{t'} - \frac{1}{t}\right) \int_0^t f(s) ds + \frac{1}{t'} \int_t^{t'} f(s) ds \\
&= \frac{1}{t} \int_0^t f(s) ds + \frac{t' - t}{t'} \left\{ \frac{1}{(t' - t)} \int_t^{t'} f(s) ds - \frac{1}{t} \int_0^t f(s) ds \right\} \\
&\leq \frac{1}{t} \int_0^t f(s) ds + \frac{t' - t}{t'} \{f(t) - f(t)\} = \frac{1}{t} \int_0^t f(s) ds,
\end{aligned}$$

where we have made repeated use of (118). The conclusion follows. $\square$

In view of the above proposition, we have that, for any $u^0 \in X$,

$$\lim_{t \downarrow 0} -\langle AS(t)u^0, S(t)u^0 \rangle = \sup_{t > 0} -\langle AS(t)u^0, S(t)u^0 \rangle \quad (119)$$

and

$$\lim_{T \downarrow 0} -\frac{1}{T} \int_0^T \langle AS(t)u^0, S(t)u^0 \rangle dt = \sup_{T > 0} -\frac{1}{T} \int_0^T \langle AS(t)u^0, S(t)u^0 \rangle dt. \quad (120)$$

The above quantities are useful to define the intermediate space $[D(A), X]_{1/2}$.

Definition 19 For any $u^0 \in X$ we set

$$|u^0|_{1/2}^2 = \sup_{t > 0} -\langle AS(t)u^0, S(t)u^0 \rangle$$

and we define

$$[D(A), X]_{1/2} = \{u^0 \in X : |u^0|_{1/2} < \infty\}. \quad (121)$$

Proposition 16 $[D(A), X]_{1/2}$ is a subspace of X containing $D(A)$ and

$$\|u^0\|_{1/2} = |u^0| + |u^0|_{1/2}$$

is a norm on $[D(A), X]_{1/2}$. Moreover,

$$|u^0|_{1/2}^2 = \sup_{T > 0} -\frac{1}{T} \int_0^T \langle AS(t)u^0, S(t)u^0 \rangle dt. \quad (122)$$

Proof. It is clear that $|u^0|_{1/2}^2 \leq |Au^0| |u^0|$ for any $u^0 \in D(A)$, so that

$$D(A) \subset [D(A), X]_{1/2}.$$

Next, for any $u^0 \in [D(A), X]_{1/2}$ we have that $|\lambda u^0|_{1/2} = |\lambda| |u^0|_{1/2}$ for all $\lambda \in \mathbb{C}$. Moreover, for any $u^0, v^0 \in [D(A), X]_{1/2}$ and any $t > 0$ let us set

$$\alpha_t(u^0, v^0) = \langle -AS(t)u^0, S(t)v^0 \rangle.$$

Then, by the Cauchy-Schwarz inequality we obtain

$$\begin{aligned}
\alpha_t(u^0 + v^0, u^0 + v^0) &= \alpha_t(u^0, u^0) + 2\Re \alpha_t(u^0, v^0) + \alpha_t(v^0, v^0) \\
&\leq \alpha_t(u^0, u^0) + 2\alpha_t(u^0, u^0)^{1/2} \alpha_t(v^0, v^0)^{1/2} + \alpha_t(v^0, v^0) \\
&= [\alpha_t(u^0, u^0)^{1/2} + \alpha_t(v^0, v^0)^{1/2}]^2.
\end{aligned}$$

Therefore, $[D(A), X]_{1/2}$ is a vector space and $|\cdot|_{1/2}$ is a semi-norm on $[D(A), X]_{1/2}$. Finally, in order to prove (122) observe that, in view of Proposition 15,

$$-\frac{1}{T} \int_0^T \langle AS(t)u^0, S(t)u^0 \rangle dt \leq \sup_{t>0} -\langle AS(t)u^0, S(t)u^0 \rangle = |u^0|_{1/2}^2 \quad \forall T > 0.$$

Similarly, for any $t > 0$ we have that

$$\begin{aligned}
-\langle AS(t)u^0, S(t)u^0 \rangle &\leq -\frac{1}{t} \int_0^t \langle AS(s)u^0, S(s)u^0 \rangle ds \\
&\leq \sup_{T>0} -\frac{1}{T} \int_0^T \langle AS(s)u^0, S(s)u^0 \rangle ds.
\end{aligned}$$

This shows that

$$|u^0|_{1/2}^2 \leq \sup_{T>0} -\frac{1}{T} \int_0^T \langle AS(s)u^0, S(s)u^0 \rangle ds,$$

thus completing the proof. $\square$

Theorem 14 *Let $A : D(A) \subset X \rightarrow X$ be a self-adjoint dissipative operator and denote by $S(t)$ the C_0 -semigroup generated by A . For any $u^0 \in [D(A), X]_{1/2}$, the solution $u(t) = S(t)u^0$ of (113) satisfies*

$$u \in H^1(0, T; X) \cap L^2(0, T; D(A)) \quad \text{for every } T > 0.$$

Moreover,

$$\int_0^\infty |AS(t)u^0|^2 dt \leq \frac{1}{2} |u^0|_{1/2}^2 \quad \forall u^0 \in [D(A), X]_{1/2}, \quad (123)$$

and

$$|S(t)u^0|_{1/2} \leq |u^0|_{1/2} \quad \forall t \geq 0, \quad \forall u^0 \in [D(A), X]_{1/2}. \quad (124)$$

Proof. Fix $u^0 \in [D(A), X]_{1/2}$ and let $u(t) = S(t)u^0$. For any $\varepsilon > 0$, the definition of $|\cdot|_{1/2}$ gives

$$-\langle Au(\varepsilon), u(\varepsilon) \rangle \leq |u^0|_{1/2}^2.$$

Since A is self-adjoint and $u' = Au$, integration over $[\varepsilon, T]$, with $T > \varepsilon$, yields

$$\begin{aligned}
\int_\varepsilon^T |Au(t)|^2 dt &= \frac{1}{2} \langle Au(T), u(T) \rangle - \frac{1}{2} \langle Au(\varepsilon), u(\varepsilon) \rangle \\
&\leq \frac{1}{2} |u^0|_{1/2}^2,
\end{aligned}$$

where we used the dissipativity of A . Letting $\varepsilon \downarrow 0$ and $T \uparrow \infty$ proves (123).

For every finite $T > 0$, the continuity of u on $[0, T]$ implies $u \in L^2(0, T; X)$. Together with (123), this gives $u \in L^2(0, T; D(A))$. Moreover, integrating $u' = Au$ over $[\varepsilon, t]$ and letting $\varepsilon \downarrow 0$, we obtain

$$u(t) - u^0 = \int_0^t Au(s) ds \quad \forall t \in [0, T].$$

Consequently, $u \in H^1(0, T; X)$.

Finally, for every $t > 0$, the semigroup property gives

$$\begin{aligned} |S(t)u^0|_{1/2}^2 &= \sup_{\tau > 0} -\langle AS(\tau + t)u^0, S(\tau + t)u^0 \rangle \\ &= \sup_{s > t} -\langle AS(s)u^0, S(s)u^0 \rangle \leq |u^0|_{1/2}^2. \end{aligned}$$

For $t = 0$, the conclusion is immediate. $\square$

Corollary 6 *Under the assumptions of Theorem 14, suppose in addition that A is uniformly strictly dissipative: there exists $\alpha > 0$ such that*

$$\langle Av, v \rangle \leq -\alpha |v|^2 \quad \forall v \in D(A).$$

Then, for every $u^0 \in [D(A), X]_{1/2}$, the solution $u(t) = S(t)u^0$ belongs to

$$H^1(0, \infty; X) \cap L^2(0, \infty; D(A)).$$

Moreover,

$$\int_0^\infty |u(t)|^2 dt \leq \frac{1}{2\alpha} |u^0|^2.$$

Proof. The dissipativity assumption gives

$$|S(t)u^0| \leq e^{-\alpha t} |u^0| \quad \forall t \geq 0.$$

Integration yields the stated estimate. Combining it with (123) and $u' = Au$ proves the asserted regularity. $\square$

Remark 9 More generally, for $u^0 \in [D(A), X]_{1/2}$, the global regularity

$$u \in H^1(0, \infty; X) \cap L^2(0, \infty; D(A))$$

holds whenever $S(\cdot)u^0 \in L^2(0, \infty; X)$. This additional integrability does not follow from dissipativity alone: if $A = 0$ and $u^0 \neq 0$, then $u(t) = u^0$ for every $t \geq 0$.

Example 25 On $X = L^2(0, \pi)$, consider the operator

$$D(A) = H^2(0, \pi) \cap H_0^1(0, \pi), \quad Af = f''.$$

We know that A is self-adjoint and dissipative. We will show that

$$[D(A), X]_{1/2} = H_0^1(0, \pi), \quad (125)$$

and identify the seminorm $|\cdot|_{1/2}$.

For any $f \in L^2(0, \pi)$, write

$$f = \sum_{n=1}^{\infty} f_n \phi_n \quad \text{in } L^2(0, \pi), \quad \phi_n(x) = \sqrt{\frac{2}{\pi}} \sin(nx),$$

where

$$f_n = \sqrt{\frac{2}{\pi}} \int_0^{\pi} f(x) \sin(nx) dx, \quad |f|_2^2 = \sum_{n=1}^{\infty} |f_n|^2.$$

By Exercise 13, for $t > 0$,

$$S(t)f = \sum_{n=1}^{\infty} e^{-n^2 t} f_n \phi_n, \quad AS(t)f = - \sum_{n=1}^{\infty} n^2 e^{-n^2 t} f_n \phi_n,$$

with convergence in $L^2(0, \pi)$. Consequently,

$$-\langle AS(t)f, S(t)f \rangle = \sum_{n=1}^{\infty} n^2 e^{-2n^2 t} |f_n|^2.$$

Letting $t \downarrow 0$ and applying monotone convergence, we obtain

$$|f|_{1/2}^2 = \sup_{t>0} (-\langle AS(t)f, S(t)f \rangle) = \sum_{n=1}^{\infty} n^2 |f_n|^2,$$

where both sides may be infinite.

If $f \in H_0^1(0, \pi)$, integration by parts gives

$$\sqrt{\frac{2}{\pi}} \int_0^{\pi} f'(x) \cos(nx) dx = n f_n \quad (n \geq 1),$$

while $\int_0^{\pi} f'(x) dx = 0$. Parseval's identity for the cosine system therefore yields

$$\int_0^{\pi} |f'(x)|^2 dx = \sum_{n=1}^{\infty} n^2 |f_n|^2. \quad (126)$$

Thus $H_0^1(0, \pi) \subset [D(A), X]_{1/2}$ and

$$|f|_{1/2}^2 = \int_0^{\pi} |f'(x)|^2 dx \quad \forall f \in H_0^1(0, \pi). \quad (127)$$

Conversely, let $f \in [D(A), X]_{1/2}$. Then

$$\sum_{n=1}^{\infty} n^2 |f_n|^2 < \infty.$$

Consider the partial sums

$$f^{(N)} = \sum_{n=1}^N f_n \phi_n.$$

Each $f^{(N)}$ belongs to $H_0^1(0, \pi)$. Moreover, for $N > M$, orthogonality gives

$$\|f^{(N)} - f^{(M)}\|_2^2 + \|(f^{(N)} - f^{(M)})'\|_2^2 = \sum_{n=M+1}^N (1 + n^2) |f_n|^2.$$

Hence $\{f^{(N)}\}$ is Cauchy in $H_0^1(0, \pi)$. Since it also converges to f in $L^2(0, \pi)$, its limit in $H_0^1(0, \pi)$ is f . This proves (125).

In particular, the norm introduced in Proposition 16 takes the form

$$\|f\|_{1/2} = \|f\|_2 + \|f'\|_2 \quad \forall f \in H_0^1(0, \pi).$$

Exercise 30 Use Theorem 13 to show that, if $A : D(A) \subset X \rightarrow X$ is a self-adjoint dissipative operator and $S(t)$ is the C_0 -semigroup of bounded linear operators on X generated by A , then:

1. $S(t)u^0 \in D(A^n)$ for all $t > 0$, all $u^0 \in X$, and all $n \in \mathbb{N}$;
2. for all $u^0 \in X$

$$|AS(t)u^0| \leq \frac{|u^0|}{t\sqrt{2}} \quad \forall t > 0. \quad (128)$$

Solution. To prove (a) it suffices to observe that for all $t > 0$ and $u^0 \in X$,

$$AS(t)u^0 = S(t/2)AS(t/2)u^0 \in D(A) \implies S(t)u^0 \in D(A^2).$$

The general case follows by induction.

Next, using the dissipativity of A we obtain

$$\frac{d}{dt} |AS(t)u^0|^2 = 2\langle A^2 S(t)u^0, AS(t)u^0 \rangle \leq 0.$$

Thus, $t \mapsto |AS(t)u^0|^2$ is nonincreasing. So, (115) yields

$$2t^2 |AS(t)u^0|^2 = 4 \int_0^t s |AS(s)u^0|^2 ds \leq 4 \int_0^t s |AS(s)u^0|^2 ds \leq |u^0|^2. \quad \square$$

3 The inhomogeneous Cauchy problem

Throughout this section, we assume that $(X, \langle \cdot, \cdot \rangle)$ is a separable Hilbert space and denote by $\{e_j\}_{j \in \mathbb{N}}$ a complete orthonormal system in X .

We study the Cauchy problem

$$\begin{cases} u'(t) = Au(t) + f(t) \\ u(0) = u^0, \end{cases} \quad (129)$$

where $f \in L^2(0, T; X)$ and $A : D(A) \subset X \rightarrow X$ is the infinitesimal generator of a C_0 -semigroup on X , $S(t)$, which satisfies the growth condition (28). For the extension of this theory to a general Banach space, we refer the reader to the classic monograph by Pazy [27] or the more recent text [23].

3.1 Notions of solution

Let $u^0 \in X$ and let $f \in L^2(0, T; X)$.

Definition 20 (Mild solutions) The function $u \in C([0, T]; X)$ defined by

$$u(t) = S(t)u^0 + \int_0^t S(t-s)f(s) ds \quad (130)$$

is called the *mild solution* of (129).

Observe that the convolution term in formula (130) for the solution u is well-defined in view of Proposition 24 in Appendix B.

Another way to relax the notion of solution is the following one.

Definition 21 (Weak solutions) A function $u \in C([0, T]; X)$ is called a weak solution of (129) if $u(0) = u^0$ and for all $v \in D(A^*)$:

- (i) $t \mapsto \langle u(t), v \rangle$ is absolutely continuous on $[0, T]$
- (ii) $\frac{d}{dt} \langle u(t), v \rangle = \langle u(t), A^*v \rangle + \langle f(t), v \rangle$ for a.e. $t \in [0, T]$

Example 26 Let us consider the inhomogeneous heat equation on $X = L^2(0, \pi)$

$$\begin{cases} \frac{\partial u}{\partial t}(t, x) = \frac{\partial^2 u}{\partial x^2}(t, x) + f(t, x) & (t, x) \in [0, T] \times]0, \pi[\\ u(t, 0) = 0 = u(t, \pi) & t \in [0, T] \\ u(0, x) = u_0(x) & x \in [0, \pi] \end{cases} \quad (131)$$

In Example 11, we showed that the operator $A : D(A) \subset X \rightarrow X$ defined in (47) is the infinitesimal generator of a C_0 -semigroup of bounded linear operators on X . By applying the Hille-Yosida theorem we conclude that A is the infinitesimal generator of a strongly continuous semigroup of contractions on X , $S(t)$. Moreover, we saw in Example 23 that A is self-adjoint. Therefore, given any

$$u_0 \in L^2(0, \pi) \quad \text{and} \quad f \in L^2([0, T] \times [0, \pi]),$$

a function $u \in C([0, T]; L^2(0, \pi))$ is a weak solution of (131) if $u(0, x) = u_0(x)$ a.e. on $[0, \pi]$ and for all $v \in H^2(0, \pi)$ satisfying $v(0) = 0 = v(\pi)$ we have that

$$t \mapsto \int_0^\pi u(t, x)v(x) dx \quad \text{is absolutely continuous on } [0, T]$$

and, for a.e. $t \in [0, T]$,

$$\frac{d}{dt} \int_0^\pi u(t, x)v(x) dx = \int_0^\pi [u(t, x)v''(x) + f(t, x)v(x)] dx.$$

Example 27 Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with boundary of class C^2 . Consider the inhomogeneous wave equation

$$\begin{cases} \frac{\partial^2 u}{\partial t^2}(t, x) = \Delta u(t, x) + f(t, x) & (t, x) \in (0, T) \times \Omega, \\ u(t, x) = 0 & (t, x) \in (0, T) \times \partial\Omega, \\ u(0, x) = u^0(x), \quad \frac{\partial u}{\partial t}(0, x) = v^0(x) & x \in \Omega, \end{cases} \quad (132)$$

where

$$(u^0, v^0) \in H_0^1(\Omega) \times L^2(\Omega), \quad f \in L^2((0, T) \times \Omega).$$

In Example 20, we showed that the adjoint of the wave operator $\mathcal{A}$, defined in (79), is $-\mathcal{A}$. Therefore, a function

$$u \in C([0, T]; H_0^1(\Omega)) \cap C^1([0, T]; L^2(\Omega))$$

is a weak solution of (132) if

$$u(0) = u^0, \quad u_t(0) = v^0,$$

and, for every

$$\varphi \in H^2(\Omega) \cap H_0^1(\Omega), \quad \psi \in H_0^1(\Omega),$$

the map

$$t \mapsto \int_\Omega (Du(t, x) \cdot D\varphi(x) + u_t(t, x)\psi(x)) dx$$

is absolutely continuous on $[0, T]$ and satisfies

$$\begin{aligned} \frac{d}{dt} \int_\Omega (Du(t, x) \cdot D\varphi(x) + u_t(t, x)\psi(x)) dx \\ = - \int_\Omega (Du(t, x) \cdot D\psi(x) + u_t(t, x)\Delta\varphi(x)) dx + \int_\Omega f(t, x)\psi(x) dx \end{aligned}$$

for a.e. $t \in [0, T]$.

The relation between the above two notions is clarified by the following result due to J. Ball [5].

Theorem 15 *There exists a unique weak solution of (129) which is given by the mild solution.*

Proof. Let u denote the mild solution of (129). Then, for any $v \in D(A^*)$ we have that

$$\langle u(t), v \rangle = \langle u^0, S(t)^* v \rangle + \int_0^t \langle f(s), S(t-s)^* v \rangle ds \quad \forall t \in [0, T]$$

Therefore, $t \mapsto \langle u(t), v \rangle$ is absolutely continuous on $[0, T]$ and

$$\begin{aligned} \frac{d}{dt} \langle u(t), v \rangle &= \langle u^0, S(t)^* A^* v \rangle + \int_0^t \langle f(s), S(t-s)^* A^* v \rangle ds + \langle f(t), v \rangle \\ &= \left\langle S(t)u^0 + \int_0^t S(t-s)f(s) ds, A^* v \right\rangle + \langle f(t), v \rangle \\ &= \langle u(t), A^* v \rangle + \langle f(t), v \rangle \quad t \in [0, T] \text{ a.e.} \end{aligned}$$

Therefore, u is a weak solution of (129).

Let us now show that (129) has at most one weak solution. Let u_1 and u_2 be weak solutions of such a problem and set $w = u_1 - u_2$. Then

$$\begin{cases} \frac{d}{dt} \langle w(t), v \rangle = \langle w(t), A^* v \rangle & \forall v \in D(A^*), t \in [0, T] \text{ a.e.} \\ w(0) = 0 \end{cases} \quad (133)$$

Set $W(t) = \int_0^t w(s) ds$. Then $W \in C^1([0, T]; X)$ and, by integrating (133), one easily checks that

$$\langle W'(t), v \rangle = \langle W(t), A^* v \rangle \quad \forall v \in D(A^*), \forall t \in [0, T]$$

So, in view of Lemma 4 we conclude that $W(t) \in D(A)$ and $AW(t) = W'(t)$. In other words, W is a classical solution of the problem

$$\begin{cases} W'(t) = AW(t) & \forall t \in [0, T] \\ W(0) = 0 \end{cases}$$

Hence, $W(t) \equiv 0$ on account of Proposition 6. This in turn implies that $w(t) \equiv 0$, thus completing the proof. $\square$

Let now $f \in C([0, T]; X)$.

Definition 22 (Classical solutions) A classical solution of (129) is a function $u \in C([0, T]; X)$ such that

[(a)]

1. $u \in C^1([0, T]; X) \cap C([0, T]; D(A))$;
2. $u(0) = u^0$;

3. $u'(t) = Au(t) + f(t)$ for all $t \in]0, T]$.

We now show that any classical solution coincides with the mild solution.

Proposition 17 *Let u be a classical solution of (129). Then u equals the mild solution given by (130).*

Proof. Let u be a classical solution of (129). Then, for any fixed $t \in]0, T]$ we have that $s \mapsto S(t-s)u(s)$ is continuous on $[0, t]$, differentiable on $]0, t[$, and

$$\frac{d}{ds} (S(t-s)u(s)) = S(t-s)f(s) \quad (s \in]0, t[).$$

By integrating over $[0, t]$ we deduce that u is given by (130). $\square$

3.2 Regularity of mild solutions

In this section, we shall give sufficient conditions for the mild solution of (129) to be classical.

Remark 10 In general, the mild solution of (129) fails to be classical assuming just $f \in C([0, T]; X)$. For a counterexample, take A to be the unbounded generator of a C_0 -group, such as the translation group on $L^2(\mathbb{R})$ or the wave group considered above. Now, let $w \in X \setminus D(A)$ and take $f(t) = S(t)w$. Then the mild solution of (129) with $u^0 = 0$ is given by

$$u(t) = tS(t)w \quad \forall t \geq 0$$

which fails to be differentiable for any $t > 0$. Indeed, differentiability at such a time would imply $S(t)w \in D(A)$ and hence $w = S(-t)S(t)w \in D(A)$, since the group preserves $D(A)$. This contradicts the choice of w .

Our first result guarantees such a property when u^0 and f have better “space regularity”.

Theorem 16 *Let $u^0 \in D(A)$ and let $f \in L^2(0, T; D(A)) \cap C([0, T]; X)$. Then the mild solution u of problem (129) is classical. Moreover,*

$$u \in C^1([0, T]; X) \cap C([0, T]; D(A)). \quad (134)$$

We begin the proof by studying the case of $u^0 = 0$.

Lemma 6 *For any $f \in L^2(0, T; D(A)) \cap C([0, T]; X)$ define*

$$F_A(t) = \int_0^t S(t-s)f(s) ds \quad (t \in [0, T]). \quad (135)$$

Then $F_A \in C^1([0, T]; X) \cap C([0, T]; D(A))$ and

$$F'_A(t) = AF_A(t) + f(t) \quad \forall t \in [0, T]. \quad (136)$$

Proof. Since A is fixed, let us simplify the notation by writing F instead of F_A . Since $f \in L^2(0, T; D(A))$ we have that, for any $t \in [0, T]$,

$$A \int_0^t S(t-s)f(s) ds = \int_0^t S(t-s)Af(s) ds.$$

So, $F \in C([0, T]; D(A))$ on account of Proposition 24 of Appendix B.

In order to prove that $F \in C^1([0, T]; X)$, fix $t \in]0, T[$ and let $0 < h < T - t$. Then

$$\begin{aligned} \frac{F(t+h) - F(t)}{h} &= \frac{1}{h} \left\{ \int_0^{t+h} S(t+h-s)f(s) ds - \int_0^t S(t-s)f(s) ds \right\} \\ &= \frac{S(h) - I}{h} F(t) + \frac{1}{h} \int_t^{t+h} S(t+h-s)f(s) ds. \end{aligned}$$

Now,

$$\lim_{h \downarrow 0} \frac{S(h) - I}{h} F(t) = AF(t)$$

because $F \in C([0, T]; D(A))$. Also,

$$\lim_{h \downarrow 0} \frac{1}{h} \int_t^{t+h} S(t+h-s)f(s) ds = f(t)$$

because $f \in C([0, T]; X)$. This shows that the right derivative $F'_+(t)$ exists for all $t \in]0, T[$ and equals $AF(t) + f(t)$.

Let us show the existence of the left derivative $F'_-(t)$. For any $0 < h < t$ we have

$$\begin{aligned} \frac{F(t-h) - F(t)}{-h} &= \frac{F(t) - F(t-h)}{h} \\ &= \frac{1}{h} \left\{ \int_0^t S(t-s)f(s) ds - \int_0^{t-h} S(t-h-s)f(s) ds \right\} \\ &= \frac{1}{h} \int_{t-h}^t S(t-s)f(s) ds + \int_0^{t-h} S(t-h-s) \frac{S(h) - I}{h} f(s) ds. \end{aligned}$$

Then

$$\lim_{h \downarrow 0} \frac{1}{h} \int_{t-h}^t S(t-s)f(s) ds = f(t)$$

as we showed above. Moreover,

$$\lim_{h \downarrow 0} \frac{S(h) - I}{h} f(s) = Af(s) \quad s \in [0, T] \text{ a.e.}$$

and

$$\left| \frac{S(h) - I}{h} f(s) \right| = \left| \frac{1}{h} \int_0^h S(\tau) A f(s) d\tau \right| \leq C |A f(s)| \quad s \in [0, T] \text{ a.e.}$$

for all $0 < h < t$ and some constant $C > 0$. Therefore, we can apply Lebesgue's dominated convergence theorem to conclude that $F'_-(t)$ exists for all $t \in]0, T[$ and equals $AF(t) + f(t)$. In sum, $F \in C^1([0, T]; X)$ and (136) holds true. $\square$

We can now complete the proof of Theorem 16.

Proof. Let u be the mild solution of problem (129). Then

$$u(t) = S(t)u^0 + F_A(t) \quad \forall t \in [0, T],$$

where F_A is defined in (135). Theorem 3 and Lemma 6 yield the conclusion. $\square$

As an application of Theorem 16 we present the following approximation result for mild solutions.

Theorem 17 *Let $u \in C([0, T]; X)$ be the mild solution of (129) and suppose $f \in C([0, T]; X)$. Then, the sequence $u_n := nR(n, A)u$, defined for all $n > \omega$, satisfies*

$$u_n \in C^1([0, T]; X) \cap C([0, T]; D(A)) \quad \text{and} \quad u_n \xrightarrow{(n \rightarrow \infty)} u \text{ in } C([0, T]; X).$$

Proof. Define

$$\begin{cases} f_n(t) = nR(n, A)f(t) \\ u_n^0 = nR(n, A)u^0 \end{cases} \quad \forall n > \omega$$

where $\omega \geq 0$ is such that (28) holds true. Then

$$u_n(t) = S(t)u_n^0 + \int_0^t S(t-s)f_n(s) ds \quad (t \in [0, T]).$$

Since $u_n^0 \in D(A)$ and $f_n \in C([0, T]; D(A))$, by Theorem 16 we conclude that

$$u_n \in C^1([0, T]; X) \cap C([0, T]; D(A)) \quad \text{and} \quad \begin{cases} u_n' - Au_n = f_n \\ u_n(0) = u_n^0. \end{cases}$$

Moreover, invoking Lemma 1 we conclude that $u_n^0 \rightarrow u^0$ as $n \rightarrow \infty$ while

$$f_n(t) \xrightarrow{(n \rightarrow \infty)} f(t) \quad \text{and} \quad |f_n(t)| \leq \frac{Mn}{n-\omega} |f(t)| \quad (\text{for all } t \in [0, T])$$

Therefore,

$$\sup_{t \in [0, T]} |u_n(t) - u(t)| \leq Me^{\omega T} (|u_n^0 - u^0| + \int_0^T |f_n(s) - f(s)| ds) \xrightarrow{(n \rightarrow \infty)} 0.$$

The conclusion follows. $\square$

We will now treat the case when f has better “time regularity”.

Theorem 18 *Let $u^0 \in D(A)$ and let $f \in H^1(0, T; X)$. Then the mild solution u of problem (129) is classical.*

The proof is similar to the one of Theorem 16. One has just to replace Lemma 6 with the following one.

Lemma 7 *For any $f \in H^1(0, T; X)$ let F_A be defined as in (135). Then $F_A \in C^1([0, T]; X) \cap C([0, T]; D(A))$ and*

$$F'_A(t) = AF_A(t) + f(t) = S(t)f(0) + \int_0^t S(t-s)f'(s)ds \quad (t \in [0, T]).$$

Proof. Since F_A can be rewritten as

$$F_A(t) = \int_0^t S(s)f(t-s)ds \quad (t \in [0, T]),$$

by differentiating the integral we conclude that

$$\begin{aligned} F'_A(t) &= S(t)f(0) + \int_0^t S(s)f'(t-s)ds \\ &= S(t)f(0) + \int_0^t S(t-s)f'(s)ds \quad \forall t \in [0, T]. \end{aligned}$$

Now, Proposition 24 of Appendix B implies that $F_A \in C^1([0, T]; X)$. Moreover, returning to definition (135), for all $t \in [0, T]$ we also have that

$$\begin{aligned} F'_A(t) &= \lim_{h \downarrow 0} \frac{1}{h} \left\{ \int_0^{t+h} S(t+h-s)f(s)ds - \int_0^t S(t-s)f(s)ds \right\} \\ &= \lim_{h \downarrow 0} \left\{ \frac{S(h) - I}{h} F_A(t) + \frac{1}{h} \int_t^{t+h} S(t+h-s)f(s)ds \right\}. \end{aligned}$$

Since $H^1(0, T; X) \subset C([0, T]; X)$, we have that

$$\lim_{h \downarrow 0} \frac{1}{h} \int_t^{t+h} S(t+h-s)f(s)ds = f(t).$$

The above identity implies that $F_A(t) \in D(A)$ and

$$AF_A(t) = F'_A(t) - f(t) \quad \forall t \in [0, T].$$

Consequently, $F_A \in C([0, T]; D(A))$ and the proof is complete. $\square$

Exercise 31 Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with boundary of class C^2 . Give conditions on $f \in L^2([0, T] \times \Omega)$, $u^0 : \Omega \rightarrow \mathbb{R}$, and $u^1 : \Omega \rightarrow \mathbb{R}$ which guarantee the existence and uniqueness of the classical solution to inhomogeneous wave equation

$$\begin{cases} \frac{\partial^2 u}{\partial t^2}(t, x) = \Delta u + f(t, x) & \text{in }]0, \infty[\times \Omega \\ u = 0 & \text{on }]0, \infty[\times \partial\Omega \\ u(0, x) = u^0(x), \quad \frac{\partial u}{\partial t}(0, x) = u^1(x) & x \in \Omega \end{cases} \quad (137)$$

Solution: let $\mathcal{A}$ be defined as in Example 17. Then, by applying Theorem 16 and Theorem 18 we conclude that the above problem has a unique classical solution if

1. $(u^0, u^1) \in D(\mathcal{A})$, that is, $u^0 \in H^2(\Omega) \cap H_0^1(\Omega)$ and $u^1 \in H_0^1(\Omega)$;
2. f satisfies any of the following conditions
 - $f \in C([0, T]; L^2(\Omega))$, $\frac{\partial f}{\partial x} \in L^2([0, T] \times \Omega)$, and $f(t, \cdot)|_{\partial\Omega} = 0$, or
 - $\frac{\partial f}{\partial t} \in L^2([0, T] \times \Omega)$. □

3.3 Maximal regularity for self-adjoint operators

We now introduce another notion of solution, which is stronger than mild solutions but better fits the presence of a forcing term $f \in L^2(0, T; X)$ than classical solutions. Hereafter, we set for brevity

$$W_T(D(A), X) = H^1(0, T; X) \cap L^2(0, T; D(A)) \quad (138)$$

Definition 23 (Strict solutions) A function $u \in W_T(D(A), X)$ is a *strict solution* of (129) if $u(0) = u^0$ and

$$u'(t) = Au(t) + f(t) \quad (t \in [0, T] \text{ a.e.})$$

We begin with a regularity result of interest in its own right.

Lemma 8 *Let $A : D(A) \subset X \rightarrow X$ be a self-adjoint dissipative operator and denote by $S(t)$ the C_0 -semigroup of bounded linear operators on X generated by A . Let $u \in W_T(D(A), X)$ be such that $u(0) = 0$. Then the function*

$$t \mapsto \langle Au(t), u(t) \rangle$$

is absolutely continuous on $[0, T]$ and

$$\frac{d}{dt} \langle Au(t), u(t) \rangle = 2\Re \langle u'(t), Au(t) \rangle \quad (\text{a.e. } t \in [0, T]). \quad (139)$$

Proof. Define $U_n(t) = \langle A_n u(t), u(t) \rangle$ ($t \in [0, T]$), where $A_n = nAR(n, A)$ is the Yosida approximation of A . Then U_n is absolutely continuous on $[0, T]$ and

$$\frac{d}{dt} \langle A_n u(t), u(t) \rangle = 2\Re \langle u'(t), A_n u(t) \rangle \quad (\text{a.e. } t \in [0, T])$$

or

$$\langle A_n u(t), u(t) \rangle = 2\Re \int_0^t \langle u'(s), A_n u(s) \rangle ds \quad \forall t \in [0, T]. \quad (140)$$

Now, since for a.e. $t \in [0, T]$

$$\begin{aligned} A_n u(t) &= nR(n, A)Au(t) \xrightarrow{(n \rightarrow \infty)} Au(t) \\ |A_n u(t)| &\leq |Au(t)|, \end{aligned}$$

we can pass to the limit as $n \rightarrow \infty$ in (140) by Lebesgue's theorem to obtain

$$\langle Au(t), u(t) \rangle = 2\Re \int_0^t \langle u'(s), Au(s) \rangle ds \quad (\text{a.e. } t \in [0, T]).$$

Since the right-hand side above is absolutely continuous on $[0, T]$, we have proved that the summable function $t \mapsto \langle Au(t), u(t) \rangle$ has an absolutely continuous representative and that (139) holds true. $\square$

We will give an existence and uniqueness result for strict solutions. As usual, we begin by studying the case of $u^0 = 0$.

Lemma 9 *Let $A : D(A) \subset X \rightarrow X$ be a densely defined self-adjoint dissipative operator. Let $f \in L^2(0, T; X)$ and define F_A as in (135). Then $F_A \in W_T(D(A), X)$ and*

$$F'_A(t) = AF_A(t) + f(t) \quad \text{a.e. } t \in [0, T]. \quad (141)$$

Moreover, $t \mapsto \langle AF_A(t), F_A(t) \rangle$ is absolutely continuous, its derivative is given by

$$\frac{d}{dt} \langle AF_A(t), F_A(t) \rangle = 2\Re \langle F'_A(t), AF_A(t) \rangle \quad \text{a.e. } t \in [0, T], \quad (142)$$

and

$$\int_0^T |AF_A(t)|^2 dt \leq \int_0^T |f(t)|^2 dt. \quad (143)$$

Proof. Assume, first, that $f \in C([0, T]; X)$ and define

$$f_n(t) = nR(n, A)f(t) \quad \text{and} \quad F_n(t) = nR(n, A)F_A(t) \quad \forall t \in [0, T].$$

Then $f_n \in C([0, T]; D(A))$ for every n and

$$F_n(t) = \int_0^t S(t-s)f_n(s) ds \quad (t \in [0, T]).$$

Owing to Lemma 6, we have that $F_n \in C^1([0, T]; X) \cap C([0, T]; D(A))$ and

$$F'_n(t) = AF_n(t) + f_n(t) \quad \forall t \in [0, T]. \quad (144)$$

Moreover,

$$2 \int_0^t \Re \langle F'_n(s), AF_n(s) \rangle ds = \langle AF_n(t), F_n(t) \rangle \leq 0 \quad \forall t \in [0, T]$$

because A is dissipative. So, by multiplying each member of (144) by $2AF_n(t)$, taking real parts, and integrating over $[0, T]$ we obtain

$$\begin{aligned} 2 \int_0^T |AF_n(t)|^2 dt &\leq -2 \int_0^T \Re \langle f_n(t), AF_n(t) \rangle dt \\ &\leq \int_0^T (|f_n(t)|^2 + |AF_n(t)|^2) dt. \end{aligned}$$

Hence

$$\int_0^T |AF_n(t)|^2 dt \leq \int_0^T |f_n(t)|^2 dt \leq \int_0^T |f(t)|^2 dt. \quad (145)$$

Thus, $\{F_n\}_n$ is bounded in $W_T(D(A), X)$. Therefore, there exists a subsequence $\{F_{n_k}\}_k$ and a function F_∞ such that

$$F_{n_k} \xrightarrow{(n \rightarrow \infty)} F_\infty \text{ in } W_T(D(A), X).$$

Recalling that $F_{n_k} \xrightarrow{(n \rightarrow \infty)} F_A$ in $C([0, T]; X)$ by Theorem 17, we conclude that $F_\infty = F_A \in W_T(D(A), X)$ and we derive (143) from (145) by the lower semicontinuity of the norm in $L^2(0, T; D(A))$.

The fact that $t \mapsto \langle AF_A(t), F_A(t) \rangle$ is absolutely continuous and satisfies (142) is guaranteed by Lemma 8.

Now, in order to recover (141), fix any $g \in L^2(0, T; X)$. Then, taking the scalar product of each member of (144) with g for $n = n_k$, we have that

$$\int_0^T \langle F'_{n_k}(t), g(t) \rangle dt = \int_0^T \langle AF_{n_k}(t) + f_{n_k}(t), g(t) \rangle dt.$$

So, in the limit as $k \rightarrow \infty$,

$$\int_0^T \langle F'_A(t) - AF_A(t) - f(t), g(t) \rangle dt = 0 \quad \forall g \in L^2(0, T; X)$$

which in turn yields $F'_A(t) = AF_A(t) + f(t)$ for a.e. $t \in [0, T]$.

Finally, we proceed to remove the extra assumption $f \in C([0, T]; X)$. Let $f \in L^2(0, T; X)$ and let $\{f_j\}_j \subset C([0, T]; X)$ be any sequence converging to f in $L^2(0, T; X)$ ³. Then

$$F_j(t) := \int_0^t S(t-s)f_j(s) ds \rightarrow F_A(t) \text{ in } C([0, T]; X).$$

³ See Lemma 12 in Appendix B.

Moreover, $F_j \in W_T(D(A), X)$ and F_j satisfies (141), (142), and (143). This implies that $\{F_j\}$ is a Cauchy sequence in $W_T(D(A), X)$ and therefore

$$F_A \in W_T(D(A), X) \quad \text{and} \quad F'_j \rightarrow F'_A, \quad AF_j \rightarrow AF_A \quad \text{in} \quad L^2(0, T; X).$$

So, F_A satisfies (141) and (143), while (142) follows once again from Lemma 8. $\square$

We are now ready to prove the main result of this section.

Theorem 19 *Let $A : D(A) \subset X \rightarrow X$ be a densely defined self-adjoint dissipative operator. Then, for any $u^0 \in [D(A), X]_{1/2}$ and $f \in L^2(0, T; X)$, the mild solution u of (129) is the unique strict solution of the problem. Moreover*

$$\left(\int_0^T |Au(t)|^2 dt \right)^{\frac{1}{2}} \leq \frac{1}{\sqrt{2}} |u^0|_{1/2} + \left(\int_0^T |f(t)|^2 dt \right)^{\frac{1}{2}} \quad (146)$$

Proof. Let u be the mild solution of problem (129). Then

$$u(t) = u_0(t) + F_A(t) \quad \forall t \in [0, T],$$

where

- (i) $u_0(t) := S(t)u^0$ belongs to $W_T(D(A), X)$ and satisfies $\frac{d}{dt} u_0(t) = Au_0(t)$ for every $t > 0$, thanks to Theorem 13, and the upper bound

$$\int_0^T |Au_0(t)|^2 dt \leq \frac{1}{2} |u^0|_{1/2}^2$$

in view of Theorem 14;

- (ii) F_A , defined in (135), belongs to $W_T(D(A), X)$ and, owing to Lemma 9, satisfies

$$F'_A(t) = AF_A(t) + f(t) \quad \text{a.e. } t \in [0, T]$$

and

$$\int_0^T |AF_A(t)|^2 dt \leq \int_0^T |f(t)|^2 dt.$$

By combining (i) and (ii) it follows that u is a strict solution of (129) and satisfies (146). The fact that u is the unique strict solution can be justified by observing that any strict solution of (129) is also a weak solution and appealing to Theorem 15. $\square$

Example 28 We can use Theorem 19 to study the problem

$$\begin{cases} \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + f(t, x) & (t, x) \in (0, T) \times (0, \pi) \text{ a.e.} \\ u(t, 0) = 0 = u(t, \pi) & t \in (0, T) \\ u(0, x) = u^0(x) & x \in (0, \pi). \end{cases} \quad (147)$$

Recalling Example 25, we conclude that for all

$$f \in L^2([0, T] \times [0, \pi]) \quad \text{and} \quad u^0 \in H_0^1(0, \pi)$$

problem (147) has a unique strict solution u . In particular, such a solution satisfies:

$$\frac{\partial u}{\partial t}, \frac{\partial^2 u}{\partial x^2} \in L^2([0, T] \times [0, \pi])$$

4 Additive control of evolution equations

In this section, we study the controllability of evolution equations by the action of an additive control. More general results can be found in [13], [21], and [28], where several aspects of control theory—including optimal control—are treated.

We assume throughout that $(X, \langle \cdot, \cdot \rangle, |\cdot|)$ is a separable Hilbert space over $\mathbb{R}$ (the *state space*) and $(F, (\cdot, \cdot), \|\cdot\|)$ is another separable real Hilbert space (the *control space*). We are interested in the control system

$$\begin{cases} u'(t) = Au(t) + Bf(t) & (t \in [0, T]) \\ u(0) = u^0 \in X \end{cases} \quad (148)$$

where:

- $0 < T < \infty$ is the *horizon*,
- $A : D(A) \subset X \rightarrow X$ is the infinitesimal generator of a C_0 -semigroup on X , $S(t)$,
- $B : F \rightarrow X$ is a bounded linear operator, called the *control operator*, and
- $f \in L^2(0, T; F)$ is a *control strategy*.

For any $u^0 \in X$ and $f \in L^2(0, T; F)$ we denote by $u^f(t) = u^f(t, u^0)$ the mild solution of the above problem, which is given by

$$u^f(t, u^0) = S(t)u_0 + \Lambda_t f \quad (t \in [0, T])$$

where $\Lambda_t : L^2(0, t; F) \rightarrow X$ stands for the control operator

$$\Lambda_t f = \int_0^t S(t-s)Bf(s) ds \quad (t \in [0, T])$$

Remark 11 We recall that, in view of Theorem 19, if A is a self-adjoint dissipative operator and $u^0 \in [D(A), X]_{1/2}$ then u^f is a strict solution of (148) and satisfies

$$\left(\int_0^T |Au^f(t)|^2 dt \right)^{\frac{1}{2}} \leq \frac{1}{\sqrt{2}} \|u^0\|_{1/2} + \|B\| \left(\int_0^T \|f(t)\|^2 dt \right)^{\frac{1}{2}}$$

Exercise 32 Compute $\Lambda_T^* : X \rightarrow L^2(0, T; F)$.

Proof. Let $v \in X$. We have that, for every $g \in L^2(0, T; F)$,

$$\begin{aligned}
\int_0^T \left((\Lambda_T^* v)(s), g(s) \right) ds &= \langle v, \Lambda_T g \rangle \\
&= \int_0^T \langle v, S(T-s)Bg(s) \rangle ds = \int_0^T \langle B^*S(T-s)^*v, g(s) \rangle ds
\end{aligned}$$

So,

$$(\Lambda_T^* v)(s) = B^*S(T-s)^*v \quad \forall s \in [0, T].$$

Equivalently, $(\Lambda_T^* v)(s) = B^*w(s)$ where w is the mild solution of the adjoint equation $w'(s) + A^*w(s) = 0$ such that $w(T) = v$. $\square$

Definition 24 The *reachable* (or, *attainable*) set from $u^0 \in X$ in time T is defined as

$$\mathcal{R}_T(u^0) = \left\{ u^f(T, u^0) \mid f \in L^2(0, T; F) \right\} \quad (149)$$

We now define two different notions of controllability.

Definition 25 System (148) is said to be:

1. *approximately controllable* in time T if $\overline{\mathcal{R}_T(u^0)} = X$ for all $u^0 \in X$;
2. *null controllable* in time T if there exists a constant $c_T > 0$ such that, for all $u^0 \in X$ and some $f \in L^2(0, T; F)$, we have that

$$u^f(T, u^0) = 0 \quad \text{and} \quad \left(\int_0^T \|f(t)\|^2 dt \right)^{\frac{1}{2}} \leq c_T |u^0|. \quad (150)$$

When (148) is null controllable in time T , one can define:

- the *control cost* in time T of any initial datum $u^0 \in X \setminus \{0\}$ as

$$c_T(u^0) = \inf \left\{ \frac{\|f\|_2}{|u^0|} \mid u^f(T, u^0) = 0 \right\} \quad (151)$$

- the *global control cost* in time T as

$$C(T) = \sup \{ c_T(u^0) \mid u^0 \in X \setminus \{0\} \} \quad (152)$$

We shall derive dual characterizations of the above controllability properties. We begin with approximate controllability.

Theorem 20 *The following properties are equivalent:*

- (a) system (148) is *approximately controllable* in time T ;
- (b) for any $v \in X$

$$B^*S(t)^*v = 0 \quad \forall t \in [0, T] \implies v = 0. \quad (153)$$

Proof. To begin with, we note that (148) is approximately controllable in time T if and only if for all $u^0, v^0 \in X$ and all $\varepsilon > 0$ there exists $f_\varepsilon \in L^2(0, T; F)$ such that

$$\left| S(T)u^0 + \Lambda_T f_\varepsilon - v^0 \right| = \left| \Lambda_T f_\varepsilon - (v^0 - S(T)u^0) \right| < \varepsilon$$

Let us set $\mathcal{I}(\Lambda_T) = \Lambda_T(L^2(0, T; F))$. Then we have that

$$(a) \iff \overline{\mathcal{I}(\Lambda_T)} = X \iff \mathcal{I}(\Lambda_T)^\perp = \{0\}.$$

On the other hand,

$$\begin{aligned} v \in \mathcal{I}(\Lambda_T)^\perp &\iff \int_0^T \langle f(s), B^* S(T-s)^* v \rangle ds = 0 \quad \forall f \in L^2(0, T; F) \\ &\iff B^* S(t)^* v = 0 \quad \forall t \in [0, T]. \end{aligned}$$

Therefore (a) and (b) are equivalent properties and the proof is complete. $\square$

Example 29 (Approximate controllability of the wave equation by a single input) Let us study the approximate controllability in time $T \in]0, \infty[$ of the control system

$$\begin{cases} \frac{\partial^2 u}{\partial t^2}(t, x) = \frac{\partial^2 u}{\partial x^2}(t, x) + f(t) \mu(x) & \text{in }]0, T[\times]0, \pi[\\ u(t, 0) = 0 = u(t, \pi) & t \in]0, T[\\ u(0, x) = u^0(x), \quad \frac{\partial u}{\partial t}(0, x) = v^0(x) & x \in]0, \pi[\end{cases} \quad (154)$$

where

- $f \in L^2(0, T)$ is the control strategy with $\mathbb{R}$ as the control space,
- $\mu \in L^\infty(0, \pi)$ is a function to be chosen later,
- $(u^0, v^0) \in H_0^1(0, \pi) \times L^2(0, \pi)$.

As usual, we recast (154) as in Example 17, that is,

$$\begin{pmatrix} u'(t) \\ v'(t) \end{pmatrix} = \mathcal{A} \begin{pmatrix} u(t) \\ v(t) \end{pmatrix} + B f(t) \quad (t \in [0, T] \text{ a.e.})$$

where

$$\begin{cases} D(\mathcal{A}) = (H^2(0, \pi) \cap H_0^1(0, \pi)) \times H_0^1(0, \pi) \\ \mathcal{A} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ A & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} v \\ Au \end{pmatrix}, \end{cases} \quad (155)$$

$A : D(A) \subset X \rightarrow X$ being defined in (47), and $B : \mathbb{R} \rightarrow H_0^1(0, \pi) \times L^2(0, \pi)$ is the multiplication operator

$$(Br)(x) = \begin{pmatrix} 0 \\ r\mu(x) \end{pmatrix} \quad (\forall r \in \mathbb{R}, x \in [0, \pi] \text{ a.e.})$$

In order to check condition (153), it is convenient to derive the Fourier representation of the wave group. For this purpose, we rewrite the homogeneous problem

$$\begin{pmatrix} u'(t) \\ v'(t) \end{pmatrix} = \mathcal{A} \begin{pmatrix} u(t) \\ v(t) \end{pmatrix} \quad (t \in \mathbb{R}) \quad \begin{pmatrix} u(0) \\ v(0) \end{pmatrix} = \begin{pmatrix} u^0 \\ v^0 \end{pmatrix}$$

as the following second order abstract differential equation

$$\begin{cases} u''(t) = Au(t) & (t \in \mathbb{R}) \\ u(0) = u^0, \quad u'(0) = v^0 \end{cases} \quad (156)$$

As we know from Exercise 13, the eigenvalues and eigenvectors of A are given by

$$\lambda_n = -n^2 \quad \phi_n(x) = \sqrt{\frac{2}{\pi}} \sin(nx) \quad (n \geq 1, x \in [0, \pi]) \quad (157)$$

Hence, the solution of (156) can be represented as

$$u(t, x) = \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} \left(u_n^0 \cos(nt) + \frac{v_n^0}{n} \sin(nt) \right) \sin(nx) \quad (t \in \mathbb{R}, x \in [0, \pi]) \quad (158)$$

where

$$u_n^0 = \sqrt{\frac{2}{\pi}} \int_0^\pi u^0(x) \sin(nx) dx, \quad v_n^0 = \sqrt{\frac{2}{\pi}} \int_0^\pi v^0(x) \sin(nx) dx.$$

In sum, the wave group can be represented as follows:

$$S(t) \begin{pmatrix} u^0 \\ v^0 \end{pmatrix} (x) = \begin{pmatrix} \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} \left(u_n^0 \cos(nt) + \frac{v_n^0}{n} \sin(nt) \right) \sin(nx) \\ \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} \left(v_n^0 \cos(nt) - n u_n^0 \sin(nt) \right) \sin(nx) \end{pmatrix} \quad (159)$$

Let us now compute $B^* : H_0^1(0, \pi) \times L^2(0, \pi) \rightarrow \mathbb{R}$: for all $r \in \mathbb{R}$ and all $(\varphi, \psi) \in H_0^1(0, \pi) \times L^2(0, \pi)$ we have that

$$\langle Br, \begin{pmatrix} \varphi \\ \psi \end{pmatrix} \rangle = r \int_0^\pi \mu(x) \psi(x) dx = r B^* \begin{pmatrix} \varphi \\ \psi \end{pmatrix}$$

So,

$$B^* \begin{pmatrix} \varphi \\ \psi \end{pmatrix} = \int_0^\pi \mu(x) \psi(x) dx \quad \forall (\varphi, \psi) \in H_0^1(0, \pi) \times L^2(0, \pi). \quad (160)$$

Since the wave group is unitary (see Exercise 29), by (159) and (160) we conclude that, for all $(u^0, v^0) \in H_0^1(0, \pi) \times L^2(0, \pi)$,

$$B^* S(t)^* \begin{pmatrix} u^0 \\ v^0 \end{pmatrix} = B^* S(-t) \begin{pmatrix} u^0 \\ v^0 \end{pmatrix} = \sum_{n=1}^{\infty} \mu_n \left(v_n^0 \cos(nt) + n u_n^0 \sin(nt) \right) \quad (161)$$

where

$$\mu_n = \sqrt{\frac{2}{\pi}} \int_0^\pi \mu(x) \sin(nx) dx.$$

Now, let $T \geq 2\pi$ and suppose

$$B^* S(t)^* \begin{pmatrix} u_0^0 \\ v_0^0 \end{pmatrix} = 0 \quad \forall t \in [0, T].$$

Then, from (161) we derive that, for any integer $k \geq 1$,

$$\begin{aligned} \int_0^{2\pi} B^* S(t)^* \begin{pmatrix} u_0^0 \\ v_0^0 \end{pmatrix} \cos(kt) dt &= \pi \mu_k v_k^0 = 0, \\ \int_0^{2\pi} B^* S(t)^* \begin{pmatrix} u_0^0 \\ v_0^0 \end{pmatrix} \sin(kt) dt &= \pi \mu_k k u_k^0 = 0. \end{aligned}$$

Therefore, if

$$\mu \in L^\infty(0, \pi) \quad \text{is such that} \quad \mu_n \neq 0 \quad \forall n \geq 1 \quad (162)$$

then $u_n^0 = 0 = v_n^0$ for all $n \geq 1$.

CONCLUSION: If μ satisfies (162), then system (154) is approximately controllable in any time $T \geq 2\pi$.

Exercise 33 Prove that (158) gives the solution of problem (156).

Proof. In view of (157) the solution of problem (156) can be represented as

$$u(t, x) = \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} \alpha_n(t) \sin(nx) \quad (t \in \mathbb{R}, x \in [0, \pi])$$

where

$$\alpha_n(t) = \sqrt{\frac{2}{\pi}} \int_0^\pi u(t, x) \sin(nx) dx \quad (n \geq 1, t \in \mathbb{R})$$

Then, on account of (156) we have that

$$\begin{cases} \alpha_n''(t) + n^2 \alpha_n(t) = 0 & (t \in \mathbb{R}) \\ \alpha_n(0) = u_n^0 & \alpha_n'(0) = v_n^0. \end{cases}$$

Since the solution of the above Cauchy problem is

$$\alpha_n(t) = u_n^0 \cos(nt) + \frac{v_n^0}{n} \sin(nt) \quad (n \geq 1, t \in \mathbb{R})$$

we have proved that u is given by formula (158). $\square$

Exercise 34 Prove that $\mu(x) = x^k$ ($x \in [0, \pi]$) satisfies (162) for all $k \geq 1$.

Proof. For any $k \geq 1$ we have that

$$\begin{aligned}
\int_0^\pi x^k \sin(nx) dx &= \left[-\frac{x^k}{n} \cos(nx) \right]_{x=0}^{x=\pi} + \frac{k}{n} \int_0^\pi x^{k-1} \cos(nx) dx \\
&= \frac{(-1)^{n+1}}{n} \pi^k + \frac{k}{n} \int_0^\pi x^{k-1} \cos(nx) dx.
\end{aligned}$$

Moreover,

$$\left| \frac{k}{n} \int_0^\pi x^{k-1} \cos(nx) dx \right| \leq \frac{k}{n} \int_0^\pi x^{k-1} |\cos(nx)| dx < \frac{\pi^k}{n}.$$

Therefore,

$$\left| \int_0^\pi x^k \sin(nx) dx \right| \geq \frac{\pi^k}{n} - \left| \frac{k}{n} \int_0^\pi x^{k-1} \cos(nx) dx \right| > \frac{\pi^k}{n} - \frac{\pi^k}{n} = 0.$$

so that $\mu_n \neq 0$ for all $n \geq 1$. $\square$

We now turn to null controllability.

Theorem 21 *The following properties are equivalent:*

- (a) *system (148) is null controllable in time T ;*
- (b) *there exists a constant $c_T > 0$ such that*

$$|S(T)^* v| \leq c_T \left(\int_0^T \|B^* S(t)^* v\|^2 dt \right)^{\frac{1}{2}} \quad \forall v \in X. \quad (163)$$

Proof. $\boxed{(a) \Rightarrow (b)}$ Fix any $v \in X$ and let $u^0 = S(T)^* v$. If $f \in L^2(0, T; F)$ is such that (150) holds, then $\langle S(T)u^0, v \rangle = -\langle \Lambda_T f, v \rangle$. Therefore,

$$\begin{aligned}
|S(T)^* v|^2 &= - \int_0^T (f(s), B^* S(T-s)^* v) ds \\
&\leq \left(\int_0^T \|f(s)\|^2 ds \right)^{\frac{1}{2}} \left(\int_0^T \|B^* S(T-s)^* v\|^2 ds \right)^{\frac{1}{2}} \\
&\leq c_T |S(T)^* v| \left(\int_0^T \|B^* S(T-s)^* v\|^2 ds \right)^{\frac{1}{2}},
\end{aligned}$$

where c_T is a constant in (150). Estimate (163) follows.

$\boxed{(b) \Rightarrow (a)}$ For any fixed $u^0 \in X$ let us consider the optimization problem

$$\min \left\{ J_\varepsilon(f) : f \in L^2(0, T; F) \right\} \quad (164)$$

where $\varepsilon > 0$ and

$$J_\varepsilon(f) = \frac{1}{2} \int_0^T \|f(t)\|^2 dt + \frac{1}{2\varepsilon} |u^f(T, u^0)|^2. \quad (165)$$

Since $J_\varepsilon(f) \rightarrow \infty$ as $\|f\|_2 \rightarrow \infty$, the existence of the above minimum can be proved by standard weak compactness arguments (see Exercise 35 below).

Let f_ε be a minimizer for J_ε and set $u_\varepsilon(t) = u_{f_\varepsilon}(t, u^0)$.

Step 1: if c_T is the constant in (163), then we claim that

$$\frac{1}{\varepsilon} |u_\varepsilon(T)|^2 + \frac{1}{2} \int_0^T \|f_\varepsilon\|^2 dt \leq \frac{c_T^2}{2} |u^0|^2 \quad \forall \varepsilon > 0 \quad (166)$$

Indeed by Fermat's rule we have that, for every $g \in L^2(0, T; F)$,

$$0 = J'_\varepsilon(f_\varepsilon)g = \int_0^T (f_\varepsilon(t), g(t)) dt + \frac{1}{\varepsilon} \langle u_\varepsilon(T), \Lambda_T g \rangle.$$

Now, the above equation can be rewritten as follows

$$\int_0^T \left(f_\varepsilon(s) + \left(\Lambda_T^* \frac{u_\varepsilon(T)}{\varepsilon} \right)(s), g(s) \right) ds = 0 \quad \forall g \in L^2(0, T; F)$$

Hence

$$f_\varepsilon = -\Lambda_T^* \frac{u_\varepsilon(T)}{\varepsilon}.$$

Therefore, in view of Exercise 32 we have that

$$f_\varepsilon(s) = -B^* S(T-s)^* \frac{u_\varepsilon(T)}{\varepsilon} =: -B^* v_\varepsilon(s) \quad \forall s \in [0, T].$$

On account of the previous identities we obtain

$$\begin{aligned} \frac{1}{\varepsilon} |u_\varepsilon(T)|^2 - \langle u^0, v_\varepsilon(0) \rangle &= \left\langle u_\varepsilon(T), \frac{u_\varepsilon(T)}{\varepsilon} \right\rangle - \left\langle u^0, S(T)^* \frac{u_\varepsilon(T)}{\varepsilon} \right\rangle \\ &= \left\langle \Lambda_T f_\varepsilon, \frac{u_\varepsilon(T)}{\varepsilon} \right\rangle = - \int_0^T \|B^* v_\varepsilon(s)\|^2 ds \end{aligned}$$

or

$$\frac{1}{\varepsilon} |u_\varepsilon(T)|^2 + \int_0^T \|B^* v_\varepsilon(s)\|^2 ds = \langle u^0, v_\varepsilon(0) \rangle.$$

Thus, by appealing to (163) to bound $|v_\varepsilon(0)|$, we conclude that

$$\begin{aligned} \frac{1}{\varepsilon} |u_\varepsilon(T)|^2 + \int_0^T \|B^* v_\varepsilon(s)\|^2 ds &= \langle u^0, v_\varepsilon(0) \rangle \\ &\leq c_T |u^0| \left(\int_0^T \|B^* v_\varepsilon(s)\|^2 ds \right)^{\frac{1}{2}}. \end{aligned}$$

Our claim (166) follows by applying the binomial inequality $2ab \leq a^2 + b^2$.

Step 2: conclusion.

From (166) it follows that there exists a sequence $\varepsilon_n \downarrow 0$ such that $f_{\varepsilon_n} \rightharpoonup f$ in $L^2(0, T; F)$, Then

$$u_{\varepsilon_n}(t) = S(t)u_0 + \Lambda_t f_{\varepsilon_n} \xrightarrow{n \rightarrow \infty} S(t)u_0 + \Lambda_t f \quad \forall t \in [0, T]$$

Consequently, $u^f(T, u^0) = 0$ and $\|f\|_2 \leq c_T |u^0|$. $\square$

Exercise 35 Prove the existence of the minimum in (164).

Proof. Fix $\varepsilon > 0$ and let $\{f_n\} \subset L^2(0, T; F)$ be a minimizing sequence:

$$J_\varepsilon(f_n) \longrightarrow \inf \{J_\varepsilon(f) : f \in L^2(0, T; F)\}.$$

Since

$$J_\varepsilon(f) \geq \frac{1}{2} \|f\|_{L^2(0, T; F)}^2,$$

the sequence $\{f_n\}$ is bounded in $L^2(0, T; F)$. Therefore, passing to a subsequence if necessary, there exists $f_\infty \in L^2(0, T; F)$ such that

$$f_n \rightharpoonup f_\infty \quad \text{in } L^2(0, T; F).$$

For every $v \in X$ and $t \in [0, T]$, we have

$$\begin{aligned} \langle \Lambda_t f_n, v \rangle &= \int_0^t \langle S(t-s) B f_n(s), v \rangle ds \\ &= \int_0^t (f_n(s), B^* S(t-s)^* v) ds \\ &\longrightarrow \int_0^t (f_\infty(s), B^* S(t-s)^* v) ds \\ &= \langle \Lambda_t f_\infty, v \rangle. \end{aligned}$$

Thus,

$$\Lambda_t f_n \rightharpoonup \Lambda_t f_\infty \quad \text{in } X \quad \forall t \in [0, T],$$

and consequently

$$u^{f_n}(t, u^0) \rightharpoonup u^{f_\infty}(t, u^0) \quad \text{in } X \quad \forall t \in [0, T].$$

By the weak lower semicontinuity of the norms in $L^2(0, T; F)$ and X ,

$$J_\varepsilon(f_\infty) \leq \liminf_{n \rightarrow \infty} J_\varepsilon(f_n) = \inf \{J_\varepsilon(f) : f \in L^2(0, T; F)\}.$$

Hence f_∞ is a minimizer of J_ε . $\square$

Inequality (163) is often referred to as an *observability inequality*, the term on the right-hand side being the *observation*.

? Question

How to compare approximate and null controllability? Is any of these notions stronger than the other one?

By combining (153) and (163) we have that

$$B^* S(t)^* v = 0 \quad \forall t \in [0, T] \implies S(T)^* v = 0.$$

If $S(t)$ is a group, then so is $S(t)^*$. So, $S(T)^* v = 0 \implies v = 0$. This implies that, if A is the infinitesimal generator of a group and the control system (148) is null controllable in time T , then the system is also approximately controllable in time T .

The property that

$$S(T)v = 0 \implies v = 0 \quad (v \in X) \quad (167)$$

is called *backward uniqueness*. Besides groups, such a property is also true for the contraction semigroup generated by a self-adjoint dissipative operator.

Theorem 22 *Let $A : D(A) \subset X \rightarrow X$ be a densely defined self-adjoint dissipative operator and let $S(t)$ be the contraction semigroup generated by A . Then, (167) holds for any $T > 0$ and $v \in X$.*

Proof. Suppose $S(T)v = 0$ for some $T > 0$ and $v \in D(A)$. Define, for any $\lambda > 0$,

$$u_\lambda(t) = e^{\frac{\lambda}{2}t^2} S(t)v \quad (t \in [0, T]).$$

Then $u_\lambda(t) \in D(A)$ for all $t > 0$, $u_\lambda(T) = 0$, and

$$u'_\lambda(t) = Au_\lambda(t) + \lambda t u_\lambda(t) \quad \forall t \in]0, T].$$

Now, taking the scalar product of both sides of the above identity with $u'_\lambda(t)$ we obtain

$$\begin{aligned} |u'_\lambda(t)|^2 &= \langle Au_\lambda(t), u'_\lambda(t) \rangle + \lambda \langle u_\lambda(t), u'_\lambda(t) \rangle \\ &= \frac{1}{2} \frac{d}{dt} \langle Au_\lambda(t), u_\lambda(t) \rangle + \frac{\lambda t}{2} \frac{d}{dt} |u_\lambda(t)|^2 \end{aligned}$$

Therefore, integrating over $[0, T]$, since $u_\lambda(T) = 0$ and $u_\lambda(0) = v$ we get

$$\int_0^T (|u'_\lambda(t)|^2 + \frac{\lambda}{2} |u_\lambda(t)|^2) dt = -\frac{1}{2} \langle Av, v \rangle.$$

Since $|S(t)v| \leq |u_\lambda(t)|$ for all $t \geq 0$, from the above equation it follows that

$$\int_0^T |S(t)v|^2 dt \leq -\frac{1}{\lambda} \langle Av, v \rangle.$$

As $\lambda \rightarrow \infty$, we conclude that $S(t)v = 0$ for all $t \geq 0$. Hence, $v = 0$.

Finally, let us remove the extra assumption $v \in D(A)$. Let $v \in X$ and let $0 < \tau < T$. Then $S(\tau) \in D(A)$ by Theorem 13 and $S(T - \tau)S(\tau)v = S(T)v = 0$. So, $S(\tau)v = 0$ by what we have just proved and the strong continuity of $S(t)$ yields $v = 0$. $\square$

Example 30 (Heat equation with distributed control) Let us consider the heat equation with distributed control $f \in L^2([0, T] \times [0, \pi])$

$$\begin{cases} \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + f(t, x) & (t, x) \in (0, T) \times (0, \pi) \text{ a.e.} \\ u(t, 0) = 0 = u(t, \pi) & t \in (0, T) \\ u(0, x) = u^0(x) & x \in (0, \pi) \end{cases} \quad (168)$$

where $u^0 \in L^2(0, \pi)$. Problem (168) can be recast in the abstract form (148) in the Hilbert space $X = L^2(0, \pi)$ with A defined as in (47) and $B = I$. Therefore, in order to prove system (168) to be null controllable, for any $v \in L^2(0, \pi)$ we can check the validity of the observability inequality

$$\int_0^\pi |w(T, x)|^2 dx \leq c_T^2 \int_0^T dt \int_0^\pi |w(t, x)|^2 dx \quad (169)$$

where w is the solution of the adjoint problem

$$\begin{cases} \frac{\partial w}{\partial t} = \frac{\partial^2 w}{\partial x^2} & (t, x) \in (0, T) \times (0, \pi) \text{ a.e.} \\ w(t, 0) = 0 = w(t, \pi) & t \in (0, T) \\ w(0, x) = v(x) & x \in (0, \pi) \end{cases} \quad (170)$$

By taking the Fourier series of w like in Example 11 we have that

$$w(t, x) = \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} e^{-n^2 t} v_n \sin(nx) \quad (x \in [0, \pi])$$

with

$$v_n = \sqrt{\frac{2}{\pi}} \int_0^\pi v(x) \sin(nx) dx.$$

Therefore,

$$\int_0^\pi |w(T, x)|^2 dx = \sum_{n=1}^{\infty} e^{-2n^2 T} v_n^2 \quad (171)$$

while

$$\int_0^T dt \int_0^\pi |w(t, x)|^2 dx = \sum_{n=1}^{\infty} v_n^2 \int_0^T e^{-2n^2 t} dt = \sum_{n=1}^{\infty} \frac{1 - e^{-2n^2 T}}{2n^2} v_n^2 \quad (172)$$

Now, from the elementary inequality $e^x \geq 1 + x$, which is true for all $x \in \mathbb{R}$, it follows that $e^{2n^2 T} \geq 1 + 2n^2 T$, or

$$\frac{1 - e^{-2n^2 T}}{2n^2} \geq T e^{-2n^2 T} \quad (n \in \mathbb{N})$$

which in turn allows to deduce (169) from (171) and (172) with $c_T = 1/\sqrt{T}$.

CONCLUSION: System (168) is null controllable in any time $T > 0$.

Example 31 (Heat equation with boundary control) Let us consider the heat equation

$$\begin{cases} \frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} & (t, x) \in (0, T) \times (0, \pi), \\ u(t, 0) = 0, \quad u(t, \pi) = f(t) & t \in (0, T), \\ u(0, x) = u^0(x) & x \in (0, \pi), \end{cases} \quad (173)$$

where $u^0 \in L^2(0, \pi)$ and the boundary control satisfies

$$f \in H^1(0, T), \quad f(0) = f(T) = 0.$$

We will prove approximate controllability in $L^2(0, \pi)$ in any time $T > 0$, even with these endpoint restrictions.

Set

$$v(t, x) = u(t, x) - \frac{x}{\pi} f(t), \quad g(t) = f'(t).$$

Then v satisfies

$$\begin{cases} \frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial x^2} - \frac{x}{\pi} g(t) & (t, x) \in (0, T) \times (0, \pi), \\ v(t, 0) = 0 = v(t, \pi) & t \in (0, T), \\ v(0, x) = u^0(x) & x \in (0, \pi), \end{cases} \quad (174)$$

where the admissible controls belong to the closed subspace

$$G_T = \left\{ g \in L^2(0, T) : \int_0^T g(s) ds = 0 \right\}.$$

Conversely, every $g \in G_T$ defines an admissible boundary control by

$$f(t) = \int_0^t g(s) ds.$$

In particular, $v(0) = u^0$ and $u(T) = v(T)$.

Problem (174) has the abstract form (148) in $X = L^2(0, \pi)$, with A defined in (47) and

$$B : \mathbb{R} \longrightarrow X, \quad (Br)(x) = -\frac{r}{\pi} x.$$

We understand v as the mild solution of this problem and define u by the above lifting. For arbitrary $u^0 \in X$, the differential equations and boundary traces hold for positive times in the corresponding local Sobolev sense.

The final state of the lifted system is

$$v(T) = S(T)u^0 + \Lambda_T g, \quad \Lambda_T g = \int_0^T S(T-s)Bg(s) ds.$$

Thus it suffices to prove that $\Lambda_T(G_T)$ is dense in X .

Following the duality argument in the proof of Theorem 20, let $w^0 \in X$ be orthogonal to $\Lambda_T(G_T)$. Then

$$0 = \langle \Lambda_T g, w^0 \rangle = \int_0^T g(s) B^* S(T-s)^* w^0 ds \quad \forall g \in G_T.$$

The orthogonal complement of G_T in $L^2(0, T)$ consists of the constant functions. Since A is self-adjoint, $S(t)^* = S(t)$. Consequently, setting $w(t) = S(t)w^0$, we obtain the following criterion:

$$B^* w(t) \text{ is constant on } [0, T] \implies w^0 = 0. \quad (175)$$

Here constancy holds everywhere by continuity, and w solves

$$\begin{cases} \frac{\partial w}{\partial t} = \frac{\partial^2 w}{\partial x^2} & (t, x) \in (0, \infty) \times (0, \pi), \\ w(t, 0) = 0 = w(t, \pi) & t > 0, \\ w(0) = w^0 \in L^2(0, \pi). \end{cases}$$

For every $\phi \in L^2(0, \pi)$, a direct computation gives

$$B^* \phi = -\frac{1}{\pi} \int_0^\pi x \phi(x) dx.$$

Suppose now that $B^* w(t)$ is constant on $[0, T]$. By the regularity of the heat semigroup, for $0 < t < T$ we may differentiate and integrate by parts to obtain

$$\begin{aligned} 0 &= \frac{d}{dt} \int_0^\pi x w(t, x) dx = \int_0^\pi x w_{xx}(t, x) dx \\ &= [x w_x(t, x)]_{x=0}^{x=\pi} - \int_0^\pi w_x(t, x) dx = \pi w_x(t, \pi). \end{aligned}$$

Moreover, $t \mapsto w_x(t, \pi)$ is analytic for $t > 0$, as follows, for instance, from the Fourier representation below. Hence

$$w_x(t, \pi) = 0 \quad \forall t > 0. \quad (176)$$

We claim that (176) forces $w^0 = 0$. Indeed,

$$w(t, x) = \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} e^{-n^2 t} w_n^0 \sin(nx), \quad w_n^0 = \sqrt{\frac{2}{\pi}} \int_0^\pi w^0(x) \sin(nx) dx.$$

For positive times, differentiation of the series yields

$$w_x(t, \pi) = \sqrt{\frac{2}{\pi}} \sum_{n=1}^{\infty} (-1)^n n e^{-n^2 t} w_n^0 = 0 \quad \forall t > 0. \quad (177)$$

If $w^0 \neq 0$, let

$$n_0 = \min\{n \geq 1 : w_n^0 \neq 0\}.$$

Multiplying (177) by $\sqrt{\pi/2} e^{n_0^2 t}$ gives

$$(-1)^{n_0} n_0 w_{n_0}^0 = - \sum_{n=n_0+1}^{\infty} (-1)^n n e^{-(n^2-n_0^2)t} w_n^0.$$

The right-hand side tends to zero as $t \rightarrow \infty$. Indeed, its absolute value is bounded by

$$\left(\sum_{n>n_0} |w_n^0|^2 \right)^{1/2} \left(\sum_{n>n_0} n^2 e^{-2(n^2-n_0^2)t} \right)^{1/2},$$

whose second factor tends to zero by dominated convergence for $t \geq 1$. Thus $w_{n_0}^0 = 0$, a contradiction.

This proves (175), and therefore $\Lambda_T(G_T)$ is dense in X . Since $u(T) = v(T)$, the same density conclusion holds for the original boundary-controlled system.

CONCLUSION: System (173) is approximately controllable in $L^2(0, \pi)$ in every time $T > 0$, using controls $f \in H^1(0, T)$ satisfying $f(0) = f(T) = 0$.

Finally, we address exact controllability.

Definition 26 System (148) is said to be: *exactly controllable* in time T if there exists a constant $k_T > 0$ such that, for all $u^0, v^0 \in X$ and some $f \in L^2(0, T; F)$, we have that

$$u^f(T, u^0) = v^0 \quad \text{and} \quad \left(\int_0^T \|f(t)\|^2 dt \right)^{\frac{1}{2}} \leq k_T (|u^0| + |v^0|). \quad (178)$$

As we did for approximate and null controllability, we can characterize exact controllability as follows.

Theorem 23 *The following properties are equivalent:*

- (a) *system (148) is exactly controllable in time T ;*
- (b) *there exists a constant $k_T > 0$ such that*

$$|v| \leq k_T \left(\int_0^T \|B^* S(t)^* v\|^2 dt \right)^{\frac{1}{2}} \quad \forall v \in X \quad (179)$$

- (c) *the bounded linear operator*

$$Q_T = \int_0^T S(t) B B^* S(t)^* dt \quad (180)$$

is invertible⁴.

Moreover, if (148) is exactly controllable in time T , then the control strategy

$$f^0(s) = B^* S(T-s)^* Q_T^{-1} (v^0 - S(T)u^0) \quad (s \in [0, T]) \quad (181)$$

satisfies the following:

- (i) $u_{f_0}(T, u^0) = v^0$
- (ii) $\int_0^T \|f^0(s)\|^2 ds = \langle Q_T^{-1} (v^0 - S(T)u^0), v^0 - S(T)u^0 \rangle$
- (iii) $\int_0^T \|f^0(s)\|^2 ds = \min \left\{ \int_0^T \|f(s)\|^2 ds \mid u^f(T, u^0) = v^0 \right\}.$

Notice that (179) is a stronger observability inequality than (163).

Proof. (a) $\Rightarrow$ (b) First, we note that (148) is exactly controllable in time T if and only if $\mathcal{I}(\Lambda_T) := \Lambda_T(L^2(0, T; F)) = X$. Fix any $v \in X$ and let $f \in L^2(0, T; F)$ be such that

$$u^f(T, 0) = \Lambda_T f = v \quad \text{and} \quad \left(\int_0^T \|f(t)\|^2 dt \right)^{\frac{1}{2}} \leq k_T |v|.$$

Then

$$\begin{aligned} |v|^2 &= \langle \Lambda_T f, v \rangle \\ &= \int_0^T \langle S(T-s) B f(s), v \rangle ds = \int_0^T \langle f(s), B^* S(T-s)^* v \rangle ds \\ &\leq \left(\int_0^T \|f(s)\|^2 ds \right)^{\frac{1}{2}} \left(\int_0^T \|B^* S(t)^* v\|^2 dt \right)^{\frac{1}{2}} \\ &\leq k_T |v| \left(\int_0^T \|B^* S(t)^* v\|^2 dt \right)^{\frac{1}{2}}, \end{aligned}$$

which in turn yields (179).

(b) $\Rightarrow$ (c) We observe that, by definition, $Q_T \in \mathcal{L}(X)$ is self-adjoint and positive semi-definite. Moreover, (179) implies that Q_T is positive definite, hence injective, because

$$\langle Q_T u, u \rangle = \int_0^T \|B^* S(t)^* u\|^2 dt \geq \frac{1}{k_T^2} |u|^2 \quad \forall u \in X. \quad (182)$$

Let us show that Q_T is onto. Fix any $v \in X$ and define the functional $J : X \rightarrow \mathbb{R}$ as

⁴ Hence, owing to the open mapping theorem, $Q_T^{-1} \in \mathcal{L}(X)$.

$$J(u) = \frac{1}{2} \langle Q_T u, u \rangle - \langle v, u \rangle \quad \forall u \in X.$$

Then J is of class $C^1(X)$ and coercive thanks to (182):

$$J(u) \geq \frac{1}{2k_T^2} |u|^2 - |v| |u| \xrightarrow{|u| \rightarrow \infty} \infty$$

Moreover, J is convex. Indeed, for all $u^0, u^1 \in X$ and $\lambda \in [0, 1]$ we have that

$$\begin{aligned} & \lambda J(u^1) + (1 - \lambda) J(u^0) - J(\lambda u^1 + (1 - \lambda) u^0) \\ &= \frac{\lambda(1 - \lambda)}{2} \{ \langle Q_T u^1, u^1 \rangle + \langle Q_T u^0, u^0 \rangle - 2 \langle Q_T u^1, u^0 \rangle \} \geq 0 \end{aligned}$$

because

$$|\langle Q_T u^1, u^0 \rangle| \leq |\langle Q_T u^1, u^1 \rangle|^{1/2} |\langle Q_T u^0, u^0 \rangle|^{1/2}$$

by the Cauchy-Schwarz inequality. Therefore, Lemma 10 below ensures that J attains its minimum on X . Let $u_v \in X$ be such that $J(u_v) = \min J$. Then

$$0 = J'(u_v) = Q_T u_v - v \quad \text{and} \quad v = Q_T u_v.$$

(c) $\Rightarrow$ (a) Fix any $u^0, v^0 \in X$ and take f^0 as in (181). Then

$$u_{f^0}(T, u^0) = S(T)u^0 + \int_0^T S(T-s) B B^* S(T-s)^* Q_T^{-1} (v^0 - S(T)u^0) dt = v^0$$

and

$$\begin{aligned} & \int_0^T \|f^0(s)\|^2 ds \\ &= \int_0^T \left\langle B^* S(T-s)^* Q_T^{-1} (v^0 - S(T)u^0), B^* S(T-s)^* Q_T^{-1} (v^0 - S(T)u^0) \right\rangle ds \\ &= \left\langle Q_T^{-1} (v^0 - S(T)u^0), v^0 - S(T)u^0 \right\rangle \leq \|Q_T^{-1}\| (|v^0| + \|S(T)\| |u^0|)^2 \end{aligned}$$

which yields exact controllability.

Finally, we have already proved that f^0 satisfies (i) and (ii). In order to show (iii), let $f \in L^2(0, T; F)$ be such that $u^f(T, u^0) = v^0$. In view of (ii) we have that

$$\begin{aligned} \int_0^T (f^0(s), f(s)) ds &= \int_0^T (B^* S(T-s)^* Q_T^{-1} (v^0 - S(T)u^0), f(s)) ds \\ &= \int_0^T \left\langle Q_T^{-1} (v^0 - S(T)u^0), S(T-s) B f(s) \right\rangle ds \\ &= \left\langle Q_T^{-1} (v^0 - S(T)u^0), v^0 - S(T)u^0 \right\rangle = \int_0^T \|f^0(s)\|^2 ds \end{aligned}$$

Therefore,

$$0 \leq \int_0^T \|f(s) - f^0(s)\|^2 ds = \int_0^T \|f(s)\|^2 ds - \int_0^T \|f^0(s)\|^2 ds,$$

which in turn yields (iii). $\square$

The following lemma is a milestone in infinite dimensional optimization.

Lemma 10 *Let $J : X \rightarrow \mathbb{R}$ be a coercive lower semicontinuous convex function. Then J has a minimum point in X .*

Proof. Let $\{u_n\} \subset X$ be such that $J(u_n) \rightarrow \inf J$. The coercivity of J yields the existence of a subsequence $\{u_{n_k}\}$ such that $u_{n_k} \rightharpoonup u_\infty \in X$. For any $\alpha > \inf J$ let

$$\Gamma_\alpha = \{u \in X \mid J(u) \leq \alpha\}.$$

Then Γ_α is convex (since J is convex), closed (since J is lower semicontinuous), and nonempty. We claim that $u_\infty \in \Gamma_\alpha$. For otherwise, by a corollary of the Hahn-Banach theorem we could separate u_∞ and Γ_α by a hyperplane:

$$\exists v \in X : \sup_{u \in \Gamma_\alpha} \langle u, v \rangle < \langle u_\infty, v \rangle$$

Since $u_{n_k} \in \Gamma_\alpha$ for k large enough, we have that

$$\limsup_{k \rightarrow \infty} \langle u_{n_k}, v \rangle < \langle u_\infty, v \rangle$$

in contrast with $\langle u_{n_k}, v \rangle \rightarrow \langle u_\infty, v \rangle$. Our claim is thus proved.

Since $u_\infty \in \Gamma_\alpha$, we have that $J(u_\infty) \leq \alpha$ for all $\alpha > \inf J$. So, $\inf J > -\infty$ and $J(u_\infty) = \inf J$. $\square$

Example 32 (Wave equation with distributed control) Let us study the exact controllability in time $T \in]0, \infty[$ of the control system

$$\begin{cases} \frac{\partial^2 u}{\partial t^2}(t, x) = \frac{\partial^2 u}{\partial x^2}(t, x) + f(t, x) & \text{in }]0, T[\times]0, \pi[\\ u(t, 0) = 0 = u(t, \pi) & t \in]0, T[\\ u(0, x) = u^0(x), \frac{\partial u}{\partial t}(0, x) = v^0(x) & x \in]0, \pi[\end{cases} \quad (183)$$

where

- $f \in L^2([0, T] \times [0, \pi])$ is the control strategy with $L^2(0, \pi)$ as the control space,
- $(u^0, v^0) \in H_0^1(0, \pi) \times L^2(0, \pi)$.

We recast (183) as in Example 29, that is,

$$\begin{pmatrix} u'(t) \\ v'(t) \end{pmatrix} = \mathcal{A} \begin{pmatrix} u(t) \\ v(t) \end{pmatrix} + B f(t) \quad (t \in [0, T] \text{ a.e.})$$

where $\mathcal{A}$ is defined in (155) and $B : L^2(0, \pi) \rightarrow H_0^1(0, \pi) \times L^2(0, \pi)$ is given by

$$(B\xi)(x) = \begin{pmatrix} 0 \\ \xi(x) \end{pmatrix} \quad (\forall \xi \in L^2(0, \pi), x \in [0, \pi] \text{ a.e.})$$

We can compute $B^* : H_0^1(0, \pi) \times L^2(0, \pi) \rightarrow L^2(0, \pi)$ as follows: since

$$\langle B\xi, \begin{pmatrix} \varphi \\ \psi \end{pmatrix} \rangle = \langle \begin{pmatrix} 0 \\ \xi \end{pmatrix}, \begin{pmatrix} \varphi \\ \psi \end{pmatrix} \rangle = \int_0^\pi \xi(x) \psi(x) dx$$

for all $\xi, \psi \in L^2(0, \pi)$ and $\varphi \in H_0^1(0, \pi)$, we have that

$$B^* \begin{pmatrix} \varphi \\ \psi \end{pmatrix} = \psi \quad \forall (\varphi, \psi) \in H_0^1(0, \pi) \times L^2(0, \pi).$$

Since the group $S(t)$ generated by $\mathcal{A}$ is unitary, recalling (159), we deduce that

$$\begin{aligned} \int_0^T \|B^* S(t)^* \begin{pmatrix} u_0^0 \\ v_0^0 \end{pmatrix}\|^2 dt &= \int_0^T \|B^* S(-t) \begin{pmatrix} u_0^0 \\ v_0^0 \end{pmatrix}\|^2 dt \\ &= \int_0^T \sum_{n=1}^{\infty} (v_n^0 \cos(nt) + n u_n^0 \sin(nt))^2 dt. \end{aligned}$$

If $T \geq 2\pi$, then

$$\begin{aligned} \int_0^T \|B^* S(t)^* \begin{pmatrix} u_0^0 \\ v_0^0 \end{pmatrix}\|^2 dt &\geq \int_0^{2\pi} \sum_{n=1}^{\infty} (v_n^0 \cos(nt) + n u_n^0 \sin(nt))^2 dt \\ &= \int_0^{2\pi} \sum_{n=1}^{\infty} (|v_n^0|^2 \cos^2(nt) + n^2 |u_n^0|^2 \sin^2(nt) + 2n u_n^0 v_n^0 \sin(nt) \cos(nt)) dt \\ &= \int_0^{2\pi} \sum_{n=1}^{\infty} (|v_n^0|^2 \frac{1 + \cos(2nt)}{2} + n^2 |u_n^0|^2 \frac{1 - \cos(2nt)}{2} + n u_n^0 v_n^0 \sin(2nt)) dt \\ &= \pi \sum_{n=1}^{\infty} (|v_n^0|^2 + n^2 |u_n^0|^2) = \pi \left| \begin{pmatrix} u_0^0 \\ v_0^0 \end{pmatrix} \right|^2, \end{aligned}$$

so that (179) is satisfied with $k_T = 1/\sqrt{\pi}$.

CONCLUSION: System (183) is exactly controllable in any time $T \geq 2\pi$.

5 Bilinear control of evolution equations

In this section, we keep the assumption that $(X, \langle \cdot, \cdot \rangle, |\cdot|)$ is a separable Hilbert space over $\mathbb{R}$ (the *state space*) and we denote by $\{e_j\}_{j \in \mathbb{N}}$ a complete orthonormal system

in X . We will study nonlinear control systems of the form

$$\begin{cases} u'(t) = Au(t) + f(t)Bu(t) + g(t) & (t \in [0, T]) \\ u(0) = u^0 \in X, \end{cases} \quad (184)$$

where $0 < T < \infty$ is the time *horizon*, under the following Standing Assumptions **(SA)**:

- (i) $A : D(A) \subset X \rightarrow X$ is a self-adjoint dissipative operator and we denote by $S(t)$ the C_0 -semigroup on X generated by A ;
- (ii) $B \in \mathcal{L}(X)$;
- (iii) $f \in L^2(0, T)$ is a *scalar control*;
- (iv) $g \in L^2(0, T; X)$ is a *forcing term*.

The above hypotheses are quite restrictive but they will allow us to prove the main results in a rather simple way. We begin by defining a mild solution of (184).

Definition 27 A mild solution of (184) is a function $u \in C([0, T]; X)$ such that

$$u(t) = S(t)u^0 + \int_0^t S(t-s)[f(s)Bu(s) + g(s)] ds \quad \forall t \in [0, T]. \quad (185)$$

The following result ensures the well-posedness of problem (184).

Theorem 24 Let $T > 0$, $u^0 \in X$, and $g \in L^2(0, T; X)$ be fixed. Then for any $f \in L^2(0, T)$ problem (184) has a unique mild solution u^f . Moreover,

$$\|u^f\|_\infty \leq (|u^0| + \|g\|_1) e^{\|B\| \|f\|_1} \quad (186)$$

Proof. We begin by constructing the solution of (184) on some time interval $[0, T_f]$ with $T_f \in]0, T]$ to be defined later. For any $u \in C([0, T_f]; X)$ let us set

$$\phi[u](t) = S(t)u^0 + \int_0^t S(t-s)[f(s)Bu(s) + g(s)] ds \quad \forall t \in [0, T].$$

Since $s \mapsto f(s)Bu(s) + g(s)$ is in $L^2(0, T_f; X)$, we have that $\phi[u] \in C([0, T_f]; X)$. So, $\phi : C([0, T_f]; X) \rightarrow C([0, T_f]; X)$. Let us choose T_f in order for ϕ to be a contraction. For any $u, v \in C([0, T_f]; X)$ we have that

$$\begin{aligned} |\phi[v](t) - \phi[u](t)| &= \left| \int_0^t f(s)S(t-s)B(v(s) - u(s))ds \right| \\ &\leq \|B\| \|f\|_1 \|v - u\|_\infty \leq \|B\| \|f\|_2 \sqrt{T_f} \|v - u\|_\infty. \end{aligned}$$

Thus, by taking $T_f \in]0, T]$ such that $\|B\| \|f\|_2 \sqrt{T_f} < 1$ we conclude that, being a contraction, ϕ possesses a unique fixed point u^f , which gives the unique mild solution of (184) on $[0, T_f]$. However, since T_f does not depend on the initial condition, by finitely many iterations of the above construction one gets to time T .

In order to prove (186), we observe that (185) yields

$$|u^f(t)| \leq |u^0| + \|g\|_1 + \|B\| \int_0^t |f(s)| |u(s)| ds \quad \forall t \in [0, T].$$

Therefore, by the Gronwall lemma (Lemma 13),

$$|u^f(t)| \leq (|u^0| + \|g\|_1) e^{\|B\| \|f\|_1} \quad \forall t \in [0, T].$$

The conclusion follows. $\square$

An interesting property of bilinear systems is that they turn weak convergence of controls into uniform convergence of trajectories, as we shall soon see.

Proposition 18 *Let $0 < T < \infty$ and let $\{f_n\}_{n \in \mathbb{N}}$ be such that $f_n \rightharpoonup f$ in $L^2(0, T)$. Then*

$$u^{f_n}(\cdot, u^0) \rightarrow u^f(\cdot, u^0) \quad \text{in } C([0, T]; X). \quad (187)$$

Proof. Let

$$u_n(t) = u^{f_n}(t, u^0) \quad \text{and} \quad u(t) = u^f(t, u^0) \quad (t \in]0, T])$$

Then

$$\begin{aligned} u_n(t) - u(t) &= \int_0^t f_n(s) S(t-s) B(u_n(s) - u(s)) ds \\ &\quad + \int_0^t (f_n(s) - f(s)) S(t-s) B u(s) ds \\ &=: U_n(t) + V_n(t). \end{aligned}$$

We will now estimate both U_n and V_n . For the former term we observe that, since $\{f_n\}_{n \in \mathbb{N}}$ is bounded in $L^2(0, T)$, by Hölder's inequality we have that

$$\begin{aligned} |U_n(t)|^2 &\leq \|B\|^2 \left(\int_0^t |f_n(s)| |u_n(s) - u(s)| ds \right)^2 \\ &\leq \|B\|^2 \int_0^t |f_n(s)|^2 ds \int_0^t |u_n(s) - u(s)|^2 ds \\ &\leq C_1 \int_0^t |u_n(s) - u(s)|^2 ds \quad (t \in]0, T]) \end{aligned} \quad (188)$$

for some constant $C_1 \geq 0$.

The study of the latter term is more complicated. First of all, we note that

$$\|V_n\|_\infty \leq \|B\| \|u\|_\infty \int_0^t |f_n(s) - f(s)| ds \leq C_2 \quad (189)$$

for some constant $C_2 \geq 0$. Similarly, thanks to Lemma 9,

$$\int_0^T |AV_n(t)|^2 dt \leq \|B\|^2 \|u\|_\infty^2 \int_0^T |f_n(t) - f(t)|^2 dt \leq C_3 \quad (190)$$

for some constant $C_3 \geq 0$. We now claim that

$$\lim_{n \rightarrow \infty} V_n(t) = 0 \quad \forall t \in [0, T]. \quad (191)$$

Indeed, fix any $t \in [0, T]$ and define

$$v(s) = S(t-s)Bu(s) \quad \text{and} \quad v_j(s) = \langle v(s), e_j \rangle \quad (s \in [0, t], j \in \mathbb{N})$$

where $\{e_j\}_{j \in \mathbb{N}}$ is a complete orthonormal system in X . Now, fix any $\varepsilon > 0$. Since

$$\int_0^t |v(s)|^2 ds = \sum_{j=1}^{\infty} \int_0^t |v_j(s)|^2 ds < \infty,$$

there exists $j_\varepsilon = j_\varepsilon(t) \in \mathbb{N}$ such that

$$\sum_{j=j_\varepsilon+1}^{\infty} \int_0^t |v_j(s)|^2 ds < \varepsilon^2.$$

Moreover, since

$$V_n(t) = \int_0^t (f_n(s) - f(s))v(s) ds = \sum_{j=1}^{\infty} \left(\int_0^t (f_n(s) - f(s))v_j(s) ds \right) e_j$$

we have that

$$\begin{aligned} |V_n(t)|^2 &= \sum_{j=1}^{j_\varepsilon} \left(\int_0^t (f_n(s) - f(s))v_j(s) ds \right)^2 \\ &\quad + \sum_{j=j_\varepsilon+1}^{\infty} \left(\int_0^t (f_n(s) - f(s))v_j(s) ds \right)^2 \\ &\leq \sum_{j=1}^{j_\varepsilon} \left(\int_0^t (f_n(s) - f(s))v_j(s) ds \right)^2 + C_4 \sum_{j=j_\varepsilon+1}^{\infty} \int_0^t |v_j(s)|^2 ds \\ &\leq \sum_{j=1}^{j_\varepsilon} \left(\int_0^t (f_n(s) - f(s))v_j(s) ds \right)^2 + C_4 \varepsilon^2. \end{aligned}$$

Now, owing to the weak convergence of $\{f_n\}_{n \in \mathbb{N}}$ there exists $n_\varepsilon = n_\varepsilon(t)$ such that

$$\left(\int_0^t (f_n(s) - f(s))v_j(s) ds \right)^2 < \frac{\varepsilon^2}{j_\varepsilon} \quad (j = 1, \dots, j_\varepsilon)$$

for all $n \geq n_\varepsilon$. In conclusion,

$$|V_n(t)|^2 \leq (1 + C_4)\varepsilon^2 \quad \forall n \geq 1. \quad (192)$$

This shows our claim (191).

Now, since $u_n(t) - u(t) = U_n(t) + V_n(t)$, in view of (188) we have that

$$|u_n(t) - u(t)|^2 \leq 2\left(|V_n(t)|^2 + C_1 \int_0^t |u_n(s) - u(s)|^2 ds\right) \quad \forall t \in [0, T].$$

So, by Gronwall's lemma (Lemma 13 below) with $w(t) = 2|V_n(t)|^2$ we conclude that

$$\begin{aligned} |u_n(t) - u(t)|^2 &\leq 4 \int_0^t \langle V_n'(s), V_n(s) \rangle e^{2C_1(t-s)} ds \\ &\leq 4e^{2C_1T} \left(\int_0^T |V_n'(s)|^2 ds \right)^{\frac{1}{2}} \left(\int_0^T |V_n(s)|^2 ds \right)^{\frac{1}{2}} \quad (t \in [0, T]) \end{aligned} \quad (193)$$

Since

$$|V_n'(t)| \leq |AV_n(t)| + \|B\| \|u\|_\infty |f_n(t) - f(t)| \quad (t \in [0, T] \text{ a.e.}),$$

owing to (190) and the fact that $\{f_n\}_{n \in \mathbb{N}}$ is bounded in $L^2(0, T)$ we have that $|V_n'|$ is also bounded in $L^2(0, T)$. Therefore, (193) yields

$$\|u_n - u\|_\infty^2 \leq C_5 \left(\int_0^T |V_n(s)|^2 ds \right)^{\frac{1}{2}}$$

for some constant $C_5 \geq 0$. Then, the conclusion follows from (189), (191), and the dominated convergence theorem. $\square$

5.1 A negative result

In this section, we address exact controllability for system (184) assuming, for simplicity, that $g \equiv 0$. The definition of reachable set for system (184) is similar to the one of Section 4.

Definition 28 The *reachable* (or, *attainable*) set from $u^0 \in X$ in time T for system (184) is defined as

$$\mathcal{R}_T(u^0) = \left\{ u^f(T, u^0) \mid f \in L^2(0, T) \right\} \quad (194)$$

The *reachable* (or, *attainable*) set from $u^0 \in X$ for system (184) is defined as

$$\mathcal{R}(u^0) = \bigcup_{T>0} \mathcal{R}_T(u^0). \quad (195)$$

The following result, obtained by Ball, Marsden, and Slemrod [6, Theorem 3.6] in more general settings than the ones we present here, establishes the failure of exact controllability for system (184) when X is infinite dimensional.

Theorem 25 *If $\dim X = \infty$, then $X \setminus \mathcal{R}(u^0)$ is dense in X for every $u^0 \in X$.*

Proof. For any $m, n \in \mathbb{N}$ let us set

$$\mathcal{R}_m^n(u^0) = \left\{ u^f(t, u^0) \mid 0 \leq t \leq m, \int_0^m |f(s)|^2 ds \leq n^2 \right\}.$$

We claim that $\mathcal{R}_m^n(u^0)$ is compact for all $m, n \in \mathbb{N}$. Indeed, fix any pair of sequences $\{t_j\}$ and $\{f_j\}$ with

$$0 \leq t_j \leq m, \quad f_j \in L^2(0, m), \quad \|f_j\|_2 \leq n \quad (j \in \mathbb{N}).$$

Then there exist subsequences $\{t_{j_k}\}$ and $\{f_{j_k}\}$ such that

$$t_{j_k} \rightarrow t_\infty \in [0, m] \quad \text{and} \quad f_{j_k} \rightharpoonup f_\infty \quad (k \rightarrow \infty)$$

with $\|f_\infty\|_2 \leq n$. Our claim follows by appealing to Proposition 18.

Since $\mathcal{R}_m^n(u^0)$ is compact and $\dim X = \infty$, we have that $\mathcal{R}_m^n(u^0)$ has empty interior. So, by Baire's lemma

$$X \setminus \mathcal{R}(u^0) = X \setminus \bigcup_{m,n \in \mathbb{N}} \mathcal{R}_m^n(u^0) = \bigcap_{m,n \in \mathbb{N}} (X \setminus \mathcal{R}_m^n(u^0))$$

is dense in X . □

5.2 Controllability to eigensolutions

As for positive results, most of the research activity has focussed on controllability to eigensolutions and the study of attainable sets. In this section, we describe—without giving detailed proofs—some recent result from [1, 2, 3] on the former topic, the latter will be outlined in the next section.

Let us consider system (184) with $g \equiv 0$, that is,

$$\begin{cases} u'(t) = Au(t) + f(t)Bu(t) & (t \in [0, T]) \\ u(0) = u^0 \in X. \end{cases} \quad (196)$$

Assumptions **(SA)** above will be reinforced as follows:

(v) $\exists \lambda_0 > 0$ such that $R(\lambda_0, A) : X \rightarrow X$ is compact.

Then, one can show that there exists a real sequence $0 \leq \lambda_j \rightarrow \infty$ and a complete orthonormal system $\{e_j\}_{j \in \mathbb{N}}$ in X such that

$$e_j \in D(A) \quad \text{and} \quad Ae_j = -\lambda_j e_j \quad (j \in \mathbb{N})$$

(see, for instance, [14] or [24]). We will call e_1 the *ground state* of A and the function

$$\phi_1(t) = e^{-\lambda_1 t} e_1 \quad (t \geq 0)$$

the *ground state solution*. In general, the functions $\phi_j(t) = e^{-\lambda_j t} e_j$ ($j \geq 1$) are called the *eigensolutions* of (196).

Exercise 36 Prove that, if assumption (v) above is satisfied, then $R(\lambda, A)$ is compact for all $\lambda > 0$.

Hint: use the resolvent identity (19).

In order to study local controllability to trajectories, the notion of j -null controllability in time $T > 0$ for the pair $\{A, B\}$ was introduced in [1]: such a property requires the existence of a constant $N > 0$ such that for any initial condition $y_0 \in X$, there exists a control $f \in L^2(0, T)$ satisfying

$$\|f\|_2 \leq N|y_0|, \quad (197)$$

and for which $y(T) = 0$, where $y(\cdot)$ is the solution of the following linear problem

$$\begin{cases} y'(t) = Ay(t) + f(t)Be_j, & t \in [0, T] \\ y(0) = y_0. \end{cases} \quad (198)$$

The best constant N satisfying (197), given as

$$N(T) := \sup_{|y_0|=1} \inf \{ \|f\|_2 : y^f(T, y_0) = 0 \}, \quad (199)$$

is called the *control cost*. The following result ensures local exact controllability for problem (196) assuming a precise behavior of the control cost for small time.

Theorem 26 ([2]) *Let $A : D(A) \subset X \rightarrow X$ be a densely defined linear operator satisfying (SA) – (v) and let $B : X \rightarrow X$ be a bounded linear operator. Assume that $\{A, B\}$ is j -null controllable in any time $T > 0$ for some $j \in \mathbb{N}$ and suppose that*

$$N(\tau) \leq e^{\nu/\tau}, \quad \forall 0 < \tau \leq T_0, \quad (200)$$

for some constants $\nu, T_0 > 0$. Then, for any $T > 0$, there exists a constant $R_T > 0$ such that, for any $u_0 \in B_{R_T}(e_j)$, there exists a control $f \in L^2(0, T)$ such that the solution u^f of (196) satisfies $u^f(T) = \phi_j(T)$. Moreover, the following estimate holds

$$\|f\|_2 \leq \frac{e^{-\pi^2 C/T}}{e^{2\pi^2 C/(3T)} - 1},$$

where C is a suitable positive constant.

The proof of the above result, for which the reader is referred to [2], is fully constructive being based on an algorithm that allows to compute all relevant constants. In particular, no use of the inverse mapping theorem is made. The main idea consists of applying a stability estimate ([1]) to a suitable sequence of time intervals of decreasing length T_j , such that $\sum_{j=1}^{\infty} T_j < \infty$. The sequence $\{T_j\}$, which can be constructed only thanks to (200), has to be carefully chosen.

In [1], sufficient conditions for j -null controllability were given. However, such hypotheses are insufficient to guarantee the validity of condition (200) for the control cost. In the result that follows, sufficient conditions for $N(T)$ to satisfy (200) are given. It would be interesting to understand if (200) is also necessary for the local exact controllability of (196).

Theorem 27 ([2]) *Let $A : D(A) \subset X \rightarrow X$ be a densely defined linear operator satisfying (SA) – (v) and suppose that there exists a constant $\alpha > 0$ for which the eigenvalues of A fulfill the gap condition*

$$\sqrt{\lambda_{k+1}} - \sqrt{\lambda_k} \geq \alpha, \quad \forall k \in \mathbb{N}. \quad (201)$$

Let $B : X \rightarrow X$ be a bounded linear operator such that there exist $b, q > 0$ for which

$$\langle Be_j, e_j \rangle \neq 0 \quad \text{and} \quad |\lambda_k - \lambda_j|^q |\langle Be_j, e_k \rangle| \geq b, \quad \forall k \neq j. \quad (202)$$

Then, the pair $\{A, B\}$ is j -null controllable in any time $T > 0$ with control cost $N(T)$ that satisfies (200).

From Theorems 26 and 27 we deduce the following Corollary.

Corollary 7 ([2]) *Let $A : D(A) \subset X \rightarrow X$ be a densely defined linear operator satisfying (SA) – (v) and suppose that there exists a constant $\alpha > 0$ for which (201) is satisfied. Let $B : X \rightarrow X$ be a bounded linear operator that verifies (202) for some $b, q > 0$. Then, problem (196) is locally controllable to the j th eigensolution ϕ_j in any time $T > 0$.*

We now show how to apply the above controllability result to a single input bilinear control problem for the heat operator.

Example 33 (Diffusion equation with Dirichlet boundary conditions) Let $I = (0, 1)$ and $X = L^2(0, 1)$. Consider the problem

$$\begin{cases} u_t(t, x) - u_{xx}(t, x) + f(t)\mu(x)u(t, x) = 0 & x \in I, t > 0, \\ u(t, 0) = 0, \quad u(t, 1) = 0 & t > 0, \\ u(0, x) = u_0(x) & x \in I. \end{cases} \quad (203)$$

Define

$$D(A) = H^2(I) \cap H_0^1(I), \quad A\varphi = \frac{d^2\varphi}{dx^2}.$$

We know that A satisfies **(SA)** – (v). Its eigenvalues and normalized eigenfunctions are

$$\lambda_k = (k\pi)^2, \quad e_k(x) = \sqrt{2} \sin(k\pi x), \quad k \geq 1,$$

with $Ae_k = -\lambda_k e_k$. In particular, (201) holds:

$$\sqrt{\lambda_{k+1}} - \sqrt{\lambda_k} = \pi \quad \forall k \geq 1.$$

Let $B : X \rightarrow X$ be the bounded multiplication operator

$$(Bu)(x) = \mu(x)u(x) \quad (x \in I),$$

where $\mu \in H^3(I)$ is real-valued. Problem (203) takes the form $u' = Au - f(t)Bu$. Replacing f by $-f$ puts it in the form (196), without changing the controllability conclusions.

Suppose that, for a fixed $j \geq 1$,

$$\mu'(1) \pm \mu'(0) \neq 0 \quad \text{and} \quad \langle \mu e_j, e_k \rangle \neq 0 \quad \forall k \geq 1. \quad (204)$$

We claim that there exists $b > 0$ such that

$$|\lambda_k - \lambda_j|^{3/2} |\langle \mu e_j, e_k \rangle| \geq b \quad \forall k \geq 1, \quad k \neq j. \quad (205)$$

Indeed, integrating by parts three times and using $e_j(0) = e_j(1) = 0$, we obtain

$$\begin{aligned} \langle \mu e_j, e_k \rangle &= \sqrt{2} \int_0^1 \mu(x) e_j(x) \sin(k\pi x) dx \\ &= \frac{\sqrt{2}}{k\pi} \int_0^1 (\mu e_j)'(x) \cos(k\pi x) dx \\ &= -\frac{\sqrt{2}}{(k\pi)^2} \int_0^1 (\mu e_j)''(x) \sin(k\pi x) dx \\ &= \frac{\sqrt{2}}{(k\pi)^3} [(\mu e_j)''(x) \cos(k\pi x)]_0^1 \\ &\quad - \frac{\sqrt{2}}{(k\pi)^3} \int_0^1 (\mu e_j)'''(x) \cos(k\pi x) dx \\ &= \frac{4j}{k^3 \pi^2} [\mu'(1)(-1)^{k+j} - \mu'(0)] \\ &\quad - \frac{\sqrt{2}}{(k\pi)^3} \int_0^1 (\mu e_j)'''(x) \cos(k\pi x) dx. \end{aligned}$$

Here we used

$$(\mu e_j)''(0) = 2\sqrt{2} j \pi \mu'(0), \quad (\mu e_j)''(1) = 2\sqrt{2} j \pi (-1)^j \mu'(1).$$

Since $(\mu e_j)''' \in L^2(0, 1) \subset L^1(0, 1)$, the Riemann–Lebesgue lemma gives

$$\int_0^1 (\mu e_j)'''(x) \cos(k\pi x) dx \longrightarrow 0 \quad \text{as } k \rightarrow \infty.$$

By (204), the number

$$d = \min \{ |\mu'(1) - \mu'(0)|, |\mu'(1) + \mu'(0)| \}$$

is positive. Consequently,

$$\liminf_{k \rightarrow \infty} k^3 |\langle \mu e_j, e_k \rangle| \geq \frac{4j}{\pi^2} d > 0.$$

Since

$$\frac{|\lambda_k - \lambda_j|^{3/2}}{k^3} \longrightarrow \pi^3,$$

the left-hand side of (205) is bounded away from zero for all sufficiently large k . For the finitely many remaining indices $k \neq j$, it is positive by (204). Taking the minimum proves (205). Together with $\langle \mu e_j, e_j \rangle \neq 0$, this verifies (202) with $q = 3/2$.

A function satisfying (204) is $\mu(x) = x^2$. Indeed, $\mu'(0) = 0$, $\mu'(1) = 2$, and

$$\langle x^2 e_j, e_k \rangle = \begin{cases} \frac{8kj(-1)^{k+j}}{\pi^2(k^2 - j^2)^2}, & k \neq j, \\ \frac{2j^2\pi^2 - 3}{6j^2\pi^2}, & k = j. \end{cases}$$

Therefore, the problem

$$\begin{cases} u_t(t, x) - u_{xx}(t, x) + f(t)x^2u(t, x) = 0 & x \in I, t > 0, \\ u(t, 0) = 0, \quad u(t, 1) = 0 & t > 0, \\ u(0, x) = u_0(x) & x \in I \end{cases} \quad (206)$$

is locally controllable to the j th eigensolution

$$\phi_j(t, x) = \sqrt{2} \sin(j\pi x) e^{-j^2\pi^2 t}$$

in any time $T > 0$. More precisely, for every $j \geq 1$ and $T > 0$, there exists $R_T > 0$ such that, whenever $u_0 \in B_{R_T}(e_j)$, some control $f \in L^2(0, T)$ steers the solution to $u(T) = \phi_j(T)$.

In [3], the above approach was extended to cover the case of an unbounded operator $B : D(B) \subset X \rightarrow X$ satisfying

$$D((-A)^{1/2}) \hookrightarrow D(B), \quad \text{and} \quad |Bu| \leq C \left(|(-A)^{1/2}u| + |u| \right),$$

for some constant $C > 0$. In this way, one obtains a local exact controllability result to the ground state solution of the Fokker-Planck equation

$$u_t(t, x) - u_{xx}(t, x) + f(t)(\mu(x)u(t, x))_x = 0$$

with either Dirichlet

$$u(t, 1) = u(t, 0) = 0$$

or partially reflecting

$$\frac{\partial}{\partial x}u(t, 1) = \frac{\partial}{\partial x}u(t, 0) = 0$$

boundary conditions.

To confirm the flexibility of the method we described in this section, let us mention the adaptation of these results to degenerate parabolic equations in [19], networks in [15], and nonlinear diffusive models in [22].

5.3 Further bibliographical comments

First of all, we would like to mention the works [8], on bilinear control of the wave equation, [10] which studies bilinear control for Schrödinger equation (see also [7] for a first result on this topic), and [18] which concerns a degenerate wave equation under a bilinear control action. The results obtained in these papers rely on linearization around the ground state, the use of the inverse mapping theorem, and a regularizing effect which takes place in both problems. Local controllability is proved for any positive time for the Schrödinger equation and for a sufficiently (optimal) large time for the wave equation, under conditions (201) and (202). Notice that the structure of the second order operator and the fact that the space dimension equals one allow the authors of [7], [10], and [18] to apply Ingham's theory ([26]) which requires a gap condition on the eigenvalues of A . We further observe that even if the genericity of assumption (202) is proved in [7, 10], only few explicit examples of operators B of multiplication type are available in the literature. We refer to [4] where a general constructive algorithm for building potentials which satisfy (202) is established.

If (202) is violated then it has been first shown by Coron [20], for a model describing a particle in a moving box, that there exists a minimal time for local exact controllability to hold. This model couples the Schrödinger equation with two ordinary differential equations modeling the speed and acceleration of the box (see also Beauchard, Coron [9] for local exact controllability for large time). A further paper by Beauchard and Morancey [12] for the Schrödinger equation extends [10] to cases for which the above condition fails, that is, when there exist integers k such that $\langle B\varphi_1, \varphi_k \rangle = 0$.

An example of controllability to trajectories for nonlinear parabolic systems is analysed in [25], where, however, additive controls are considered. In such an example, one can obtain controllability to free trajectories by Carleman estimates and inverse mapping arguments. Such a strategy seems hard to adapt to bilinear control.

A paper which has strong connections with the aforementioned works [1, 2, 3], is the one by Beauchard and Marbach [11], where the authors study small-time null

controllability for a scalar-input heat equation in one space dimension, with nonlinear lower order terms. Among the results of such paper, we mention null-controllability to the first eigenstate of a heat equation with bilinear control. From this result it would be possible to deduce local controllability only to the first eigenstate of the heat equation subject to Neumann boundary conditions. It is worth noting that [11] addresses a specific parabolic equation and the methods developed therein rely on the so-called source term procedure.

Finally, we observe that the bilinear controls we have considered so far, are just scalar functions of time. This fact explains why applications mainly concern problems in low space dimension, like the results in [7, 8, 9, 10, 12, 11, 20]. A stronger control action could be obtained by letting controls depend on time and space. We refer the reader to [16, 17] for more on this subject.

Appendix A: the Cauchy integral on $C([a, b]; X)$

We recall the construction of the Riemann integral for a continuous $f : [a, b] \rightarrow X$, where X is a Banach space and $-\infty < a < b < \infty$.

Let us consider the family of partitions of $[a, b]$

$$\Pi(a, b) = \left\{ \pi = \{t_i\}_{i=0}^n : n \geq 1, a = t_0 < t_1 < \dots < t_n = b \right\}$$

and define

$$\text{diam}(\pi) = \max_{1 \leq i \leq n} (t_i - t_{i-1}) \quad (\pi \in \Pi(a, b)).$$

For any $\pi \in \Pi(a, b)$, $\pi = \{t_i\}_{i=0}^n$, we set

$$\Sigma(\pi) = \left\{ \sigma = (s_1, \dots, s_n) : s_i \in [t_{i-1}, t_i], 1 \leq i \leq n \right\}.$$

Finally, for any $\pi \in \Pi(a, b)$, $\pi = \{t_i\}_{i=0}^n$, and $\sigma \in \Sigma(\pi)$, $\sigma = (s_1, \dots, s_n)$, we define

$$S_\pi^\sigma(f) = \sum_{i=1}^n f(s_i)(t_i - t_{i-1}).$$

Lemma 11 *For any $\varepsilon > 0$ there exists $\delta > 0$ such that for all $\pi, \pi' \in \Pi(a, b)$ with $\pi \subseteq \pi'$ we have that*

$$\text{diam}(\pi) < \delta \implies |S_\pi^\sigma(f) - S_{\pi'}^{\sigma'}(f)| < \varepsilon$$

for all $\sigma \in \Sigma(\pi)$ and $\sigma' \in \Sigma(\pi')$.

Proof. Since f is uniformly continuous, for any $\varepsilon > 0$ there exists $\delta > 0$ such that for all $t, s \in [a, b]$

$$|t - s| < \delta \implies |f(t) - f(s)| < \frac{\varepsilon}{b - a}. \quad (207)$$

Let

$$\begin{cases} \pi = \{t_i\}_{i=0}^n, & \sigma = (s_1, \dots, s_n) \\ \pi' = \{t'_j\}_{j=0}^m, & \sigma' = (s'_1, \dots, s'_m) \end{cases}$$

be such that $\pi \subseteq \pi'$ and $\text{diam}(\pi) < \delta$. Then there exist positive integers

$$0 = j_0 < j_1 < \dots < j_n = m$$

such that $t'_{j_i} = t_i$ for all $i = 0, \dots, n$. For any such i , it holds that

$$t_i - t_{i-1} = t'_{j_i} - t'_{j_{i-1}} = \sum_{j=j_{i-1}+1}^{j_i} (t'_j - t'_{j-i}).$$

Then

$$\begin{aligned}
S_{\pi}^{\sigma}(f) - S_{\pi'}^{\sigma'}(f) &= \sum_{i=1}^n f(s_i)(t_i - t_{i-1}) - \sum_{j=1}^m f(s'_j)(t'_j - t'_{j-1}) \\
&= \sum_{i=1}^n \sum_{j=j_{i-1}+1}^{j_i} (f(s_i) - f(s'_j))(t'_j - t'_{j-1})
\end{aligned}$$

Since for all $i = 1, \dots, n$ we have that

$$s_i, s'_j \in [t_{i-1}, t_i] \quad \forall j_{i-1} + 1 \leq j \leq j_i,$$

from (207) it follows that

$$\begin{aligned}
|S_{\pi}^{\sigma}(f) - S_{\pi'}^{\sigma'}(f)| &\leq \sum_{i=1}^n \sum_{j=j_{i-1}+1}^{j_i} |f(s_i) - f(s'_j)|(t'_j - t'_{j-1}) \\
&\leq \frac{\varepsilon}{b-a} \sum_{i=1}^n (t_i - t_{i-1}) = \varepsilon.
\end{aligned}$$

The proof is complete. $\square$

Theorem 28 *The limit*

$$\lim_{\text{diam}(\pi) \downarrow 0} S_{\pi}^{\sigma}(f) =: \int_a^b f(t) dt$$

exists uniformly for $\sigma \in \Sigma(\pi)$.

Proof. For any given $\varepsilon > 0$ let δ be as in Lemma 11. Let $\pi, \pi' \in \Pi(a, b)$ be such that $\text{diam}(\pi) < \delta$ and $\text{diam}(\pi') < \delta$. Finally, let $\sigma \in \Sigma(\pi)$ and $\sigma' \in \Sigma(\pi')$. Define $\pi'' = \pi \cup \pi'$ and fix any $\sigma'' \in \Sigma(\pi'')$. Then

$$|S_{\pi}^{\sigma}(f) - S_{\pi'}^{\sigma'}(f)| \leq |S_{\pi}^{\sigma}(f) - S_{\pi''}^{\sigma''}(f)| + |S_{\pi''}^{\sigma''}(f) - S_{\pi'}^{\sigma'}(f)| < 2\varepsilon.$$

This completes the proof since ε is arbitrary. $\square$

Proposition 19 *For any $f, g \in C([a, b]; X)$ and $\lambda \in \mathbb{C}$ we have that*

$$\begin{aligned}
\int_a^b (f(t) + g(t)) dt &= \int_a^b f(t) dt + \int_a^b g(t) dt \\
\int_a^b \lambda f(t) dt &= \lambda \int_a^b f(t) dt \\
\left| \int_a^b f(t) dt \right| &\leq \int_a^b |f(t)| dt.
\end{aligned}$$

Moreover, for any $\phi \in X^*$ we have that

$$\left\langle \phi, \int_a^b f(t) dt \right\rangle = \int_a^b \langle \phi, f(t) \rangle dt. \quad (208)$$

and, for any $\Lambda \in \mathcal{L}(X)$ we have that

$$\Lambda \int_a^b f(t) dt = \int_a^b \Lambda f(t) dt. \quad (209)$$

Proof. Exercise. □

Proposition 20 For any $f \in C^1([a, b]; X)$ we have that

$$\int_a^b f'(t) dt = f(b) - f(a) \quad (210)$$

Proof. By (208) above, for any $\phi \in X^*$ we have that

$$\left\langle \phi, \int_a^b f'(t) dt \right\rangle = \int_a^b \langle \phi, f'(t) \rangle dt.$$

On the other hand, the function $t \mapsto \langle \phi, f(t) \rangle$ is continuously differentiable on $[a, b]$ with derivative equal to $\langle \phi, f'(t) \rangle$. Therefore, for any $\phi \in X^*$,

$$\int_a^b \langle \phi, f'(t) \rangle dt = \langle \phi, f(b) - f(a) \rangle.$$

Since X^* separates points, the above identity yields (210). □

Corollary 8 Let $f \in C^1([a, b]; X)$ be such that $f'(t) = 0$ for all $t \in [a, b]$. Then f is constant.

Appendix B: the Lebesgue integral on $L^2(a, b; H)$

In this appendix, $(H, \langle \cdot, \cdot \rangle)$ is a separable Hilbert space and $\{e_j\}_{j \in \mathbb{N}}$ is a complete orthonormal system in H .

5.4 The Hilbert space $L^2(a, b; H)$

Definition 29 A function $f : [a, b] \rightarrow H$ is said to be Borel (resp. Lebesgue) measurable if so is the scalar function $t \mapsto \langle f(t), u \rangle$ for every $u \in H$.

Remark 12 Let $f : [a, b] \rightarrow H$.

1. Since, for any $x \in H$,

$$\langle f(t), x \rangle = \sum_{j=1}^{\infty} \langle f(t), e_j \rangle \overline{\langle x, e_j \rangle} \quad (t \in [a, b]),$$

we conclude that f is Borel (resp. Lebesgue) measurable if and only if so is the scalar function $t \mapsto \langle f(t), e_j \rangle$ for every $j \in \mathbb{N}$.

2. Since

$$|f(t)|^2 = \sum_{j=1}^{\infty} |\langle f(t), e_j \rangle|^2 \quad (t \in [a, b]),$$

we have that, if f is Borel (resp. Lebesgue) measurable, then so is the scalar function $t \mapsto |f(t)|$.

Definition 30 We denote by $L^2(a, b; H)$ the space of all Lebesgue measurable functions $f : [a, b] \rightarrow H$ such that

$$\|f\|_2 := \left(\int_a^b |f(t)|^2 dt \right)^{\frac{1}{2}} < \infty,$$

where two functions f and g are identified if $f(t) = g(t)$ for a.e. $t \in [a, b]$.

As in the scalar case, $L^2(a, b; H)$ can be equipped with a Hilbert space structure.

Proposition 21 $L^2(a, b; H)$ is a Hilbert space with the hermitian product

$$(f|g)_0 = \int_a^b \langle f(t), g(t) \rangle dt \quad (f, g \in L^2(a, b; H)).$$

Proof. We only prove completeness. Let $\{f_n\}$ be a Cauchy sequence in $L^2(a, b; H)$. Then $\{f_n\}$ is bounded:

$$\|f_n\|_2^2 = \int_a^b \sum_{j=1}^{\infty} |\langle f_n(t), e_j \rangle|^2 dt \leq M \quad \forall n \in \mathbb{N} \quad (211)$$

Moreover, for any $\varepsilon > 0$ there exists $\nu \in \mathbb{N}$ such that, for all $m, n \geq \nu$,

$$\|f_n - f_m\|_2^2 = \int_a^b \sum_{j=1}^{\infty} |\langle f_n(t) - f_m(t), e_j \rangle|^2 dt \leq \varepsilon \quad (212)$$

Therefore, $t \mapsto \langle f_n(t), e_j \rangle$ is a Cauchy sequence in $L^2(a, b)$ for all $j \in \mathbb{N}$. So, there exists functions $\phi_j \in L^2(a, b)$ such that $\langle f_n(\cdot), e_j \rangle \rightarrow \phi_j$ in $L^2(a, b)$ for all $j \in \mathbb{N}$. Thus, by Fatou's lemma,

$$\int_a^b \sum_{j=1}^{\infty} |\phi_j(t)|^2 dt \leq M \quad \text{and} \quad \int_a^b \sum_{j=1}^{\infty} |\langle f_n(t), e_j \rangle - \phi_j(t)|^2 dt \leq \varepsilon \quad (\forall n \geq \nu).$$

So, we conclude that

$$f(t) := \sum_{j=1}^{\infty} \phi_j(t) e_j \in H \quad t \in [a, b] \text{ a.e.}$$

as well as $f \in L^2(a, b; H)$ and

$$\int_a^b |f_n(t) - f(t)|^2 dt \leq \varepsilon \quad (\forall n \geq \nu),$$

or, $f_n \rightarrow f$ in $L^2(a, b; H)$. □

Remark 13 For any $f \in L^2(a, b; H)$ we have that

$$\sum_{j=1}^{\infty} \left| \int_a^b \langle f(t), e_j \rangle dt \right|^2 \leq (b-a) \sum_{j=1}^{\infty} \int_a^b |\langle f(t), e_j \rangle|^2 dt < \infty.$$

Therefore

$$\sum_{j=1}^{\infty} e_j \int_a^b \langle f(t), e_j \rangle dt \in H.$$

Moreover, given any complete orthonormal system $\{\tilde{e}_k\}_{k \in \mathbb{N}}$ in H , we have that

$$\tilde{e}_k = \sum_{j=1}^{\infty} \langle \tilde{e}_k, e_j \rangle e_j \quad \text{and} \quad \langle f(t), \tilde{e}_k \rangle = \sum_{j=1}^{\infty} \langle f(t), e_j \rangle \overline{\langle \tilde{e}_k, e_j \rangle} \quad t \in [a, b] \text{ a.e.}$$

Thus

$$\begin{aligned}
\sum_{k=1}^{\infty} \tilde{e}_k \int_a^b \langle f(t), \tilde{e}_k \rangle dt &= \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \langle \tilde{e}_k, e_j \rangle e_j \sum_{i=1}^{\infty} \int_a^b \langle f(t), e_i \rangle \overline{\langle \tilde{e}_k, e_i \rangle} dt \\
&= \sum_{j=1}^{\infty} e_j \sum_{i=1}^{\infty} \int_a^b \langle f(t), e_i \rangle dt \sum_{k=1}^{\infty} \langle \tilde{e}_k, e_j \rangle \overline{\langle \tilde{e}_k, e_i \rangle} \\
&= \sum_{j=1}^{\infty} e_j \sum_{i=1}^{\infty} \langle e_i, e_j \rangle \int_a^b \langle f(t), e_i \rangle dt = \sum_{j=1}^{\infty} e_j \int_a^b \langle f(t), e_j \rangle dt.
\end{aligned}$$

The above remark motivates the following definition.

Definition 31 For any $f \in L^2(a, b; H)$ we define

$$\int_a^b f(t) dt = \sum_{j=1}^{\infty} e_j \int_a^b \langle f(t), e_j \rangle dt.$$

The proof of the following proposition is left as an exercise to the reader.

Proposition 22 For any $f \in L^2(a, b; H)$ the following properties hold true.

1. For any $u \in H$ we have that

$$\left\langle u, \int_a^b f(t) dt \right\rangle = \int_a^b \langle u, f(t) \rangle dt$$

2. $\left| \int_a^b f(t) dt \right| \leq \int_a^b |f(t)| dt$

3. For any $\Lambda \in \mathcal{L}(H)$ we have that

$$\Lambda \left(\int_a^b f(t) dt \right) = \int_a^b \Lambda f(t) dt.$$

The following is a familiar density result.

Lemma 12 The space $C([a, b]; X)$ is dense in $L^2(a, b; H)$.

Proof. Let $f \in L^2(a, b; H)$ and recall that

$$\|f\|_2^2 = \int_a^b \sum_{j=1}^{\infty} |f_j(t)|^2 dt \quad \text{where} \quad f_j(t) = \langle f(t), e_j \rangle.$$

Fix any $\varepsilon > 0$ and let $j_\varepsilon \in \mathbb{N}$ be such that

$$\int_a^b \sum_{j=j_\varepsilon+1}^{\infty} |f_j(t)|^2 dt < \varepsilon. \quad (213)$$

For each $j = 1, \dots, j_\varepsilon$ there exists a function $\tilde{f}_j \in C([a, b])$ such that

$$\int_a^b |f_j(t) - \tilde{f}_j(t)|^2 dt < \frac{\varepsilon}{2^j}. \quad (214)$$

So, defining

$$\tilde{f}_\varepsilon(t) = \sum_{j=1}^{j_\varepsilon} \tilde{f}_j(t) e_j \quad (t \in [a, b])$$

we have that $\tilde{f}_\varepsilon \in C([a, b]; X)$. Moreover, (213) and (214) yield

$$\int_a^b |f(t) - \tilde{f}_\varepsilon(t)|^2 dt < \sum_{j=1}^{j_\varepsilon} \frac{\varepsilon}{2^j} + \varepsilon < 2\varepsilon$$

thus completing the proof. $\square$

Proposition 23 *Let $A : D(A) \subset H \rightarrow H$ be the infinitesimal generator of a C_0 -semigroup on H , $S(t)$. Then for any $f \in L^2(a, b; D(A))$ we have that*

$$\int_a^b f(t) dt \in D(A) \quad \text{and} \quad A\left(\int_a^b f(t) dt\right) = \int_a^b Af(t) dt.$$

Proof. For any $v \in D(A^*)$ we have that

$$\left\langle \int_a^b f(t) dt, A^*v \right\rangle = \int_a^b \langle f(t), A^*v \rangle dt = \int_a^b \langle Af(t), v \rangle dt = \left\langle \int_a^b Af(t) dt, v \right\rangle$$

The conclusion follows by Lemma 4. $\square$

Proposition 24 *Let $A : D(A) \subset H \rightarrow H$ be the infinitesimal generator of a C_0 -semigroup on H , $S(t)$. Then, for any $f \in L^2(a, b; H)$, we have that:*

- (a) *for any $t \in [a, b]$ the function $s \mapsto S(t-s)f(s)$ belongs to $L^2(a, t; H)$;*
- (b) *the function*

$$F_A(t) = \int_a^t S(t-s)f(s) ds \quad (t \in [a, b])$$

belongs to $C([a, b]; H)$.

Proof. In order to check measurability for $s \mapsto S(t-s)f(s)$ it suffices to observe that, for all $u \in H$ and a.e. $s \in [0, t]$,

$$\langle S(t-s)f(s), u \rangle = \langle f(s), S(t-s)^*u \rangle = \sum_{j=1}^{\infty} \langle f(s), e_j \rangle \overline{\langle S(t-s)^*u, e_j \rangle}.$$

Since $s \mapsto \langle S(t-s)^*u, e_j \rangle$ is continuous and $s \mapsto \langle f(s), e_j \rangle$ is measurable for all $j \in \mathbb{N}$, the measurability of $s \mapsto S(t-s)f(s)$ follows. Moreover, by (28) we have that

$$|S(t-s)f(s)| \leq M e^{\omega(t-s)} |f(s)| \quad (s \in [a, t] \text{ a.e.}),$$

which completes the proof of (a).

In order to prove point (b), fix $t \in]a, b[$ and let $t_n \rightarrow t$. Fix $\delta \in]0, t - a[$ and let $n_\delta \in \mathbb{N}$ be such that $t_n > t - \delta$ for all $n \geq n_\delta$. Then we have that

$$\begin{aligned} & |F_A(t_n) - F_A(t)| \\ & \leq \int_a^{t-\delta} |[S(t_n - s)f(s) - S(t - s)]f(s)| ds \\ & \quad + \int_{t-\delta}^{t_n} |S(t_n - s)f(s)| ds + \int_{t-\delta}^t |S(t - s)f(s)| ds. \end{aligned}$$

To complete the proof it suffices to observe that

$$\lim_{n \rightarrow \infty} \int_a^{t-\delta} |[S(t_n - s)f(s) - S(t - s)]f(s)| ds = 0$$

by the dominated convergence theorem, while the remaining terms on the right-hand side of the above inequality are small with δ . $\square$

Sobolev spaces can be defined as follows for vector-valued functions.

Definition 32 $H^1(a, b; H)$ is the space of all functions $u \in C([a, b]; H)$ such that

1. $u'(t)$ exists for a.e. $t \in [a, b]$;
2. $u' \in L^2(a, b; H)$;
3. $u(t) - u(a) = \int_a^t u'(s) ds \quad t \in [a, b]$ a.e.

Remark 14 $H^1(a, b; H)$ is a Hilbert space with the scalar product

$$(u|v)_1 = \int_a^b [\langle u(t), v(t) \rangle + \langle u'(t), v'(t) \rangle] dt \quad (u, v \in H^1(a, b; H)).$$

Appendix C: Gronwall's lemma

Lemma 13 *Let $0 < T < \infty$ and let $u, v, w \in L^2(0, T)$ be such that*

$$v(t) \geq 0 \quad \text{and} \quad u(t) \leq w(t) + \int_0^t v(s)u(s) \, ds \quad (t \in [0, T] \text{ a.e.}) \quad (215)$$

Then

$$u(t) \leq w(t) + \int_0^t w(s)v(s)e^{\int_s^t v(\tau)d\tau} \, ds \quad (t \in [0, T] \text{ a.e.}) \quad (216)$$

Moreover, if $w \in AC([0, T])$, then

$$u(t) \leq w(0)e^{\int_0^t v(s) \, ds} + \int_0^t w'(s)e^{\int_s^t v(\tau)d\tau} \, ds \quad (t \in [0, T] \text{ a.e.}) \quad (217)$$

Proof. Define

$$U(t) = \int_0^t v(s)u(s) \, ds \quad (t \in [0, T])$$

Then U is absolutely continuous on $[0, T]$ and satisfies the following

$$\begin{aligned} \frac{d}{dt} \left(U(t)e^{-\int_0^t v(s) \, ds} \right) &= \left(v(t)u(t) - v(t) \int_0^t v(s)u(s) \, ds \right) e^{-\int_0^t v(s) \, ds} \\ &\leq w(t)v(t)e^{-\int_0^t v(s) \, ds} \quad (t \in [0, T] \text{ a.e.}) \end{aligned}$$

owing to (215). Therefore, by integrating the above inequality we obtain

$$U(t) \leq \int_0^t w(s)v(s)e^{\int_s^t v(\tau)d\tau} \, ds \quad (t \in [0, T]),$$

which, inserted into (215), gives (216).

Next, in order to prove (217) observe that, if $w \in AC([0, T])$, then

$$\begin{aligned} \int_0^t w(s)v(s)e^{\int_s^t v(\tau)d\tau} \, ds &= - \int_0^t w(s) \frac{d}{ds} e^{\int_s^t v(\tau)d\tau} \, ds \\ &= \left[-w(s) e^{\int_s^t v(\tau)d\tau} \right]_{s=0}^{s=t} + \int_0^t w'(s) e^{\int_s^t v(\tau)d\tau} \, ds \\ &= w(0) e^{\int_0^t v(\tau)d\tau} - w(t) + \int_0^t w'(s) e^{\int_s^t v(\tau)d\tau} \, ds. \end{aligned}$$

Plugging the above identity into (215) yields (217). $\square$

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